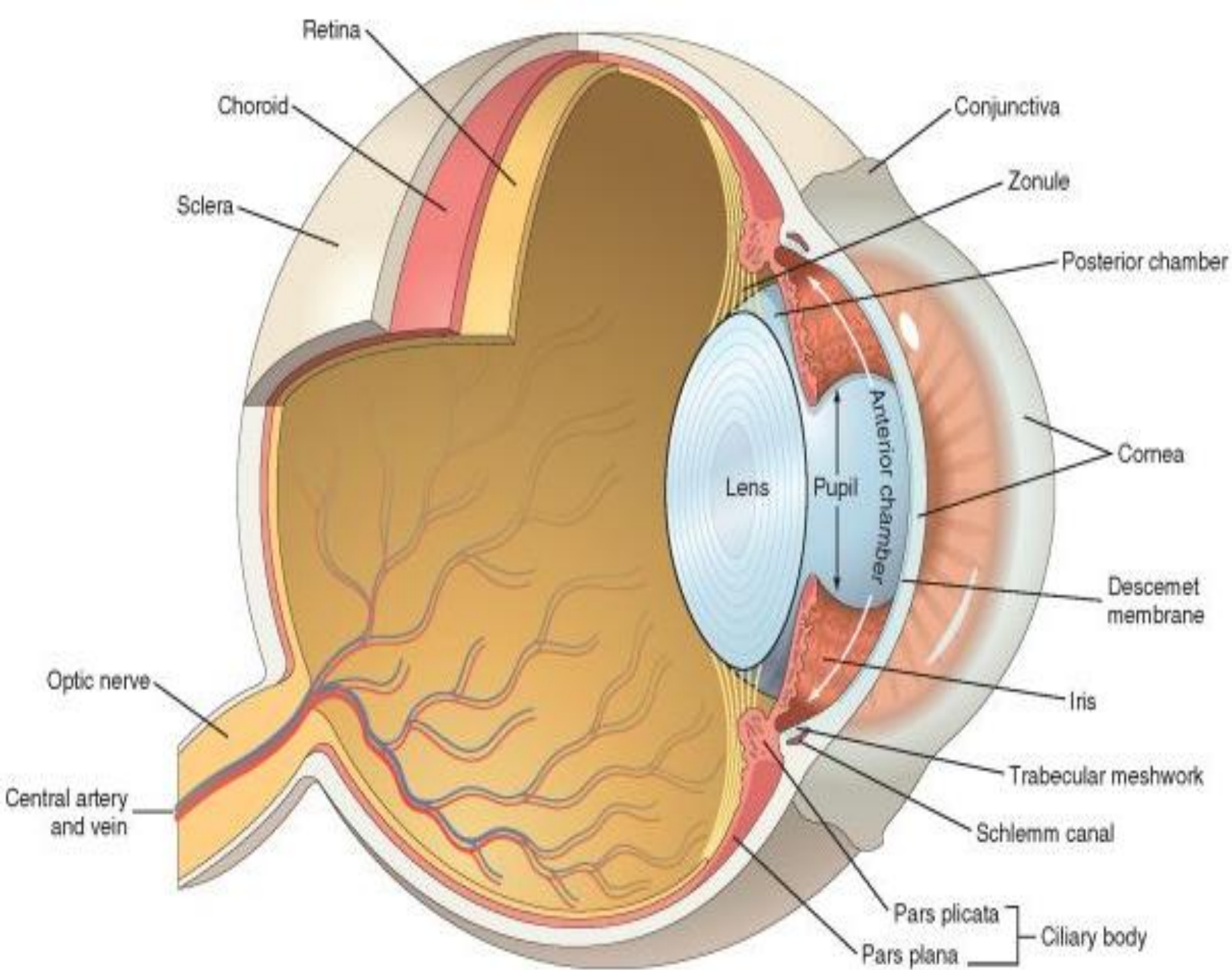


Eye

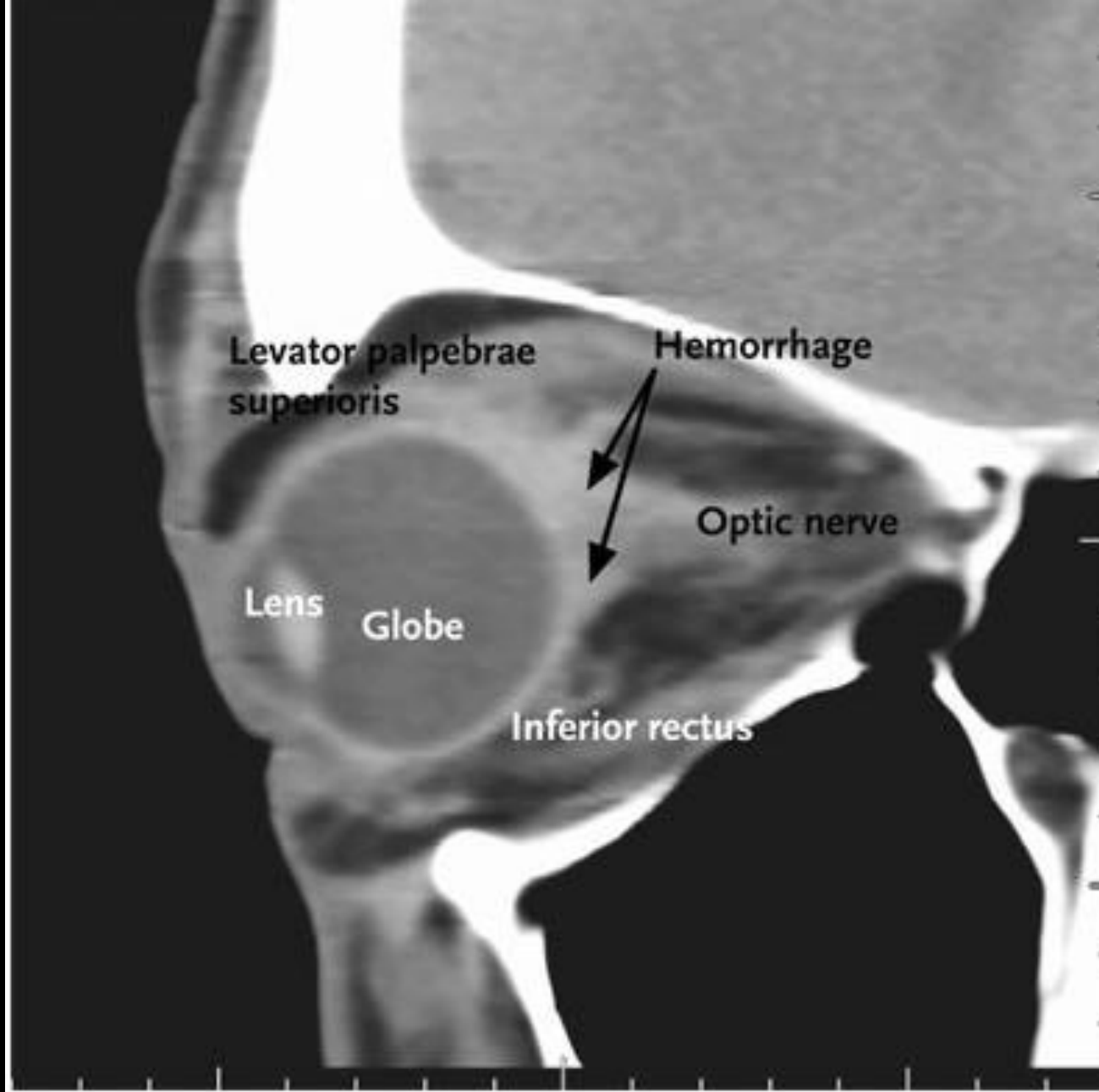


Blindness in Ethiopia

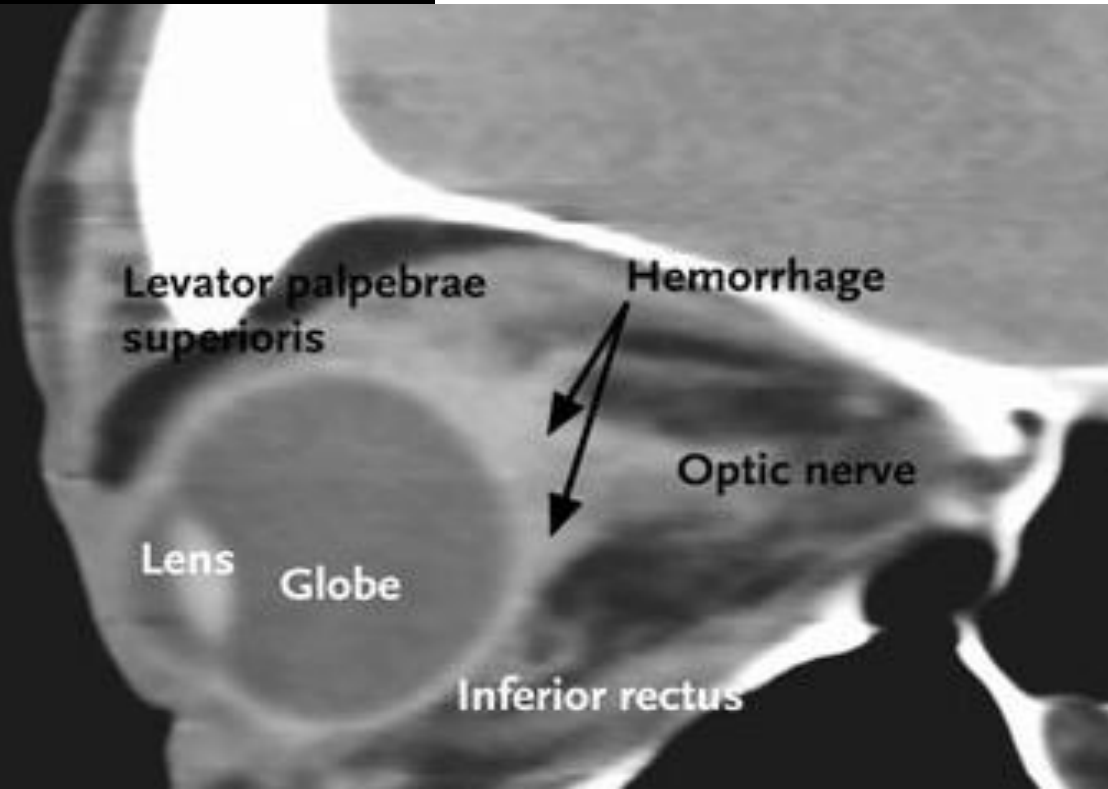
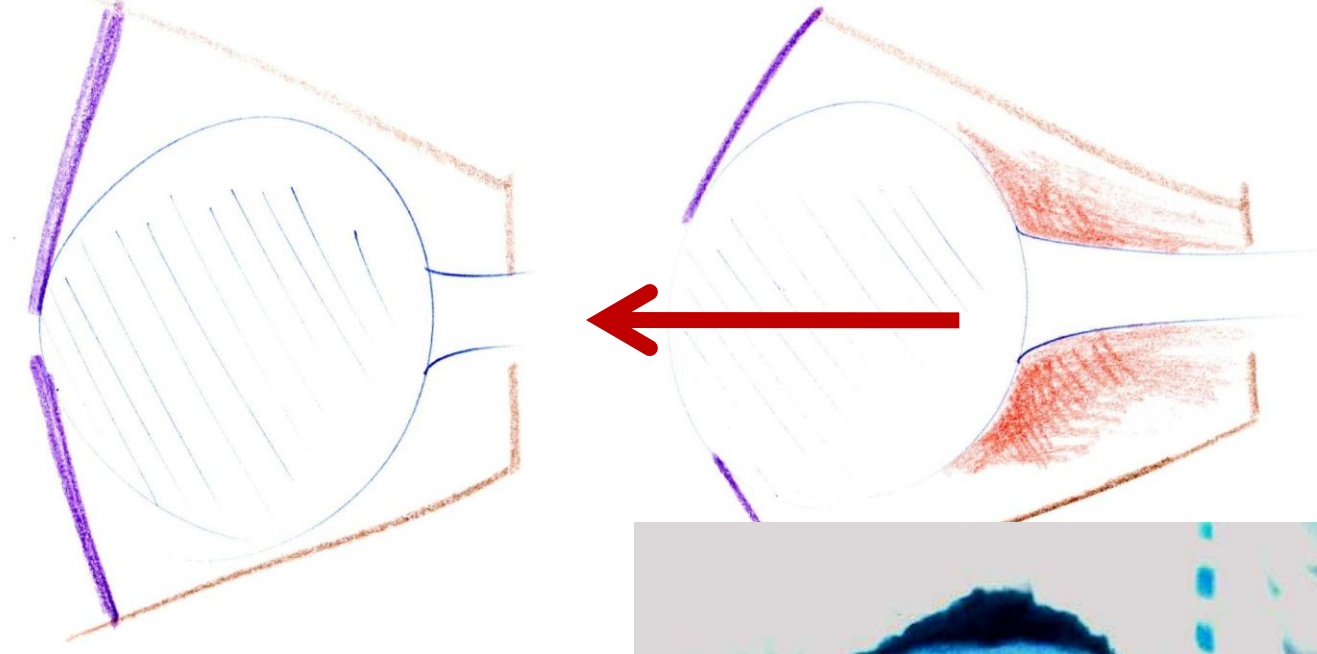
Monitor Oct. 13, 2008:

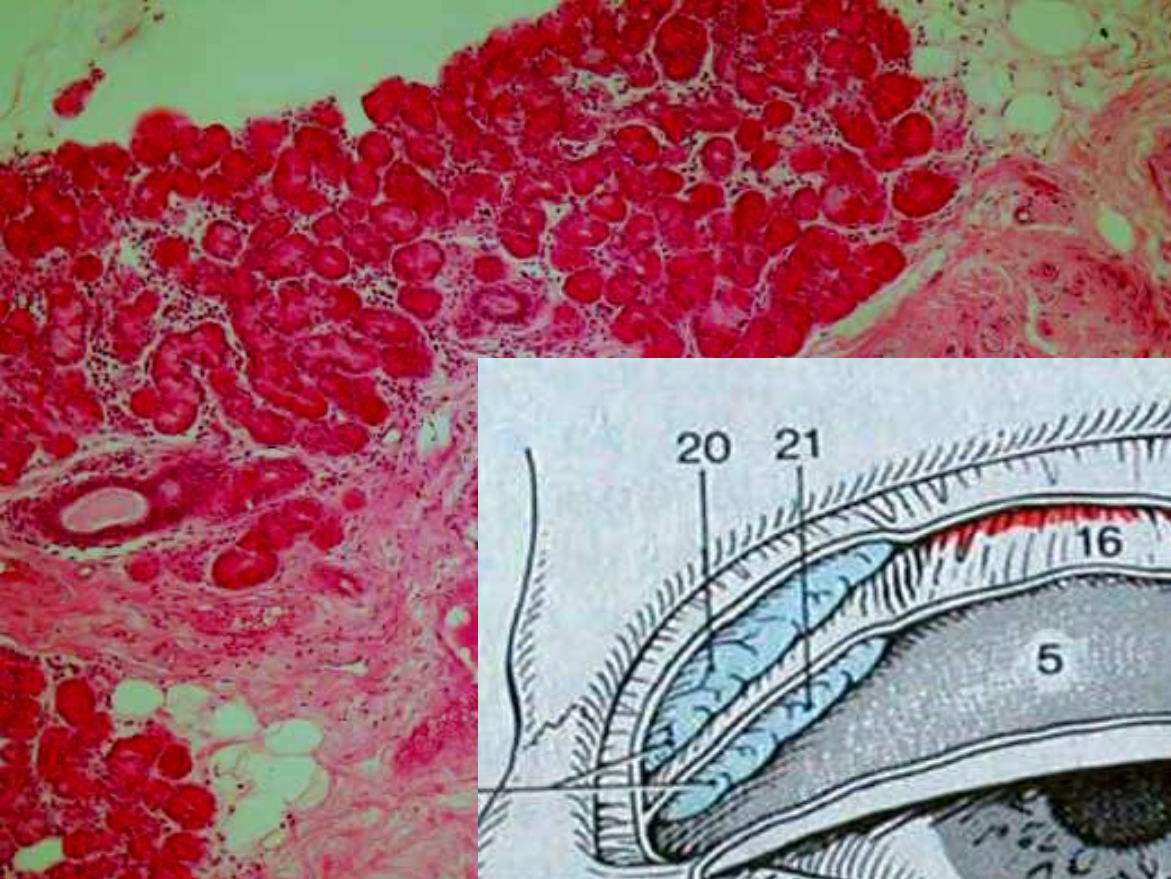
- **National prevalence of blindness: 1.6% (1.28 mio) and low vision 3.7% (2.96 mio)**
- **Major causes:**
 - Cataract, the leading cause of blindness in the world, also the leading cause of blindness in ET, accounting for 49.9%
 - Trachoma: 11.5%
 - Corneal opacity: 7.8%
 - Refractive error: 7.8%
 - Glaucoma: 5.2%
- **About 90% of blindness is preventable**

Orbit

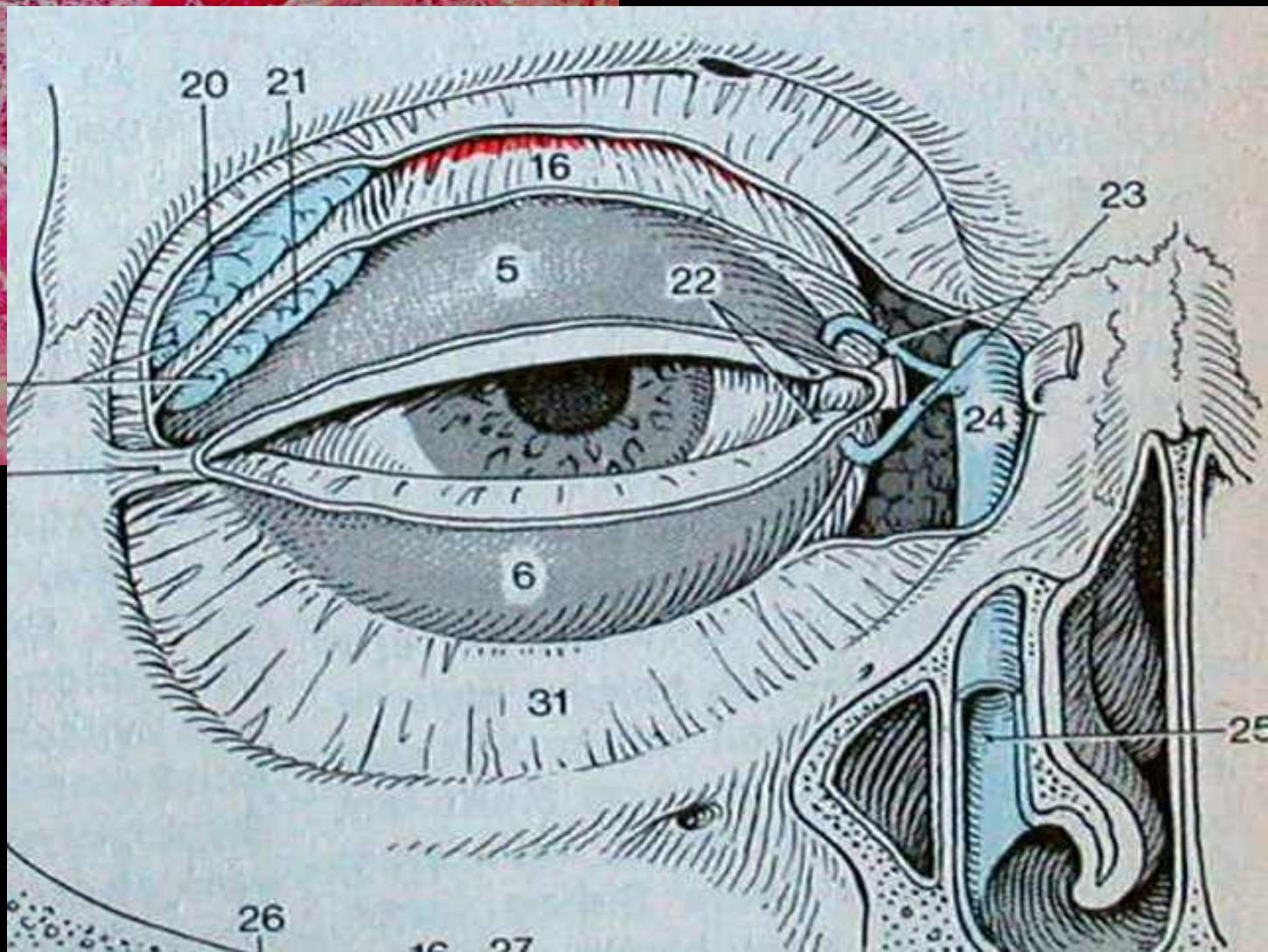


Proptosis Exophthalmos





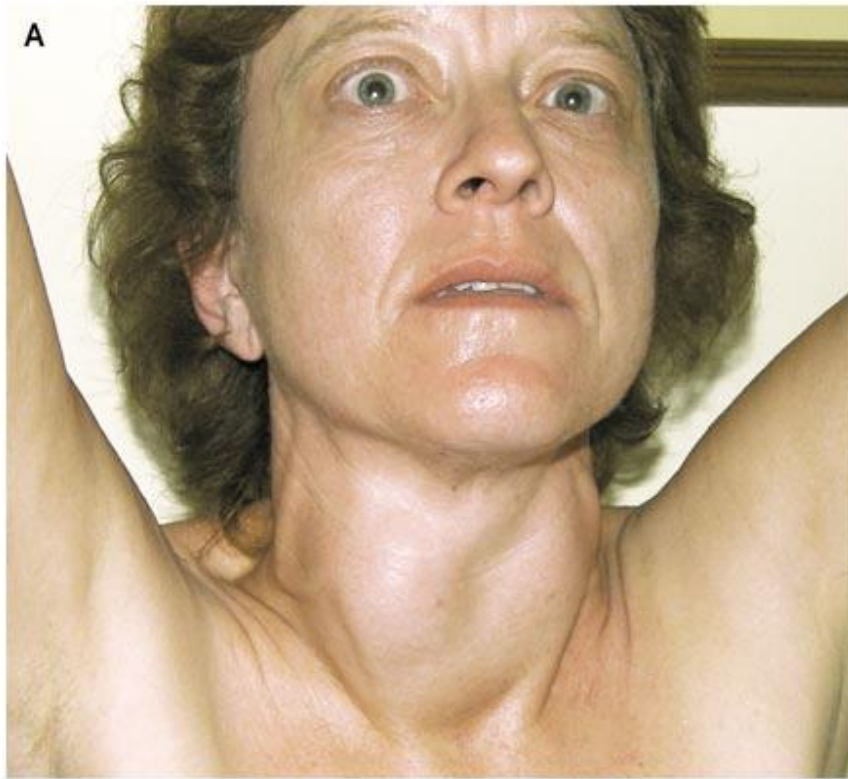
lacrimal gland



Orbital diseases in Adults

- **Thyroid orbitopathy** **60%**
- Neoplasia 12%
- Inflammation 8%

A 42-year-old woman presented with palpitations, anxiety, tremor, and weight loss

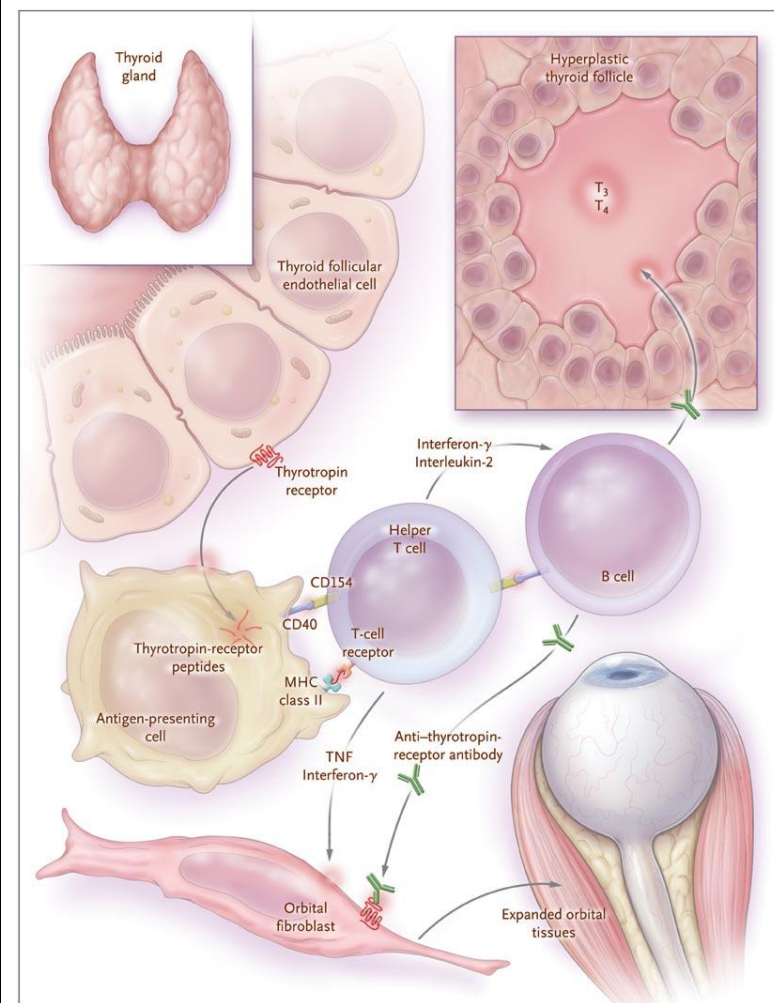


Iraci G and Fux-Otta C. N Engl J Med 2009;360:e31



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Patients with Graves' Ophthalmopathy



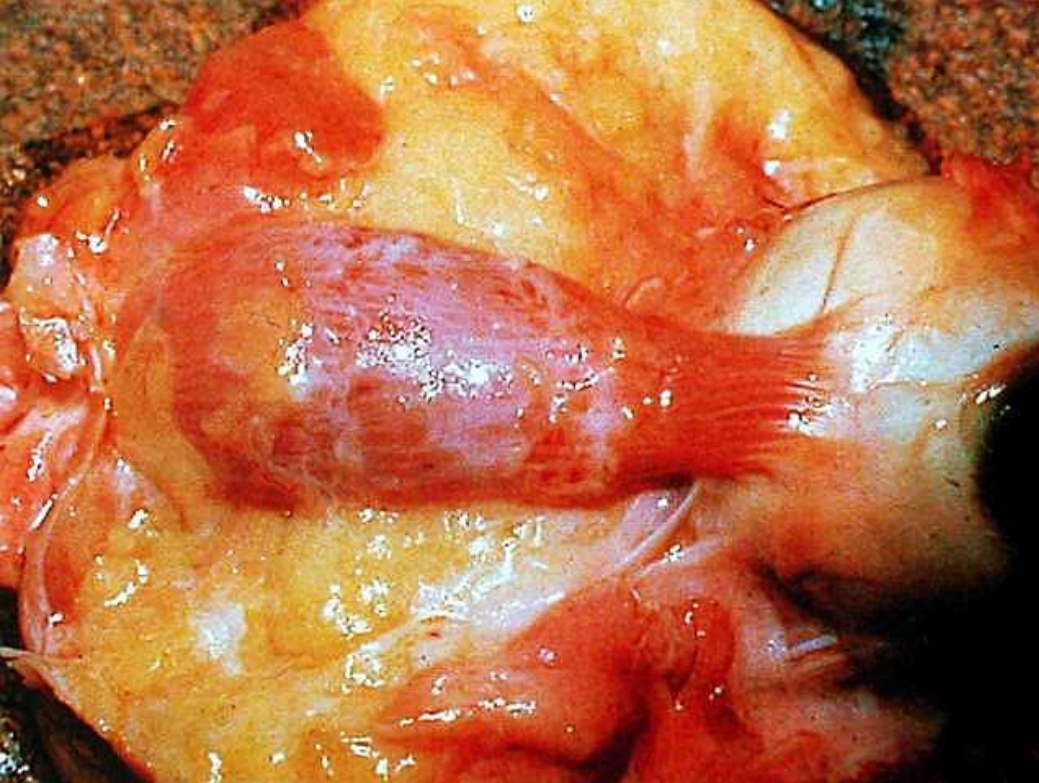
Bahn R. N Engl J Med 2010;362:726-738



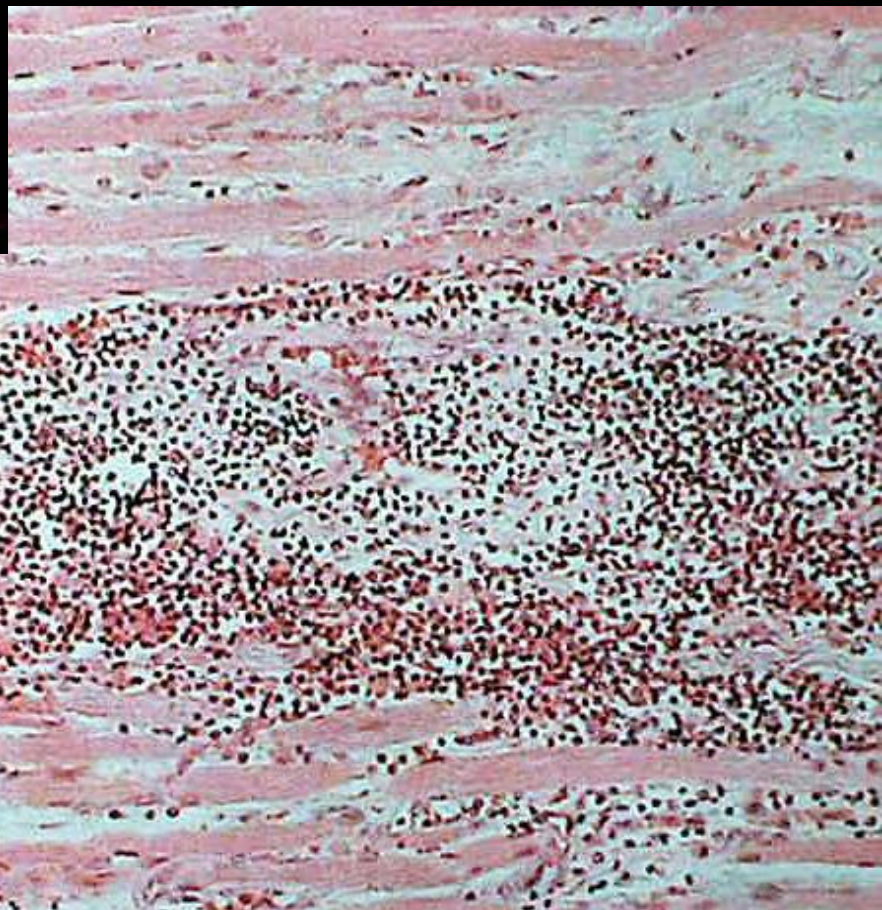
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Graves Ophthalmopathy

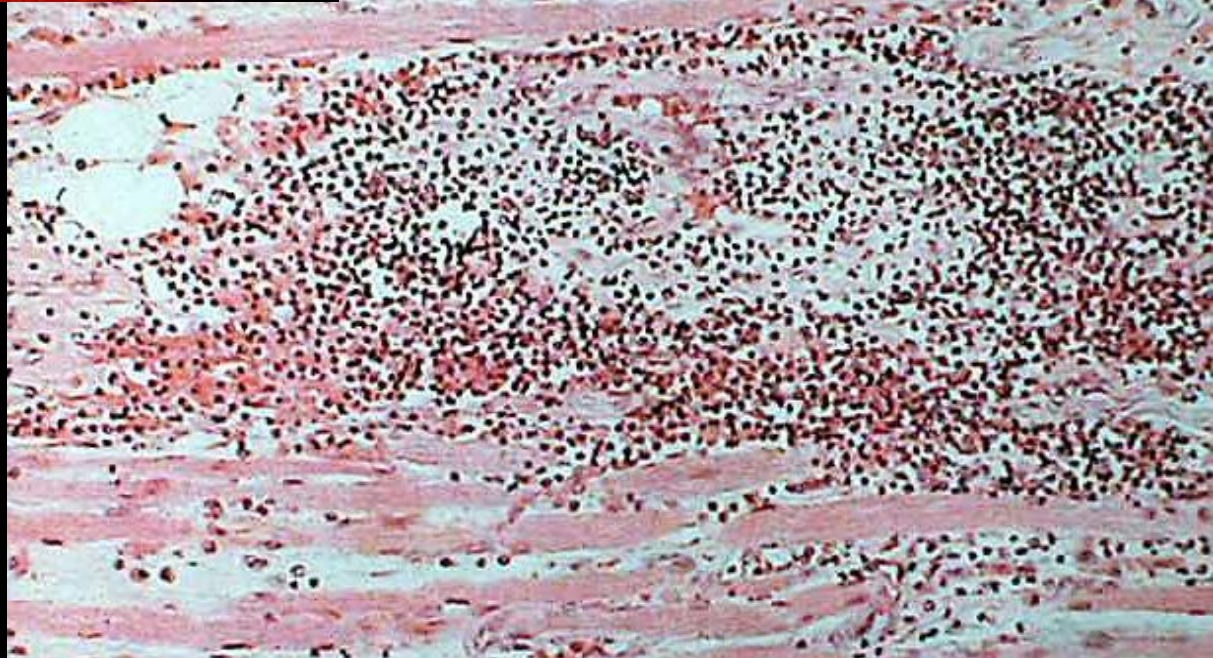
- Accumulation of T cells
- Edema of extraocular muscles.
- Deposition of proteoglycans(eg.hyaluronic acid) by the fibroblasts which are stimulated by Antibodies to TSH
- Increased number of adipocytes.



Graves' disease

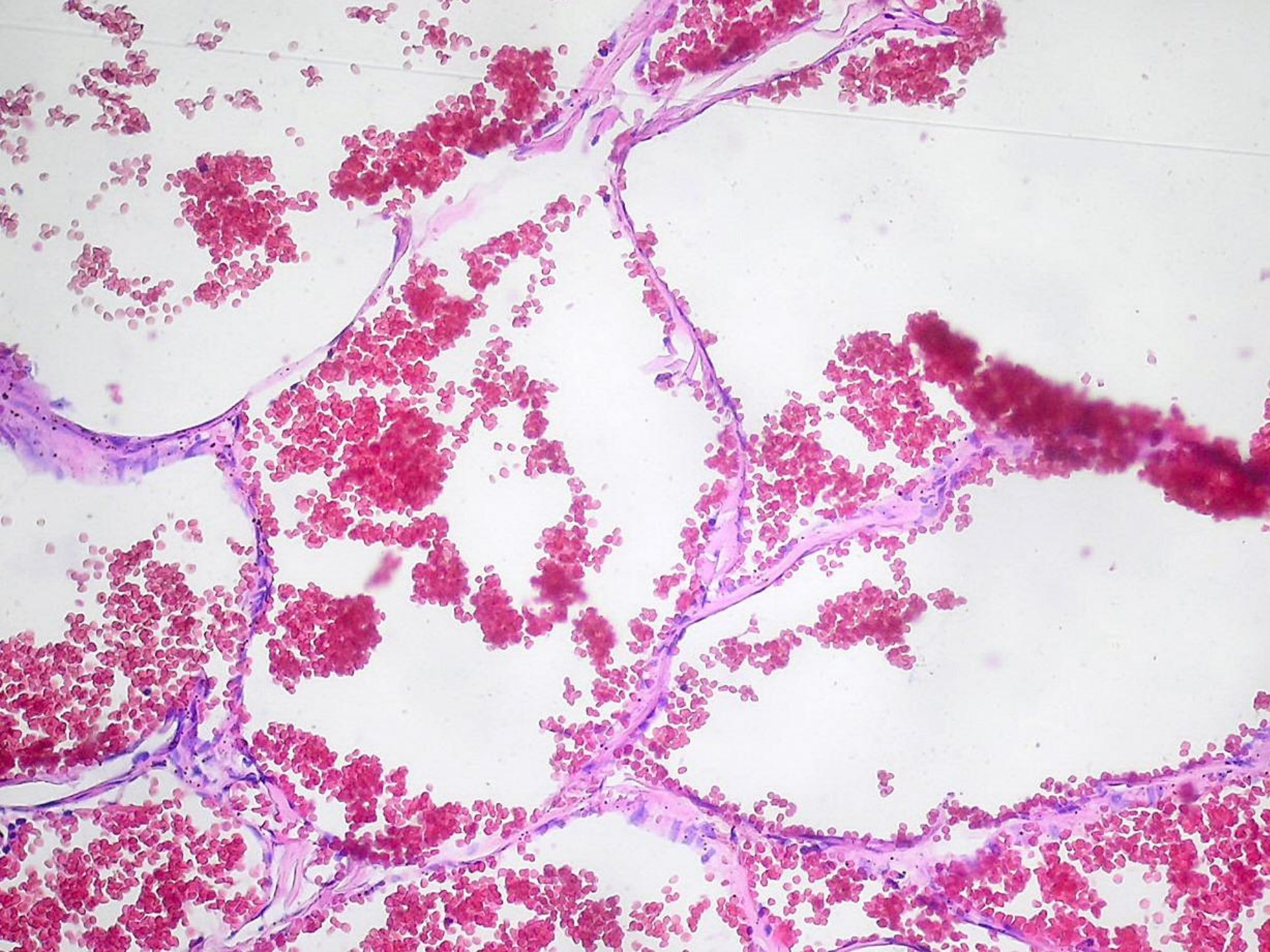


Thyroid orbitopathy



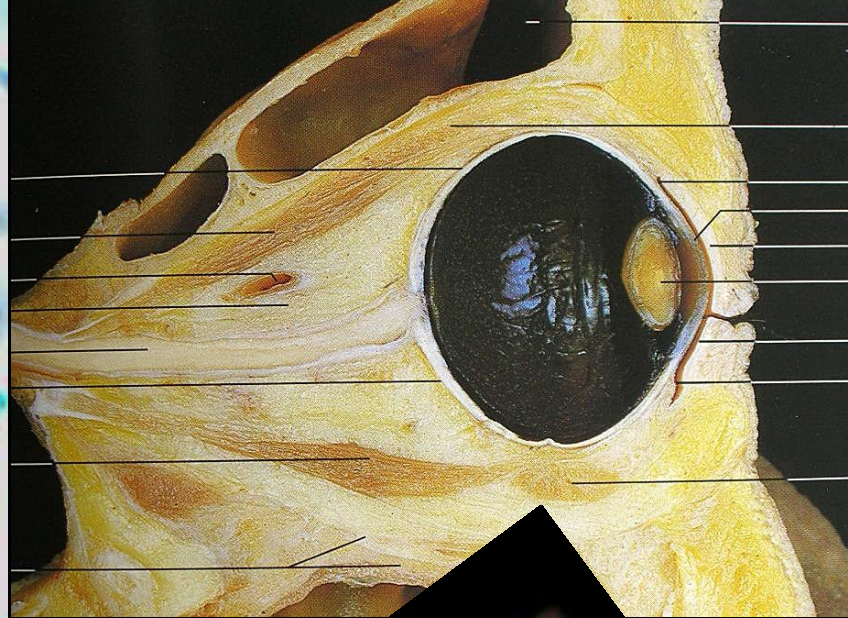
Orbital diseases in Adults and Children

- Thyroid orbitopathy 60%
- **Neoplasia** **12%**
 - Cavernous hemangioma
 - Schwannoma
 - Rhabdomyosarcoma
 - non-Hodgkin lymphoma
 - Larimal Gland Tumors(Pleomorphic adenoma, adenoid cystic carcinoma,
- Inflammation 8%



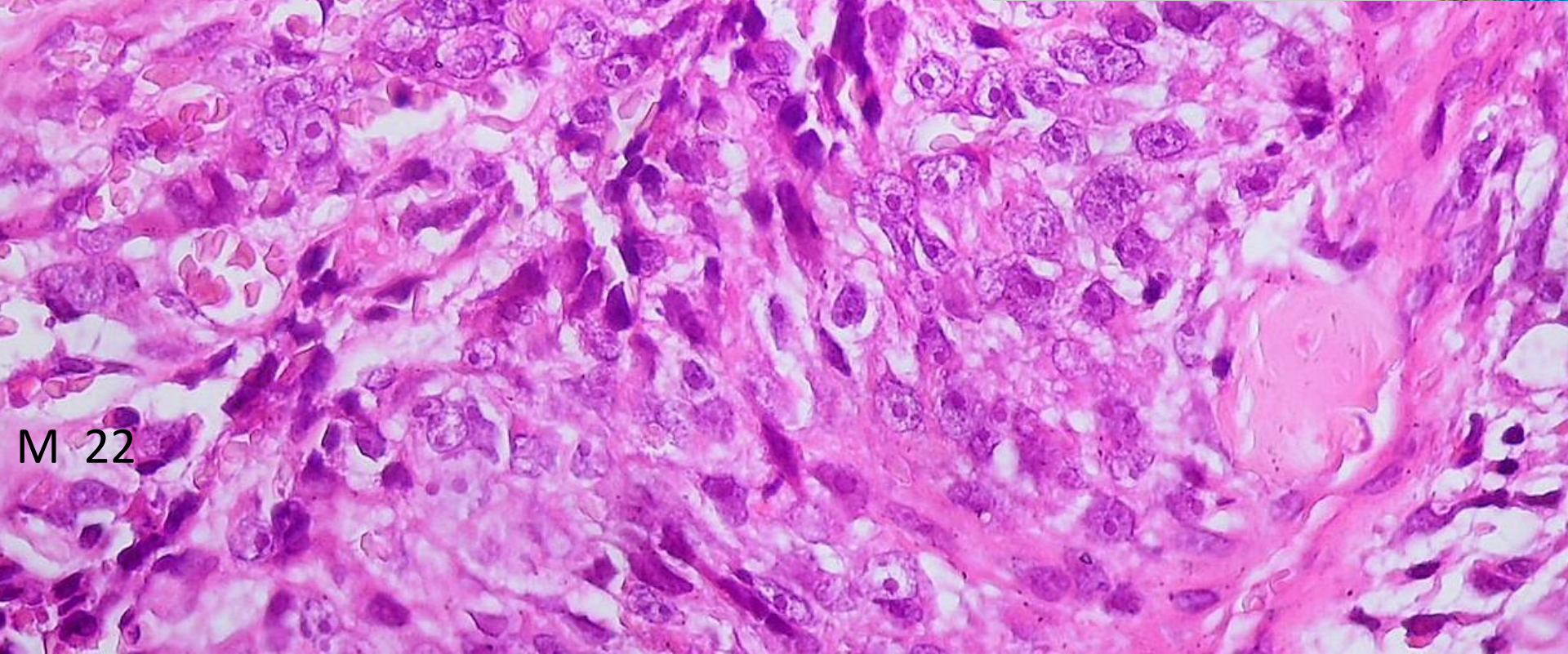
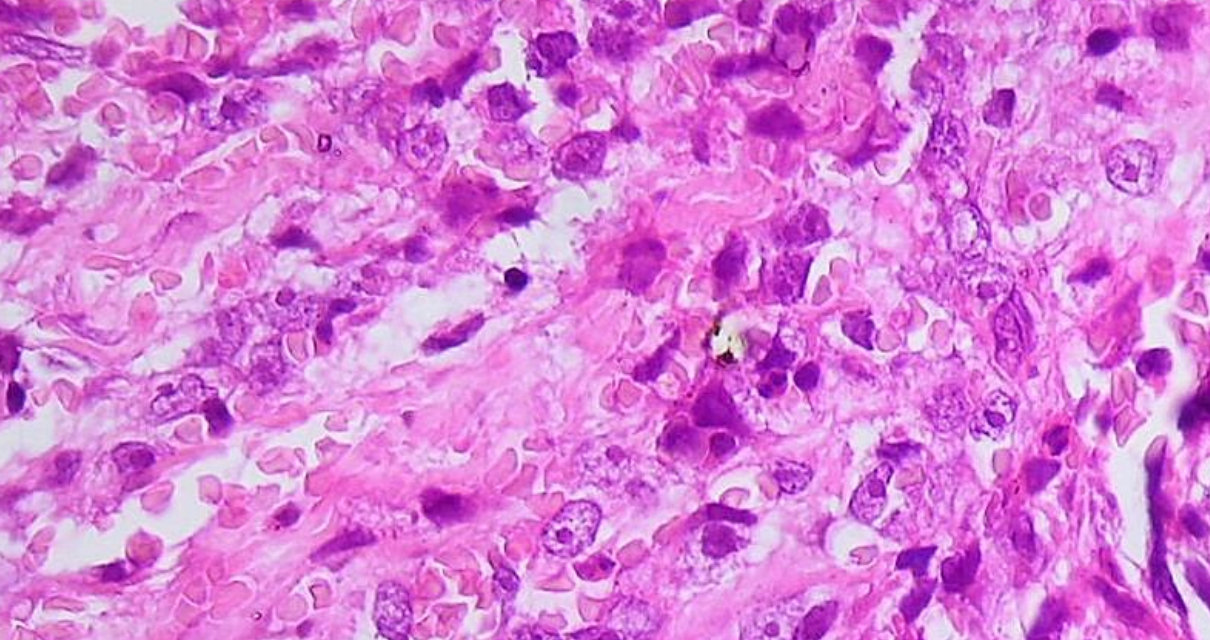
Proptosis

Exophthalmus



Schwannoma

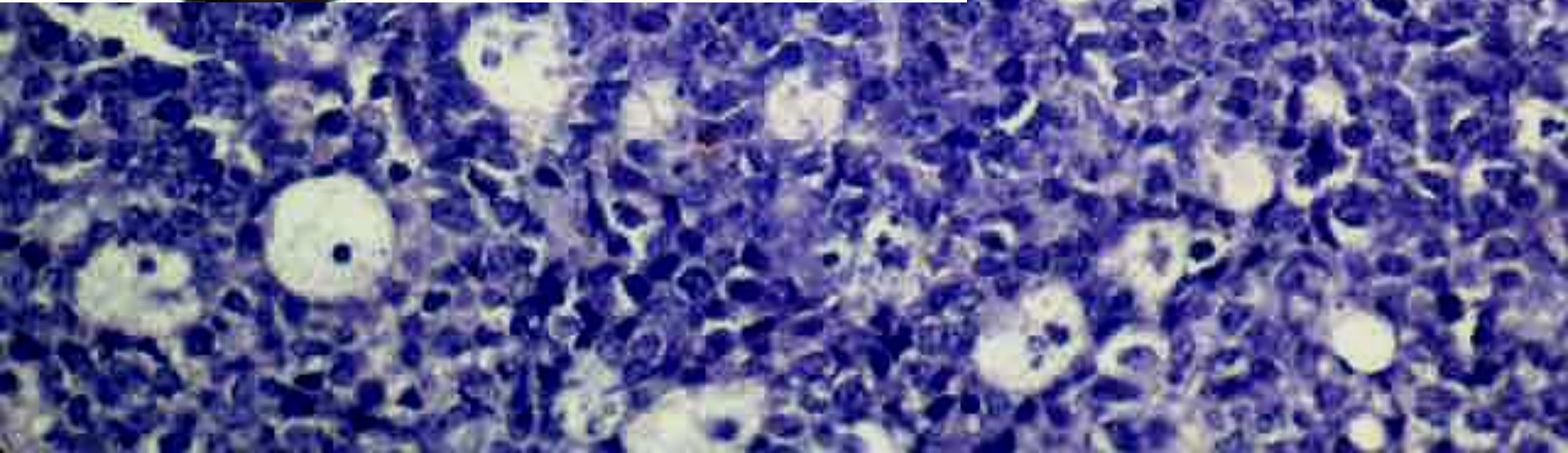
Rhabdomyosarcoma



M 22

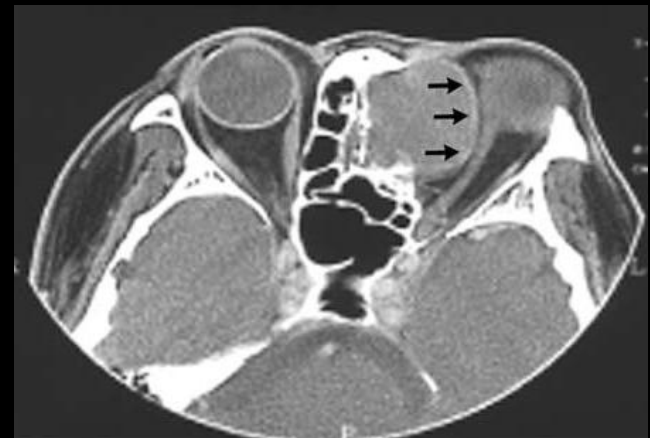
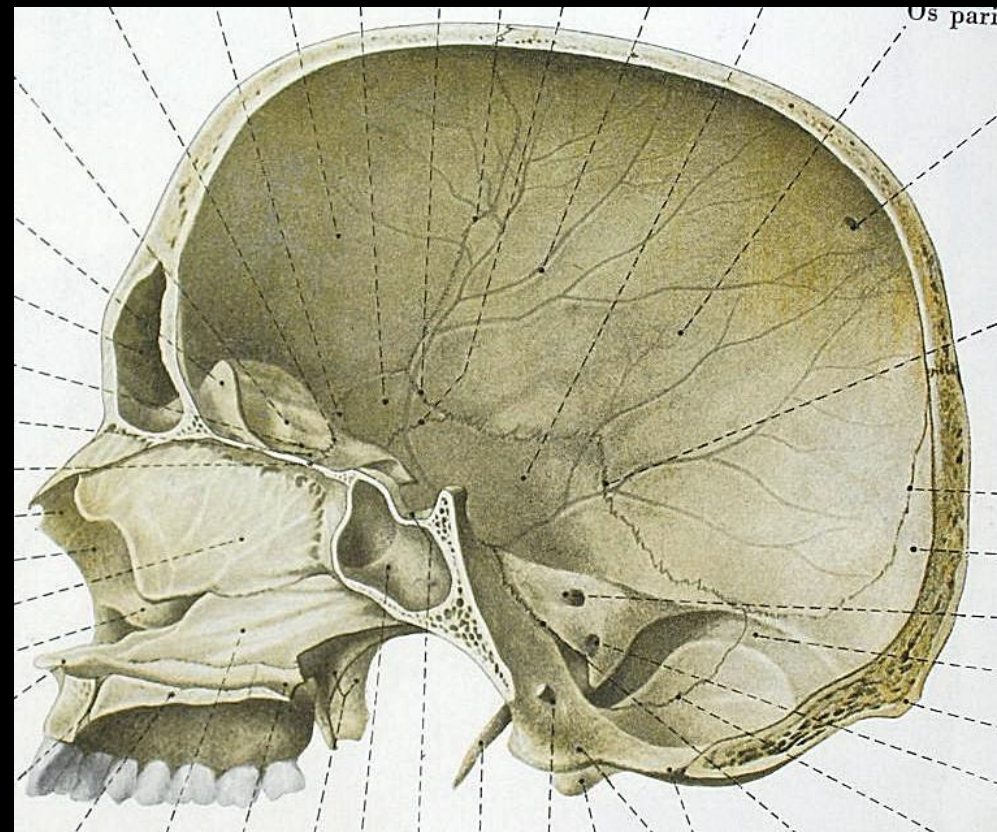
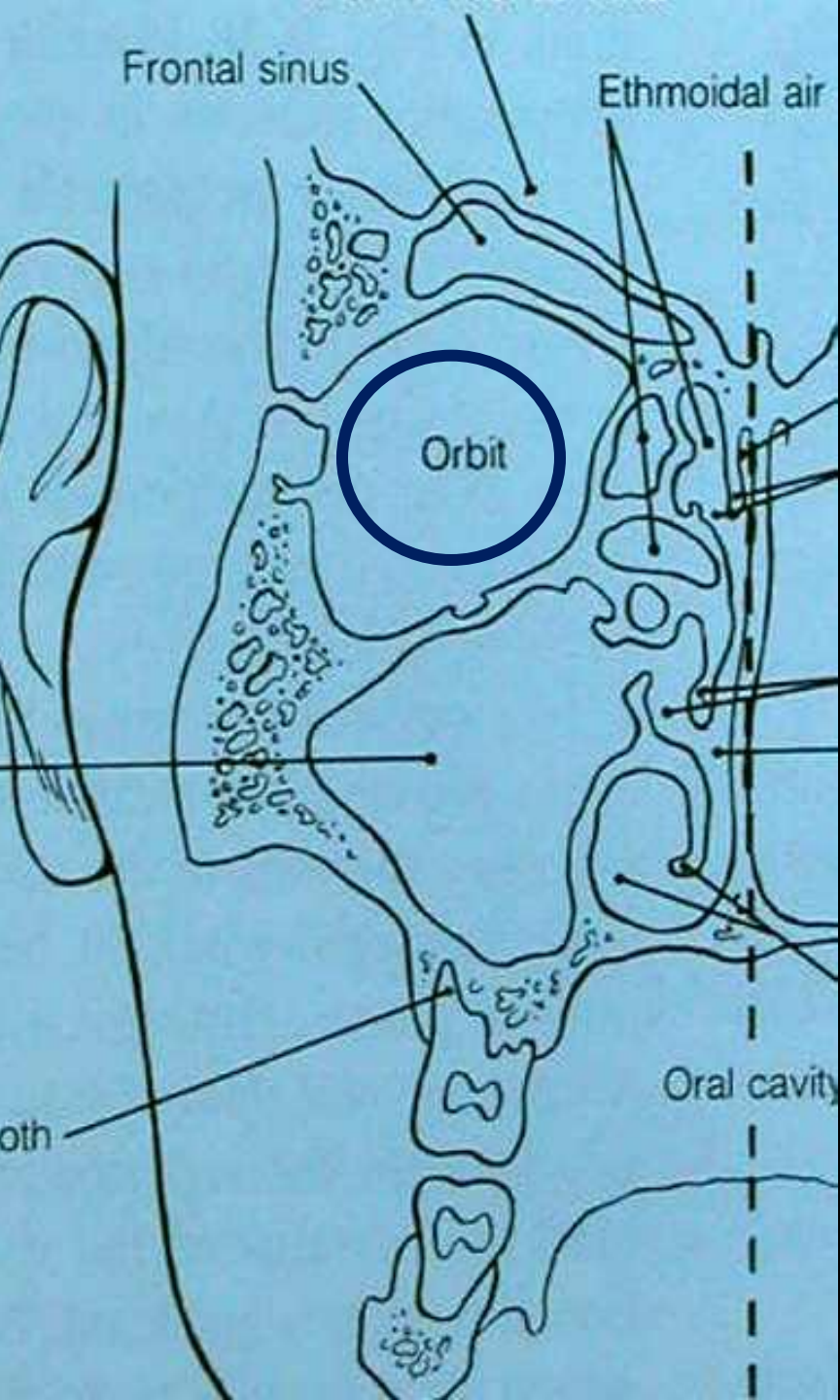


Non-Hodgkin lymphoma

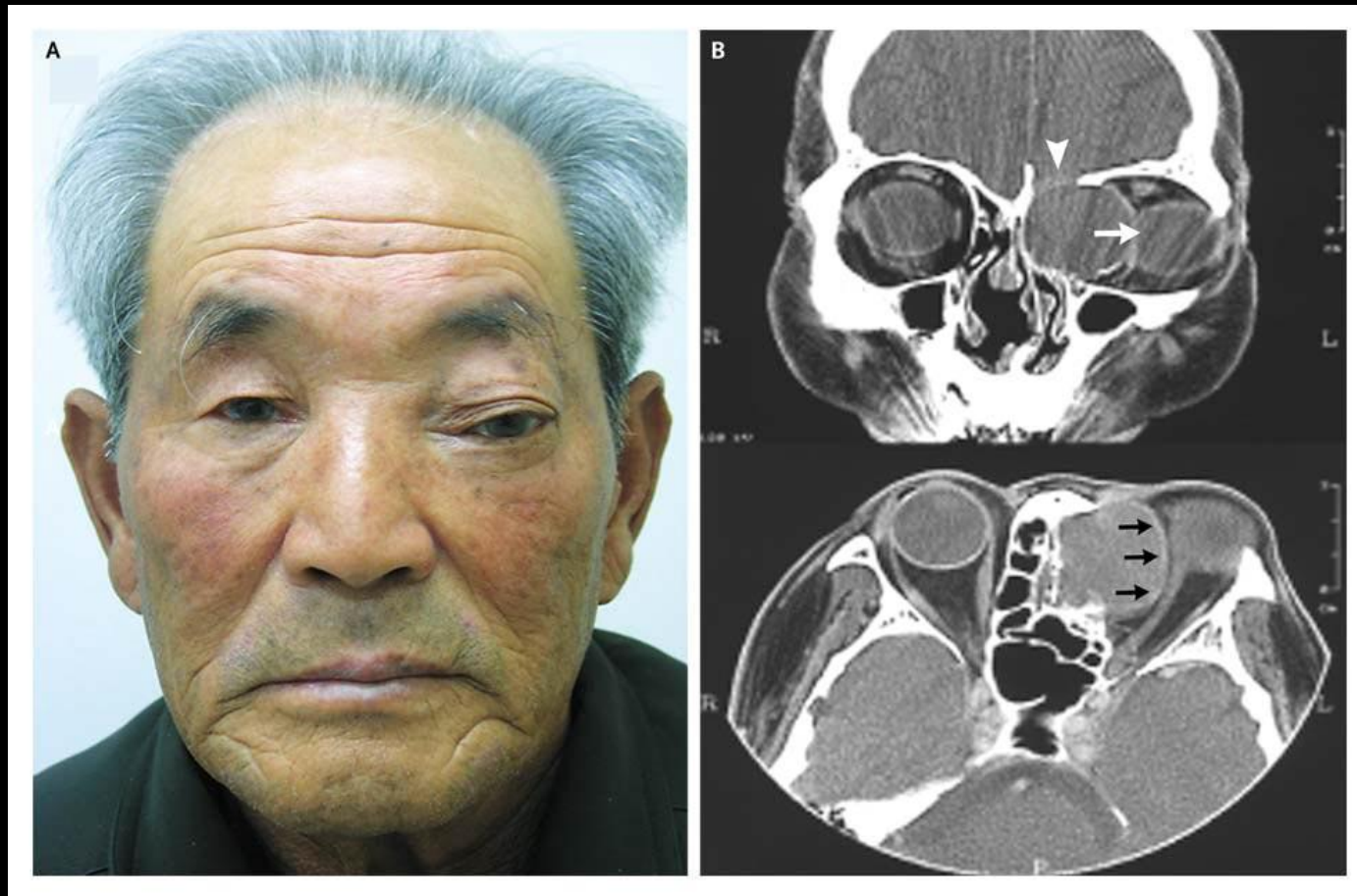


Orbital diseases in Adults

- Thyroid orbitopathy 60%
- Neoplasia 12%
 - Schwannoma
 - Rhabdomyosarcoma
 - non-Hodgkin lymphoma
- **Inflammation 8%**



A 74-year-old man was referred because of left exophthalmos and outward displacement of the orbit (Panel A)

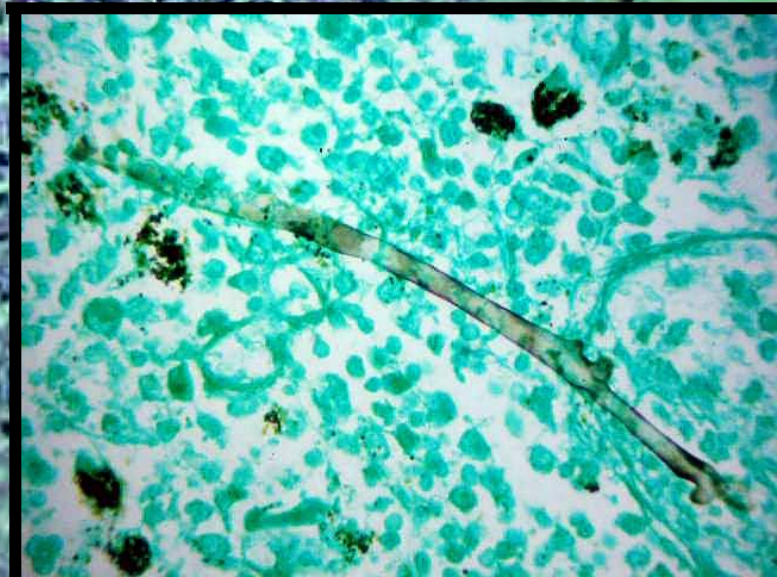
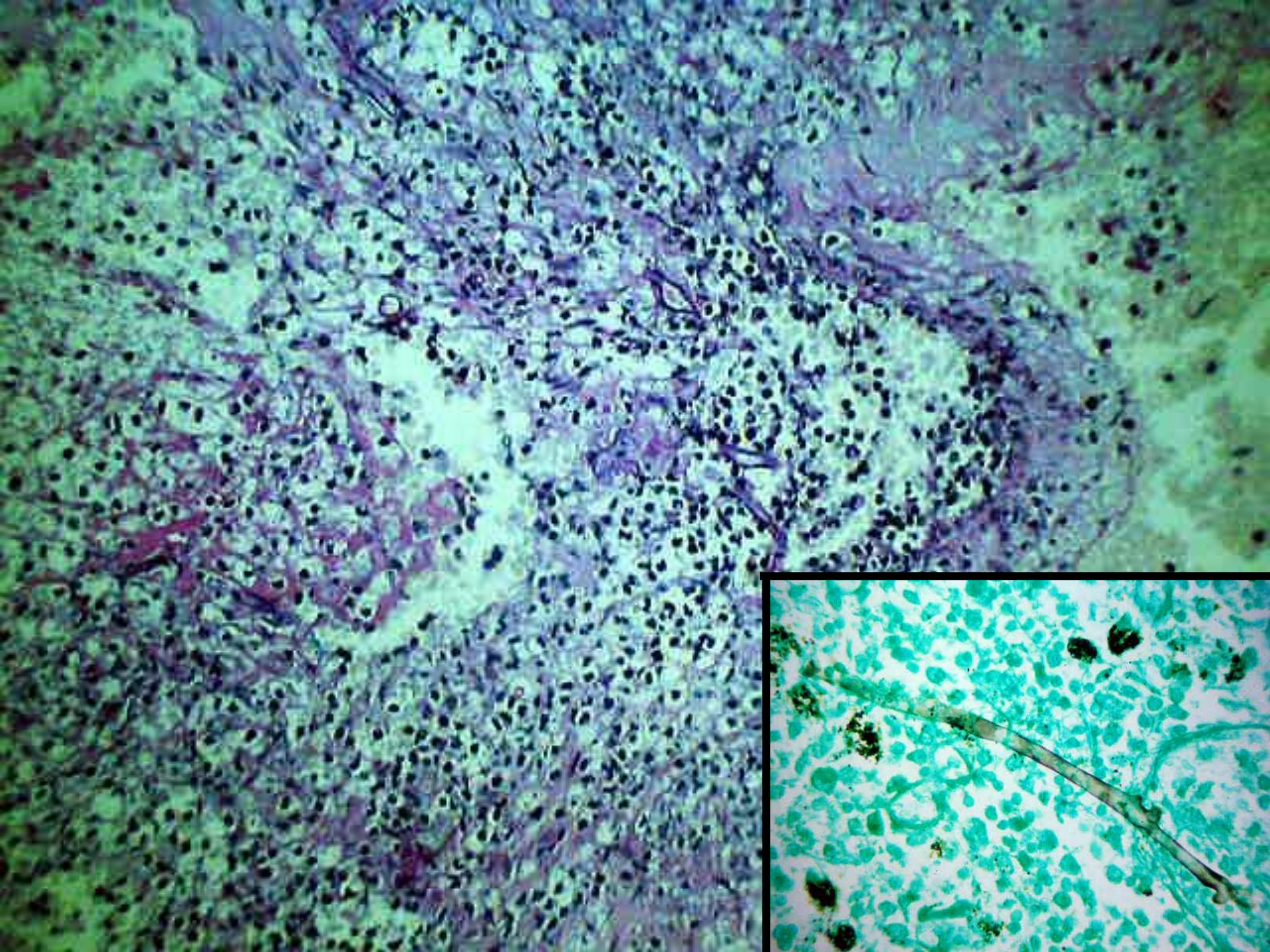


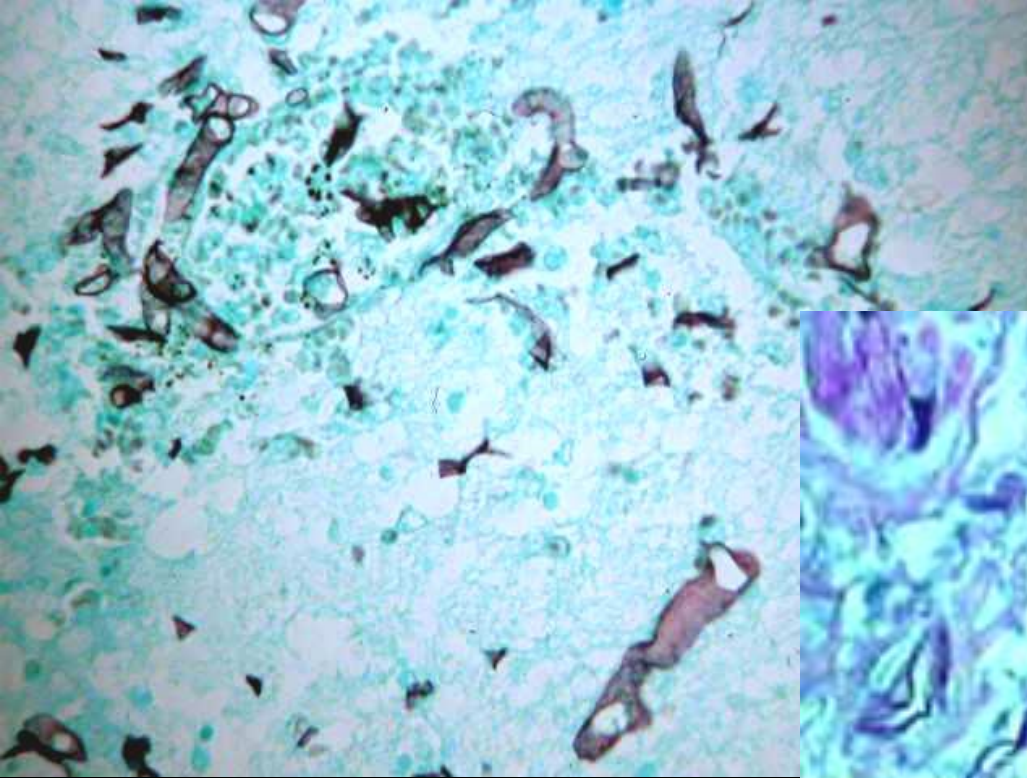
Lee J. N Engl J Med 2008;358:2495

Proptosis

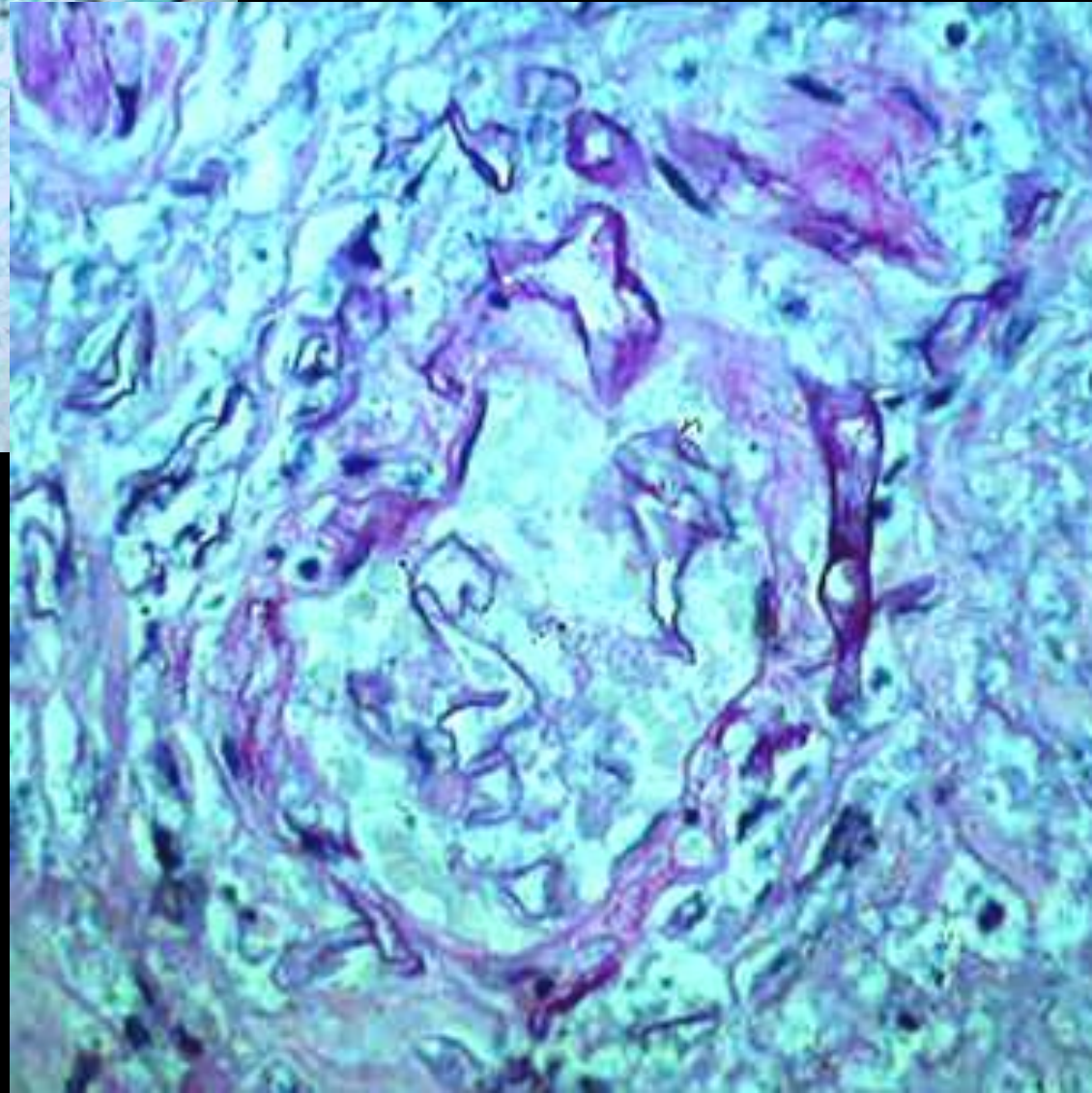


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Diabetes



Mucor mycosis

Diabetes: Consequences

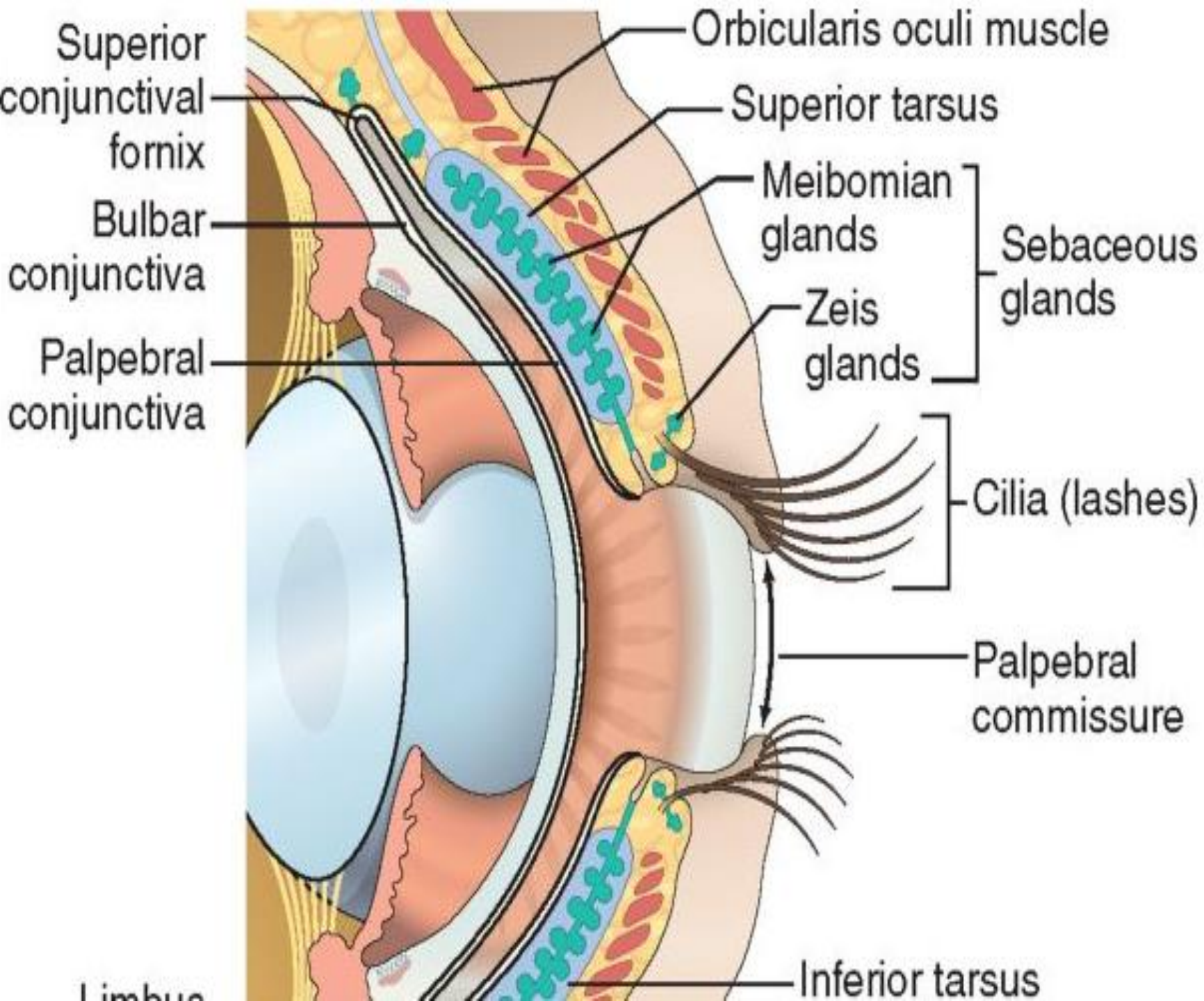
- **Microangiopathy**
 - Nephropathy 30-40%
 - Retinopathy > 50 %
 - Neuropathy 70 %
 - Arteriolosclerosis
- **Macroangiopathy = Atherosclerosis**
- **Infection: increased susceptibility**
 - Impaired leukocyte function
 - Poor blood supply [micro+macroangiopathy]

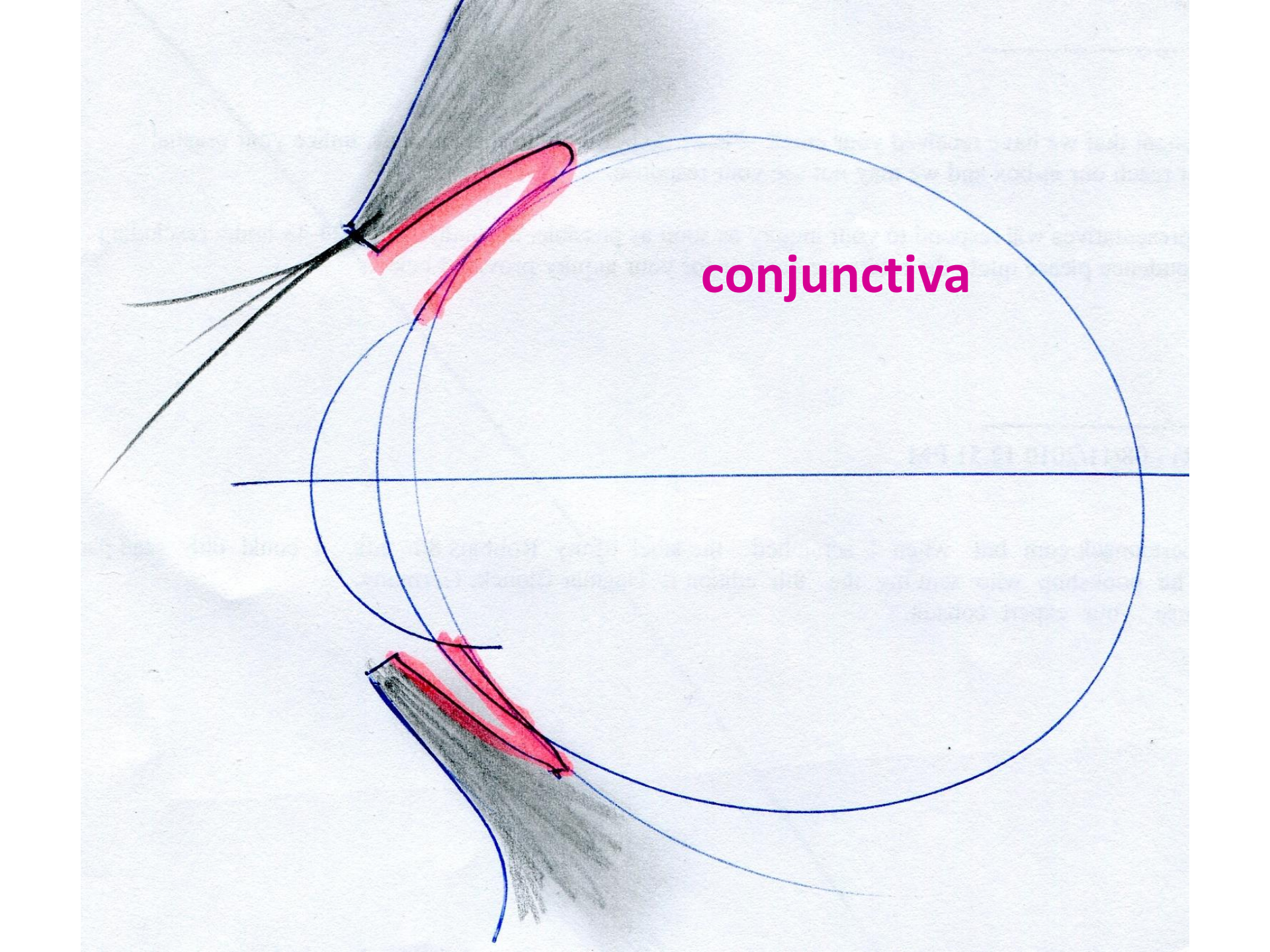
Orbital diseases in Adults

- Thyroid orbitopathy 60%
- Neoplasia 12%
 - Rhabdomyosarcoma
 - Non-Hodgkin lymphoma
- **Inflammation 8%**
 - Sinusitis
 - Diabetes: Mucor

Chalazion

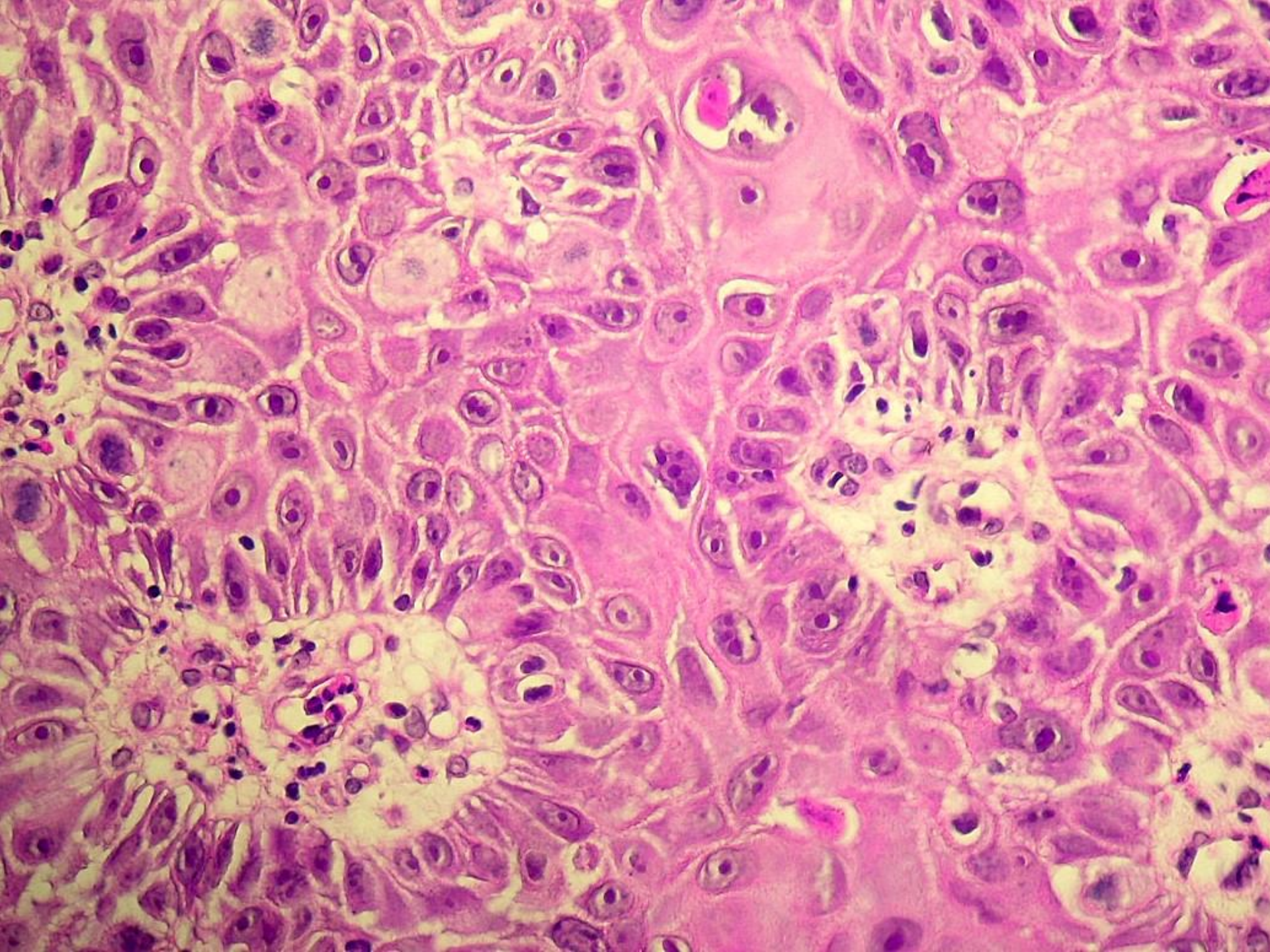
- Granulomatous reaction to extravasated lipid on the eyelid.
- Occurs when drainage of the sebaceous glands is obstructed by chronic inflammation at the eyelid margin(Blepharitis)





A hand-drawn geometric diagram on a light blue background. It features a large circle with a horizontal diameter line. Two smaller circles are positioned on the left side, overlapping the large circle. The upper-left small circle is shaded with grey pencil strokes. The lower-left small circle is also shaded with grey pencil strokes. Two regions, one on the upper-left small circle and one on the lower-left small circle, are highlighted with red pencil strokes. The word "conjunctiva" is written in magenta text in the upper-right quadrant of the large circle.

conjunctiva



**National
prevalence of
blindness is 1.6 %**

**(World Sight Day
October 2012)**

Currently over 138,000 Ethiopians suffer from advanced trachoma: MoH

By Fikremariam Tesfaye

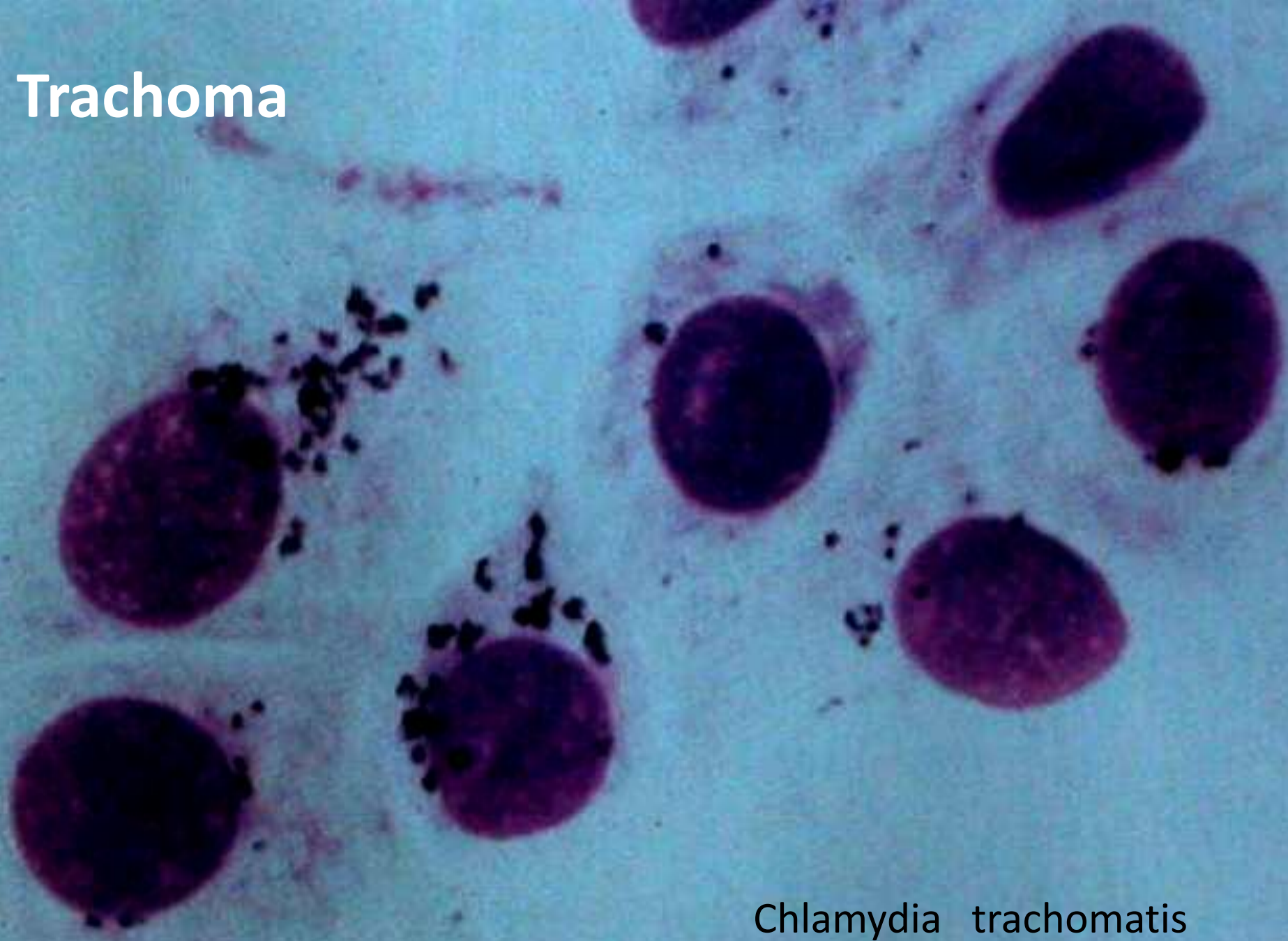
ADDIS ABABA- The Federal Ministry of Health (MoH) disclosed that over 138,000 Ethiopians are living with the advanced stage of trachoma that could lead to blindness.

During a World Sight Day that was marked yesterday at Minilik II Memorial Hospital here, Health Promotion, Disease Prevention and Eye Care Program focal person with MoH, Sisay Endale said, that the National prevalence of blindness is 1.6 percent where active trachoma (in children aged from 1-9 years old) is 40 percent.

Even if, trachoma is an

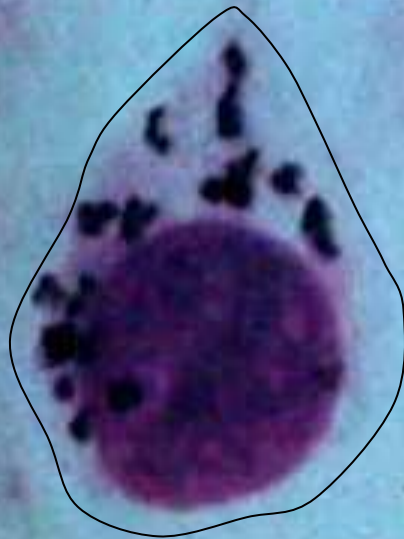
***“Currently the
prevalence of
trachoma
is decreasing due to
awareness raising
and change in life
standard of
the public. But
more effort
needed mainly
in rural society,”***

Trachoma



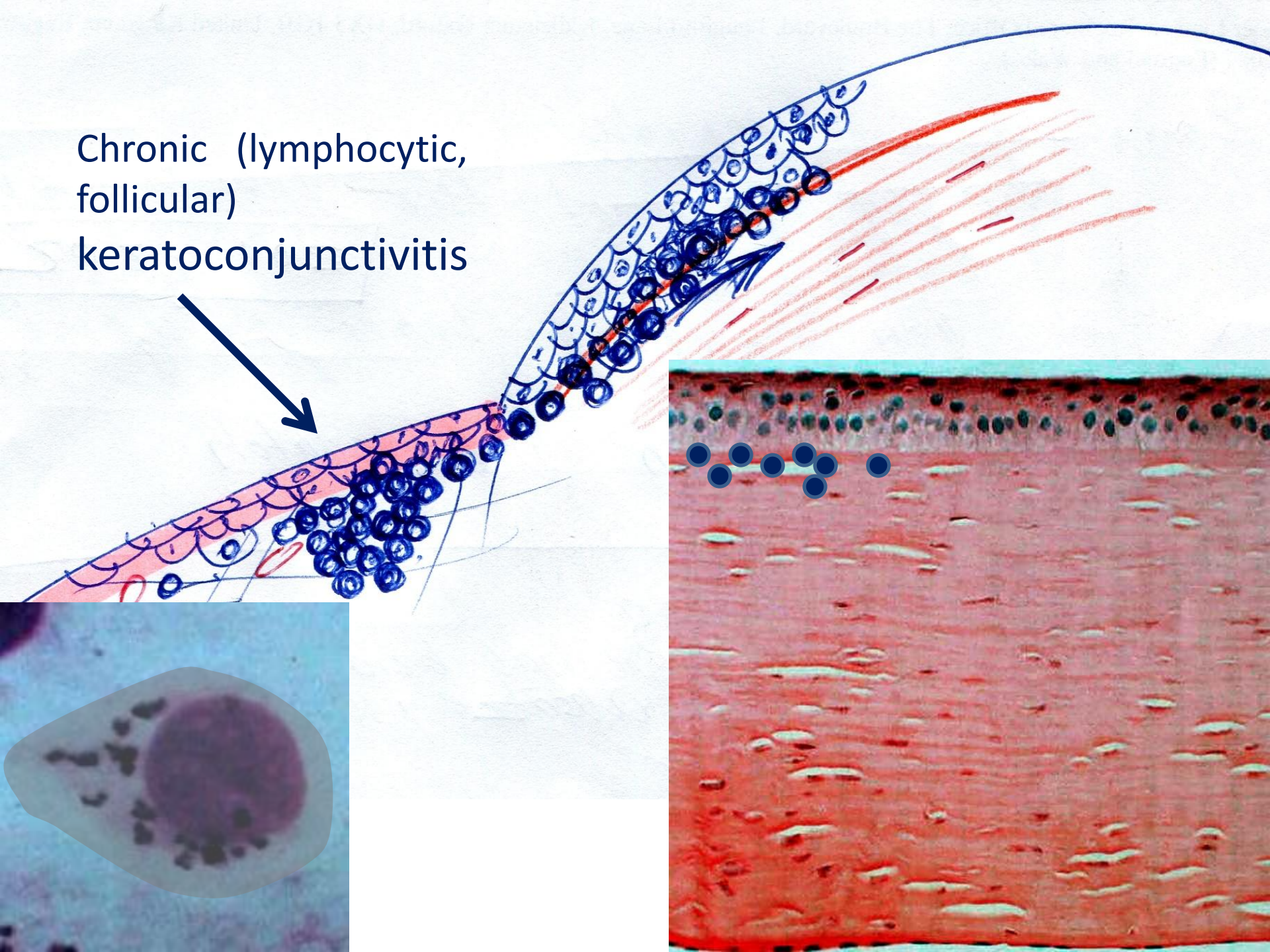
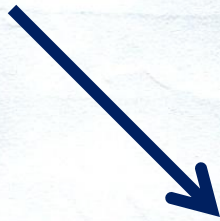
Chlamydia trachomatis

Chlamydia trachomatis

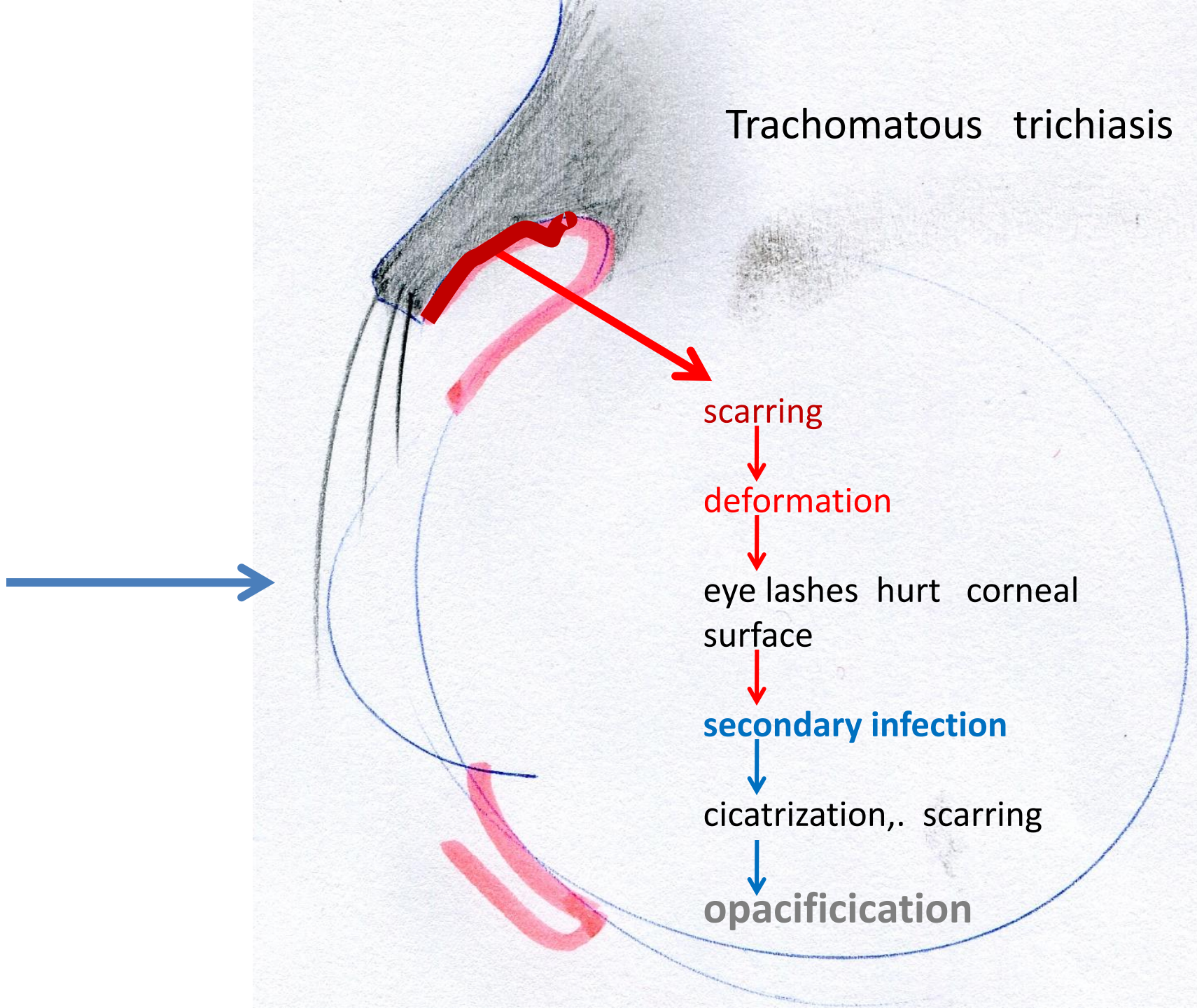


intracellular

Chronic (lymphocytic,
follicular)
keratoconjunctivitis



Trachomatous trichiasis

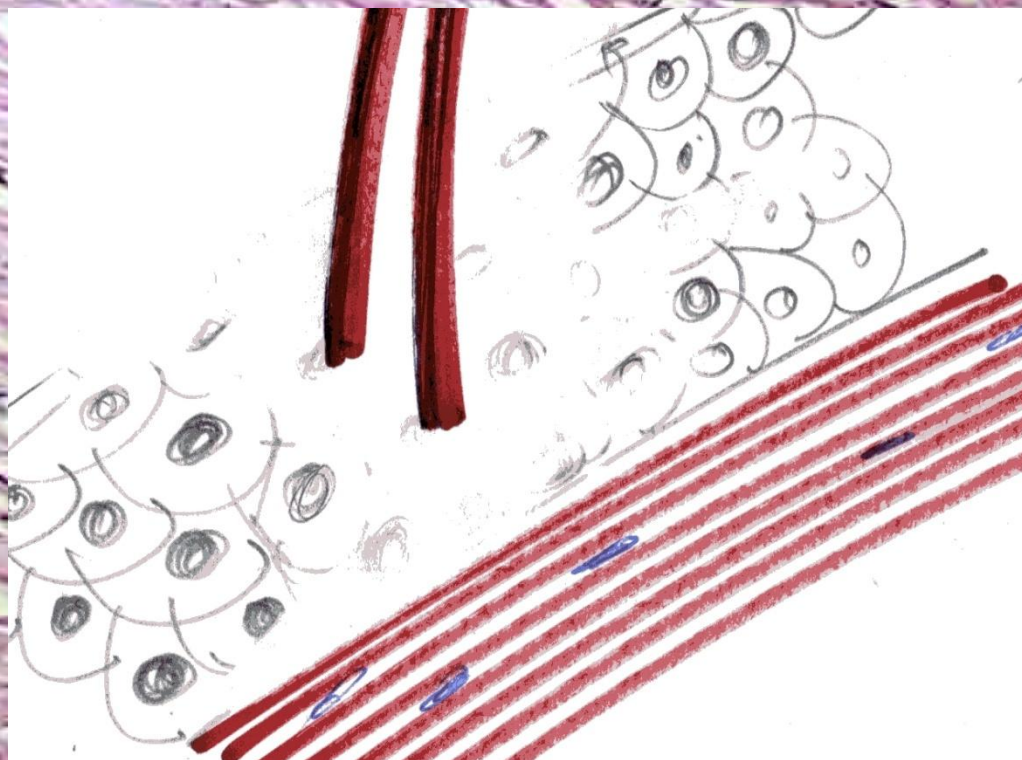
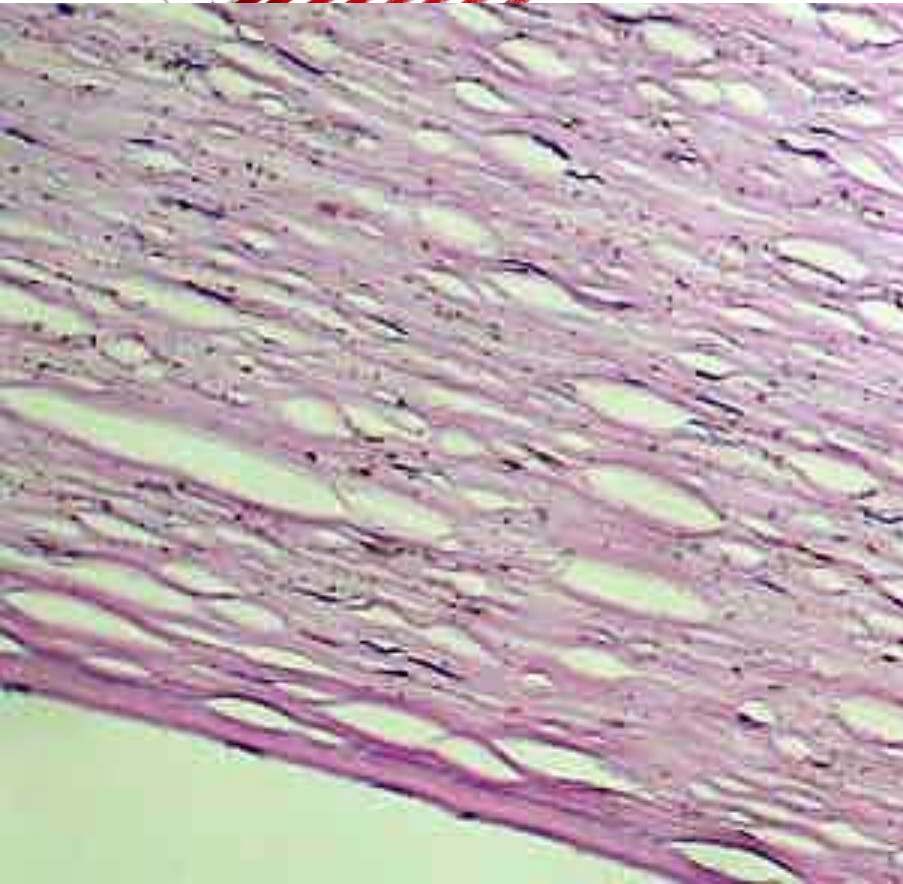
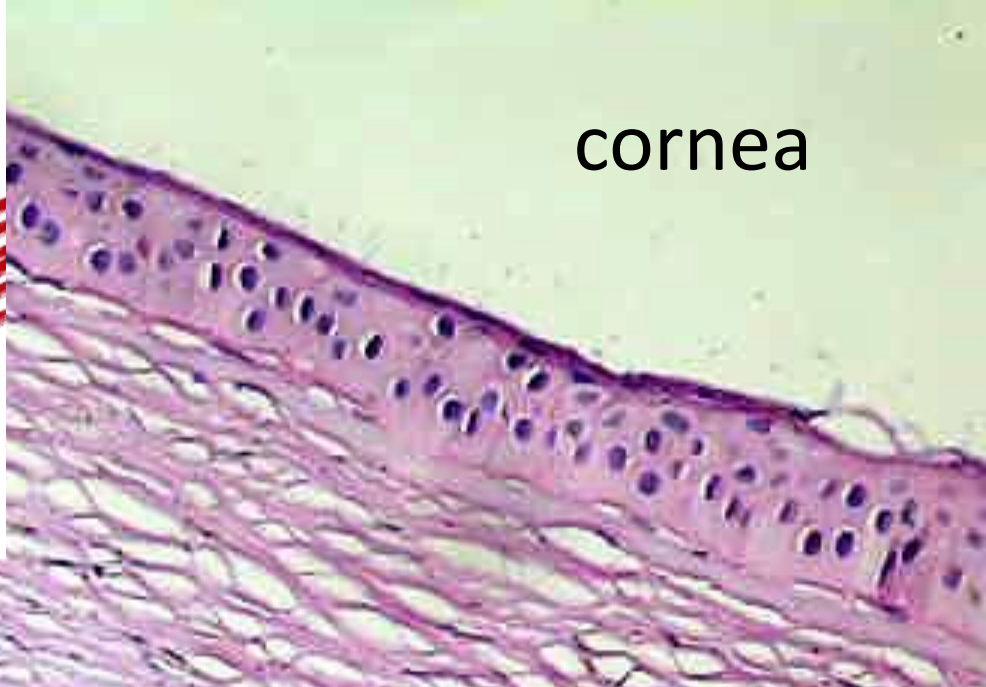
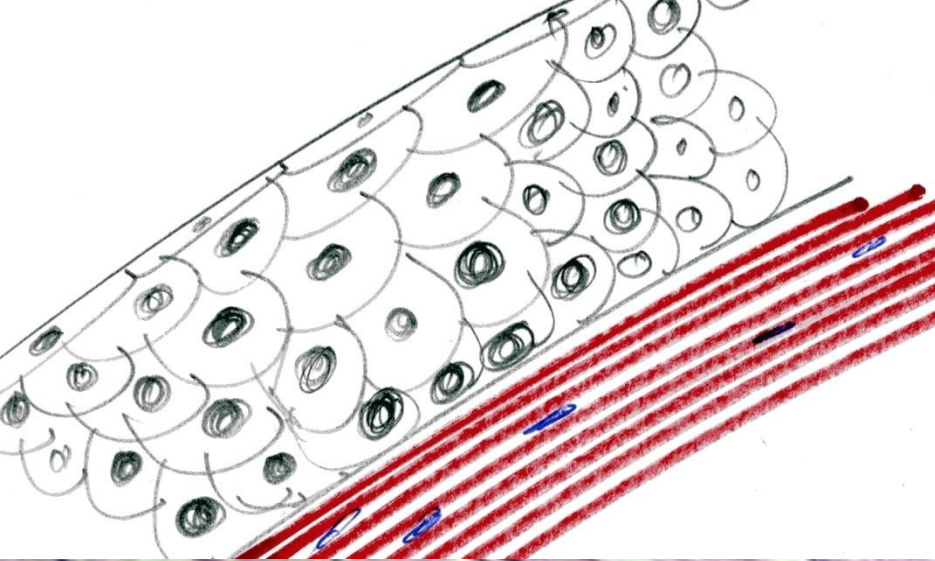


- Pinguecula
 - Submucosal growth of degenerated elastic tissue
 - Due to Exposure to the sun.
 - Starts at the sun exposed area of conjunctiva(in the fissure between the eyelids)
- Pterygium
 - Grows into the cornea

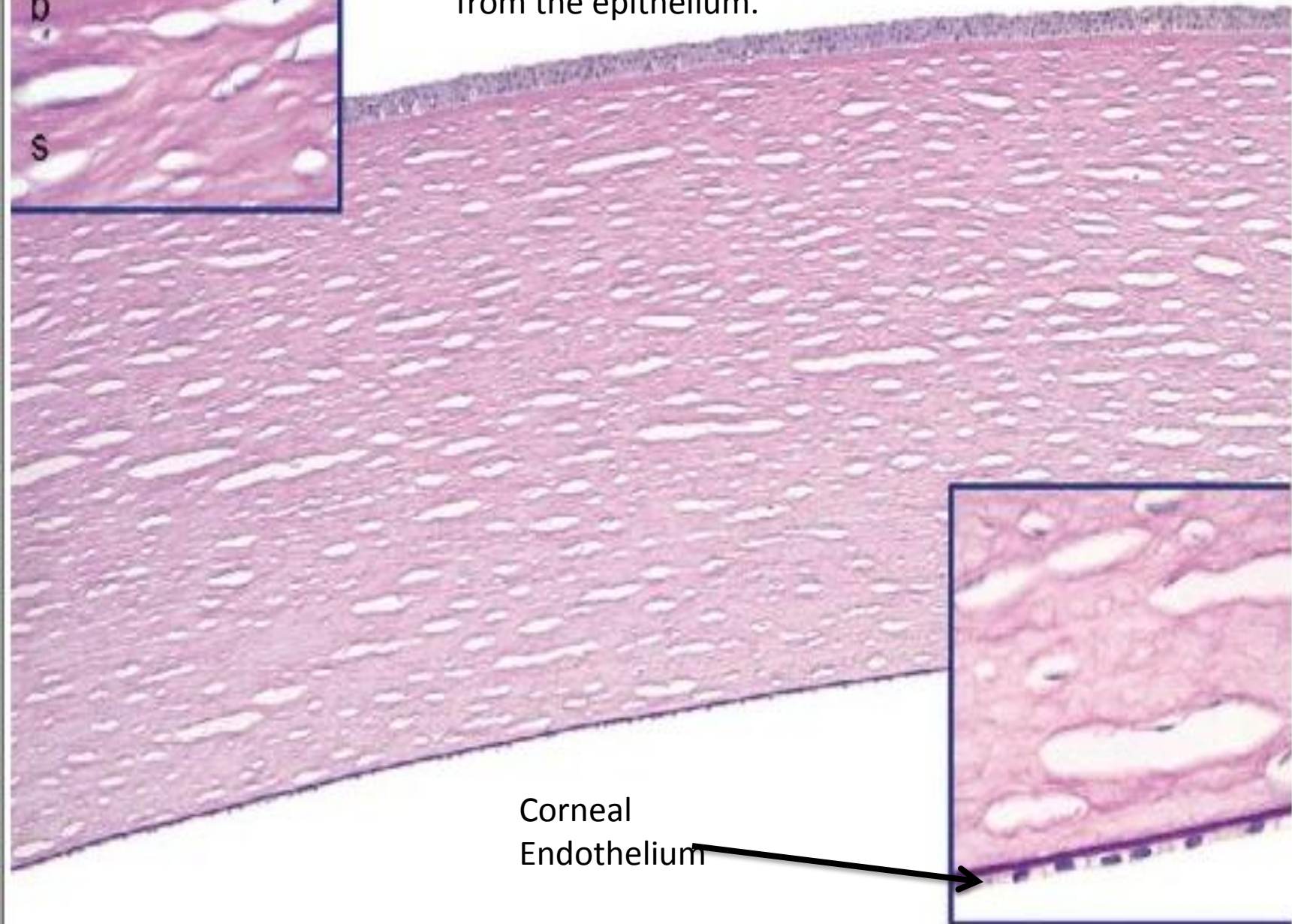
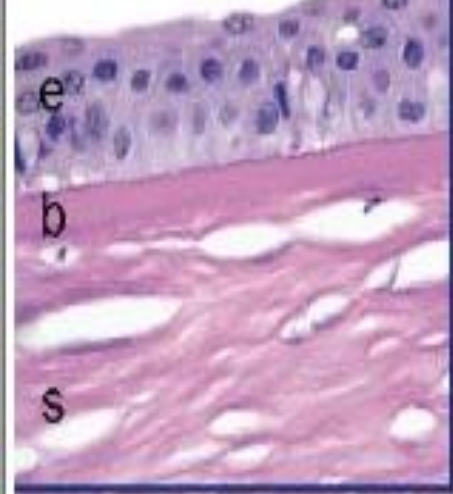
Neoplasms

- Benign
 - Conjunctiva Nevi- melanocytes
- Malignant
 - Squamous Cell Carcinoma
 - Mucoepidermoid Carcinoma
 - Melanoma

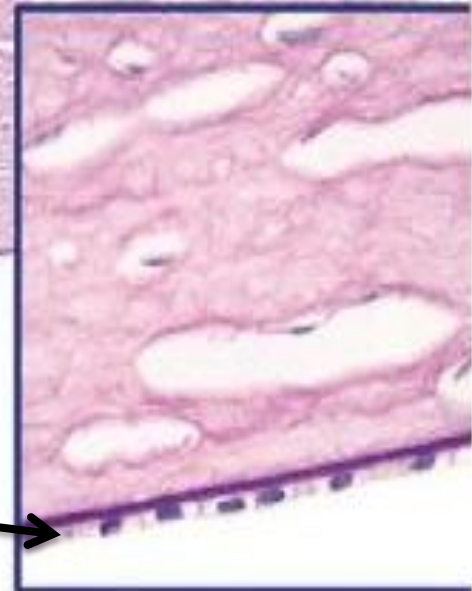
cornea

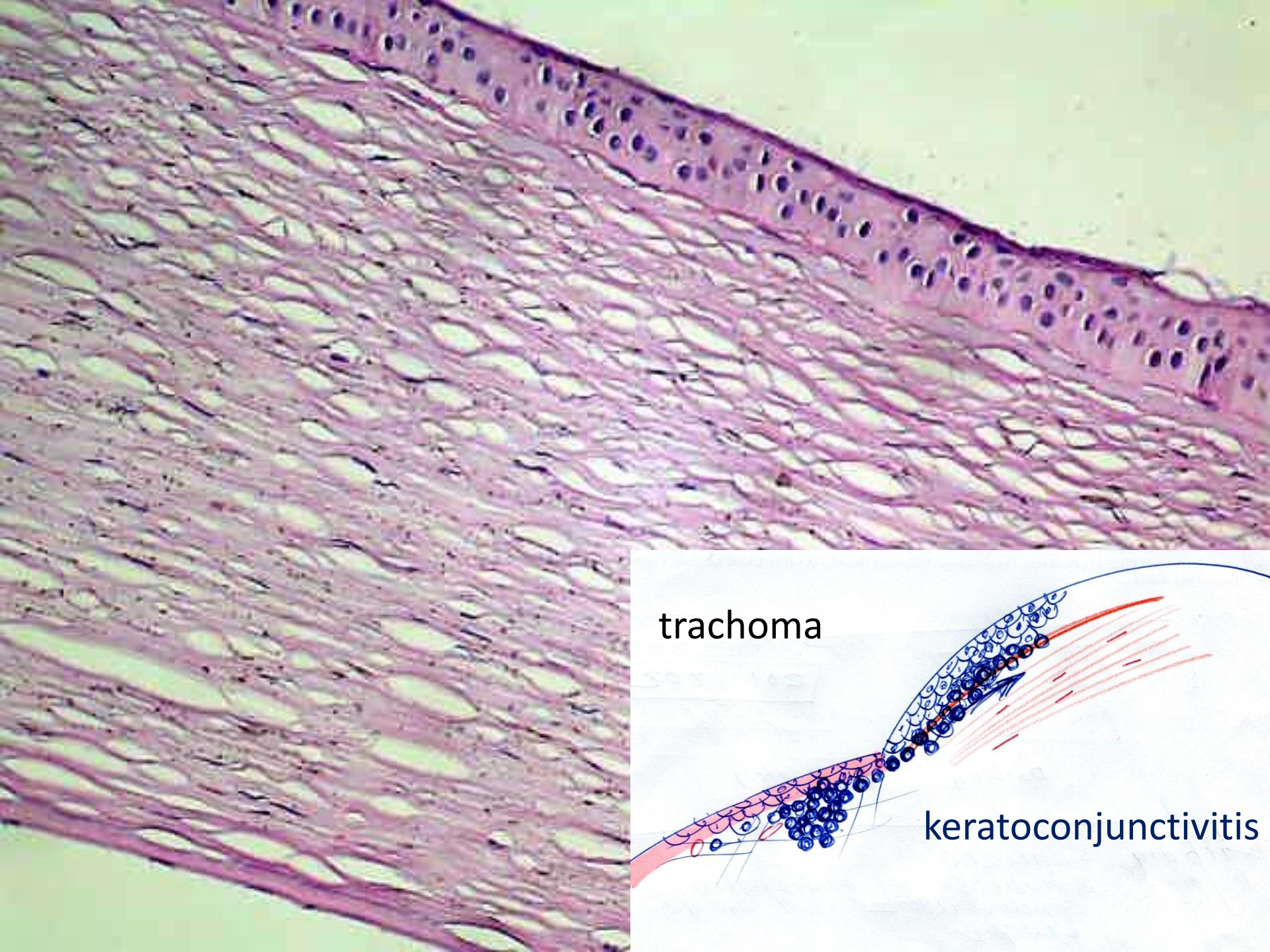


b. Bowman's Layer- Acellular
barrier against
potentially malignant cells
from the epithelium.

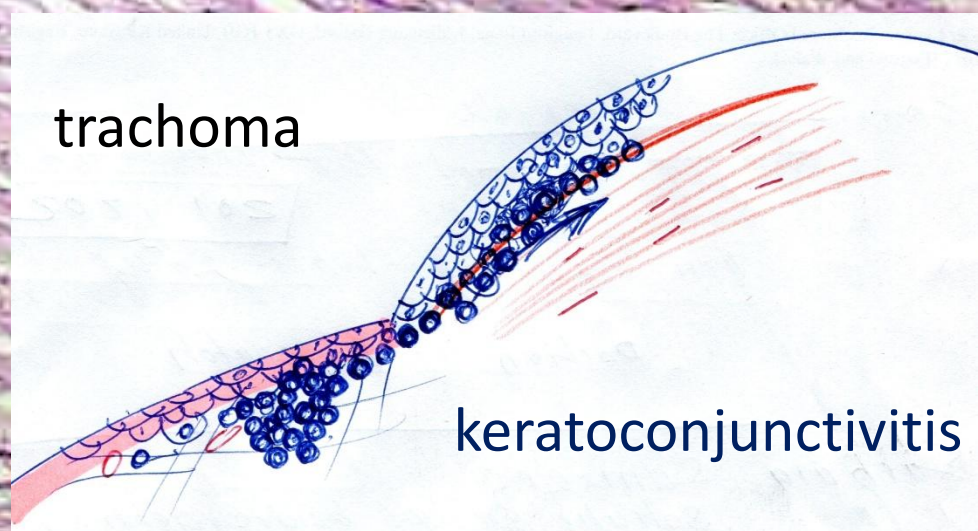


Corneal
Endothelium



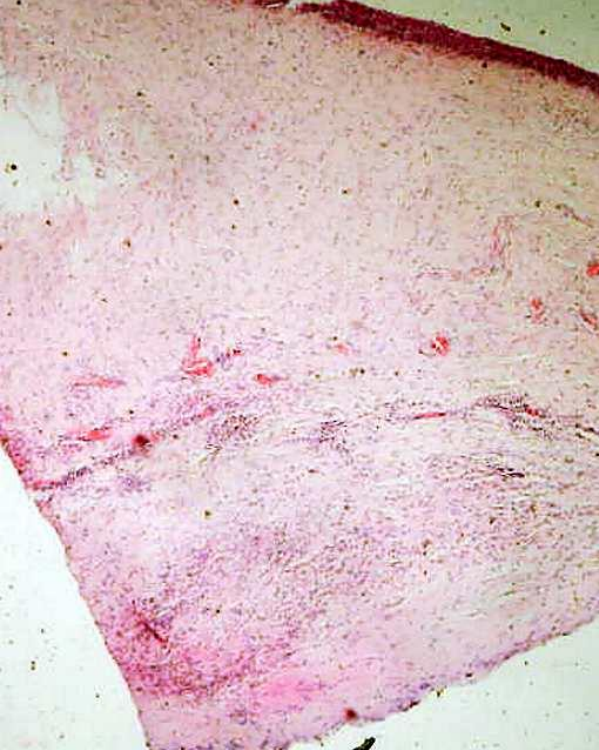


trachoma

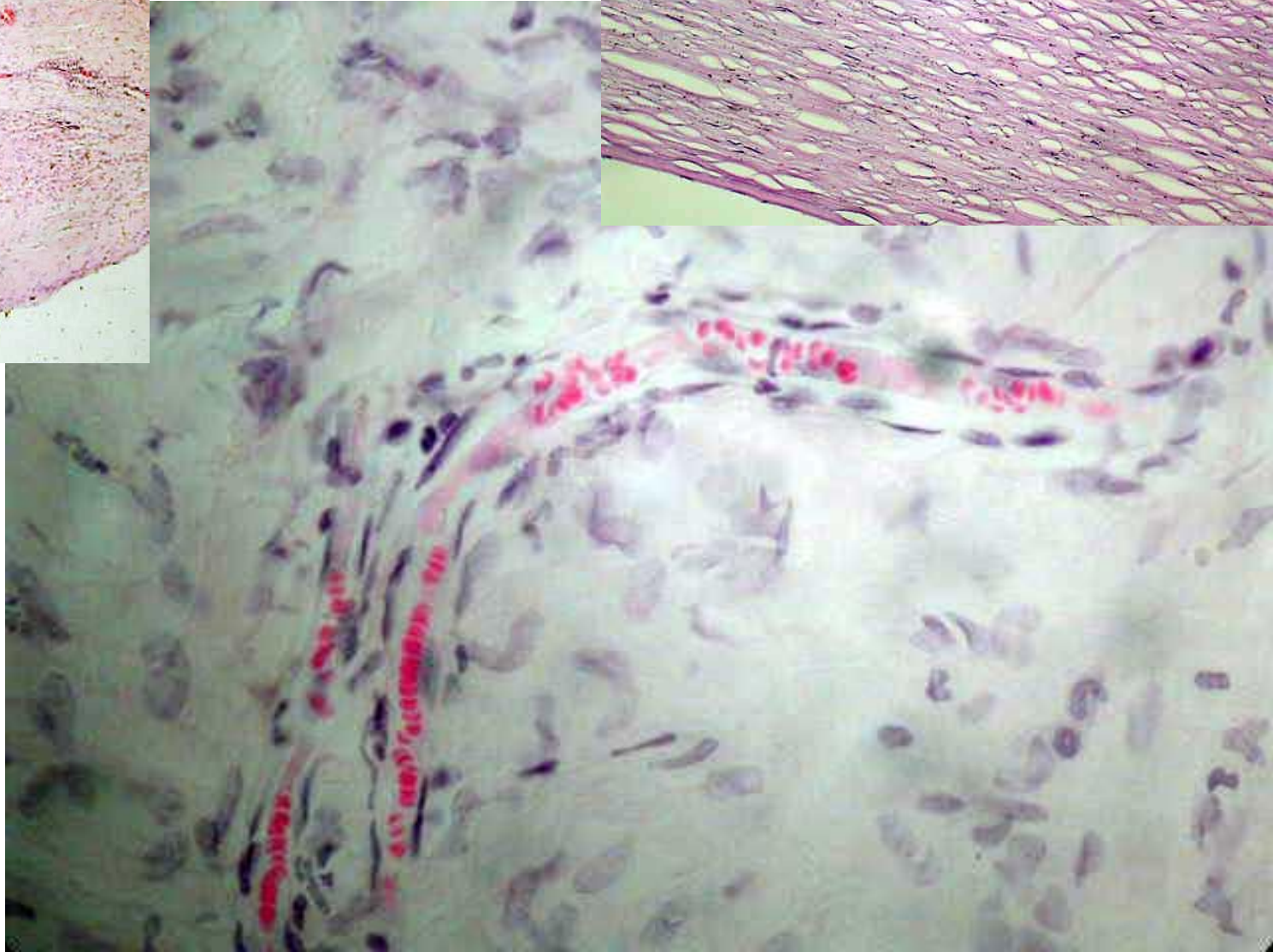


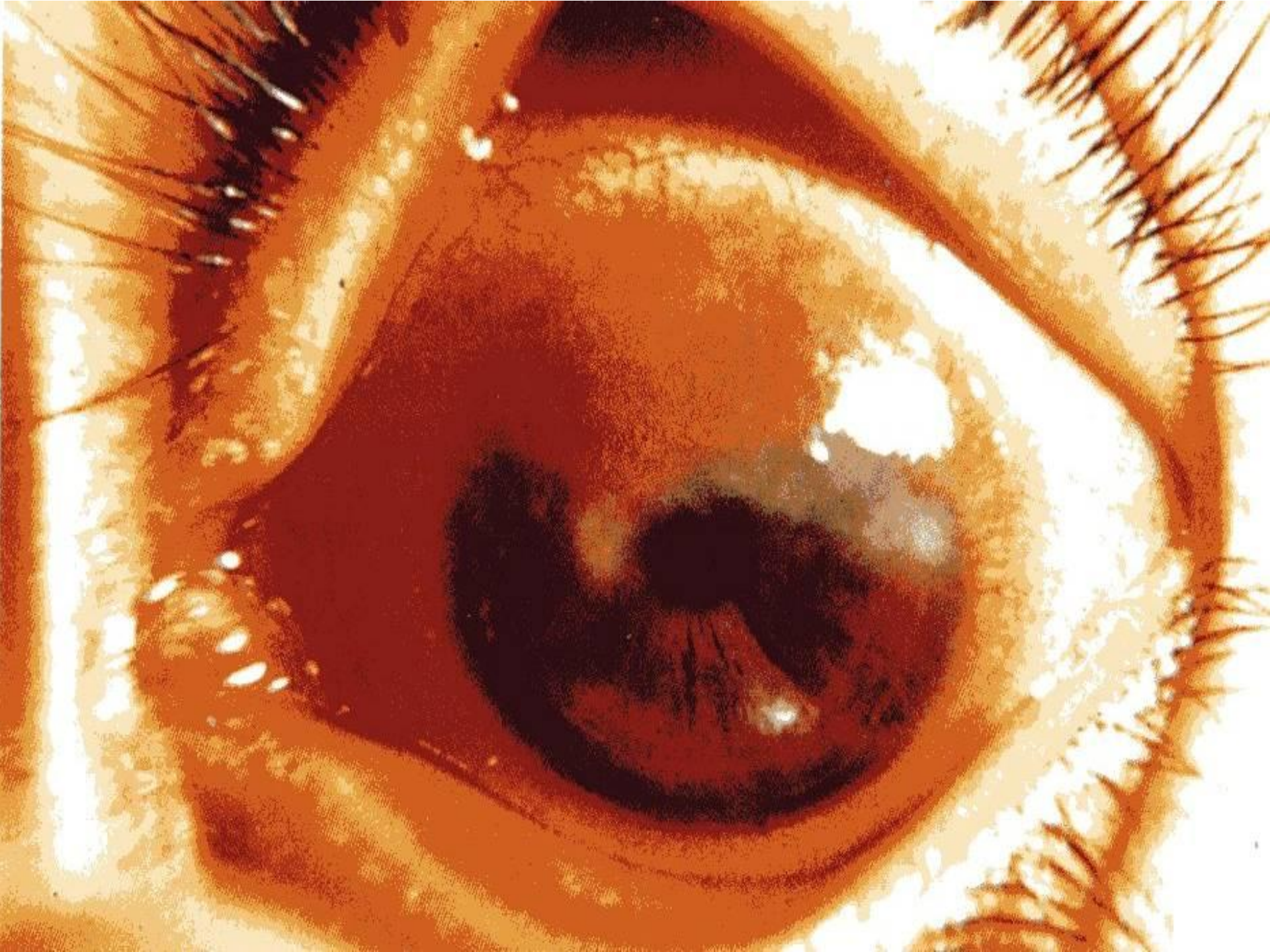
keratoconjunctivitis

Cornea

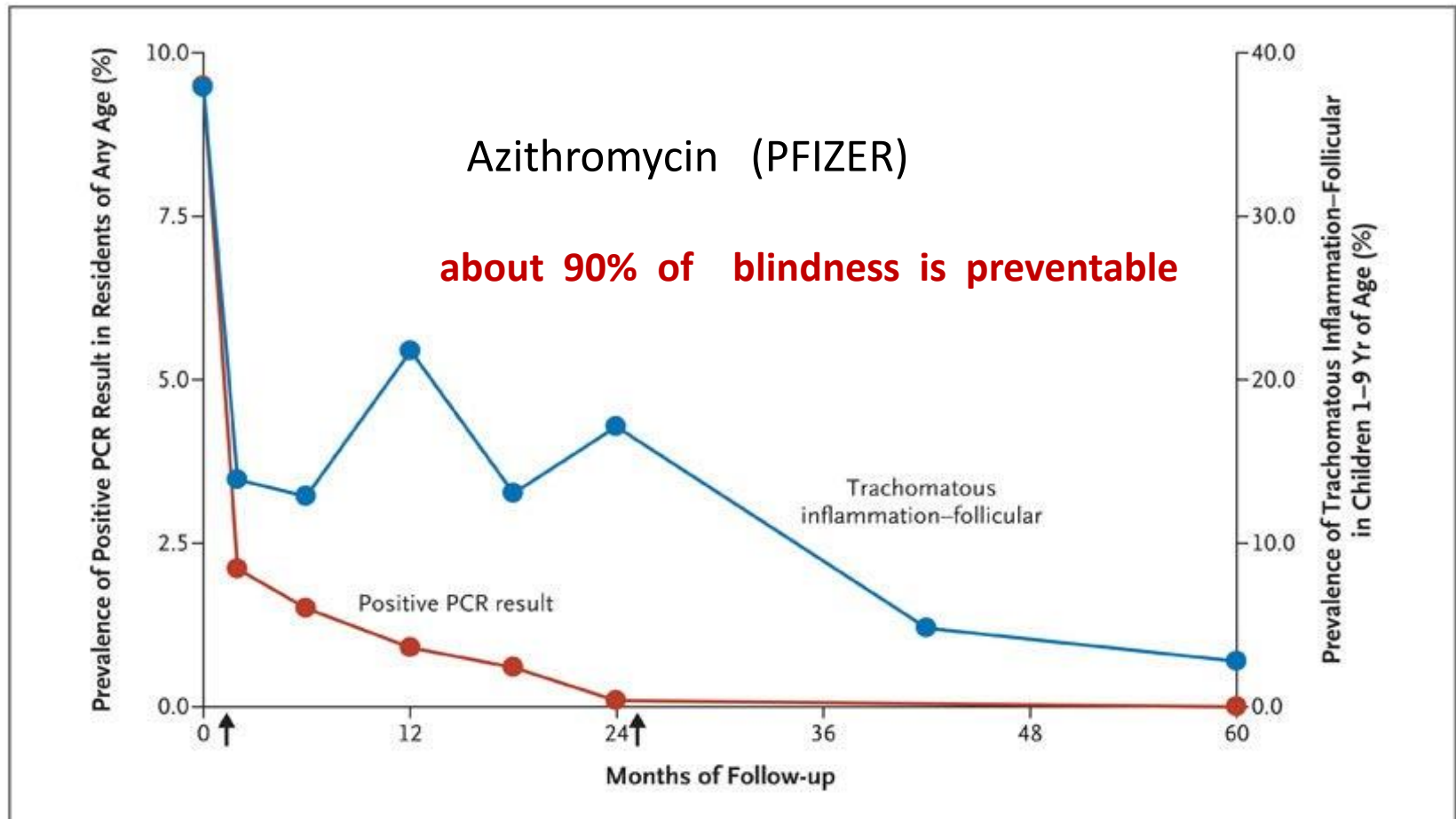


capillarization

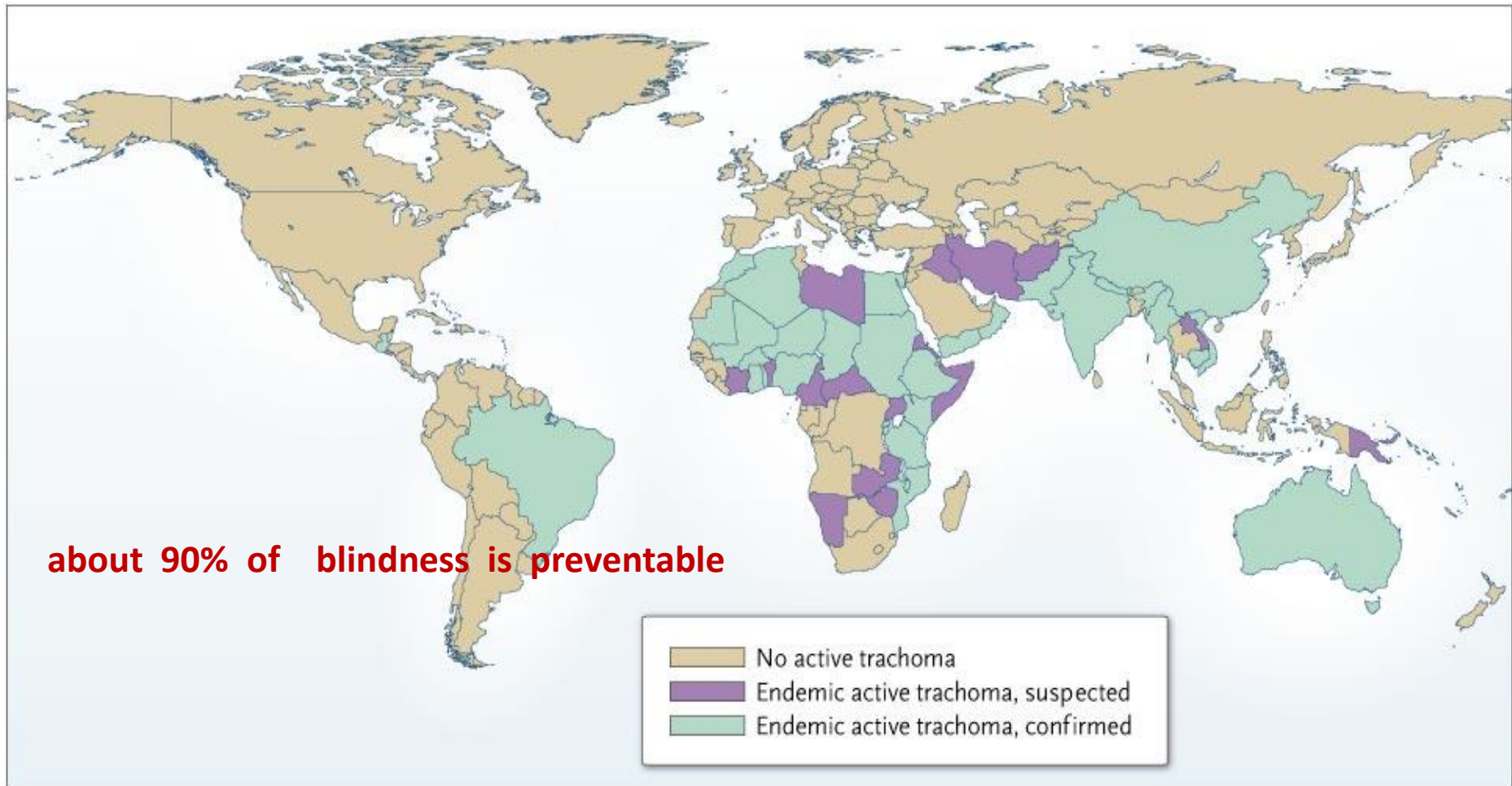




Prevalences of Ocular Chlamydia trachomatis Infection (Positive Result on Polymerase-Chain-Reaction Assay [PCR]) and Disease (Trachomatous Inflammation–Follicular) during Follow-up in a Tanzanian Community.



Global Distribution of Trachoma in 2005



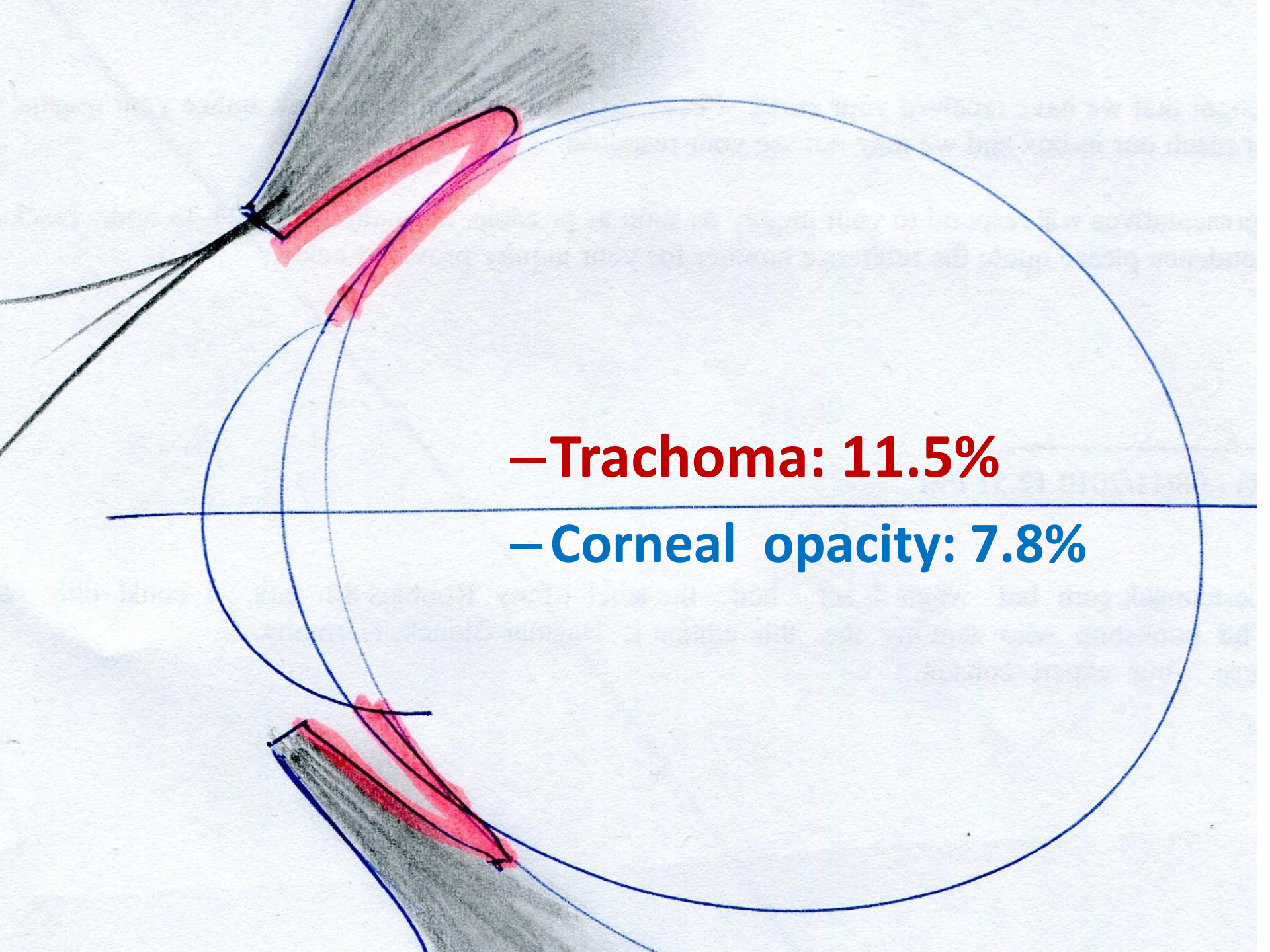
1996: 100 mio infected

2008: 43 mio

Cook J. N Engl J Med 2008;358:1777-1779



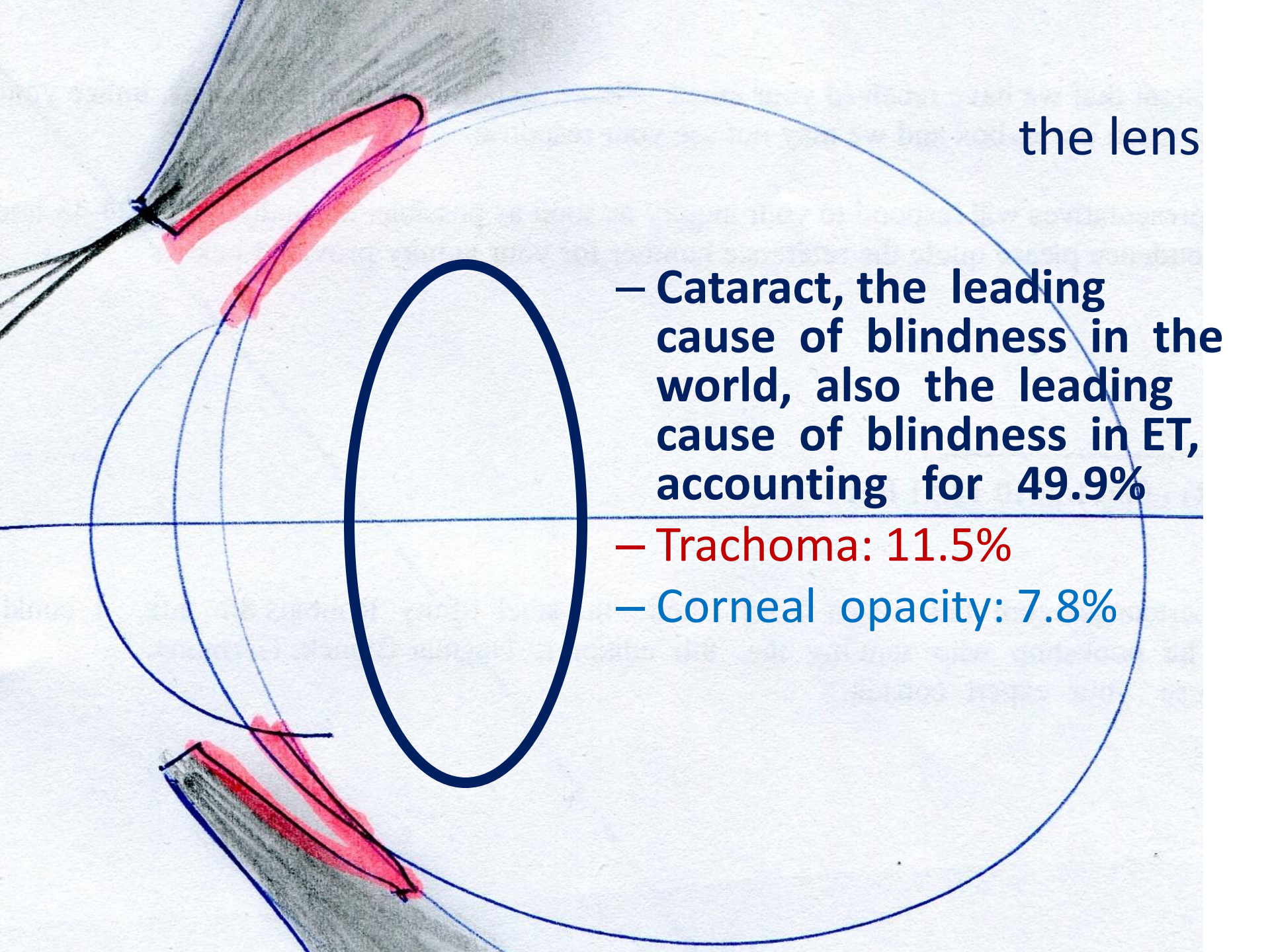
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A hand-drawn diagram of a human eye. The eye is represented by a large circle with a horizontal line through its center. Two smaller circles are drawn on the left side, representing the cornea and sclera. Two red, elongated, curved shapes are drawn on the cornea and sclera, one above and one below the horizontal line. These shapes are outlined in black and filled with red. The background of the eye is shaded with grey pencil strokes.

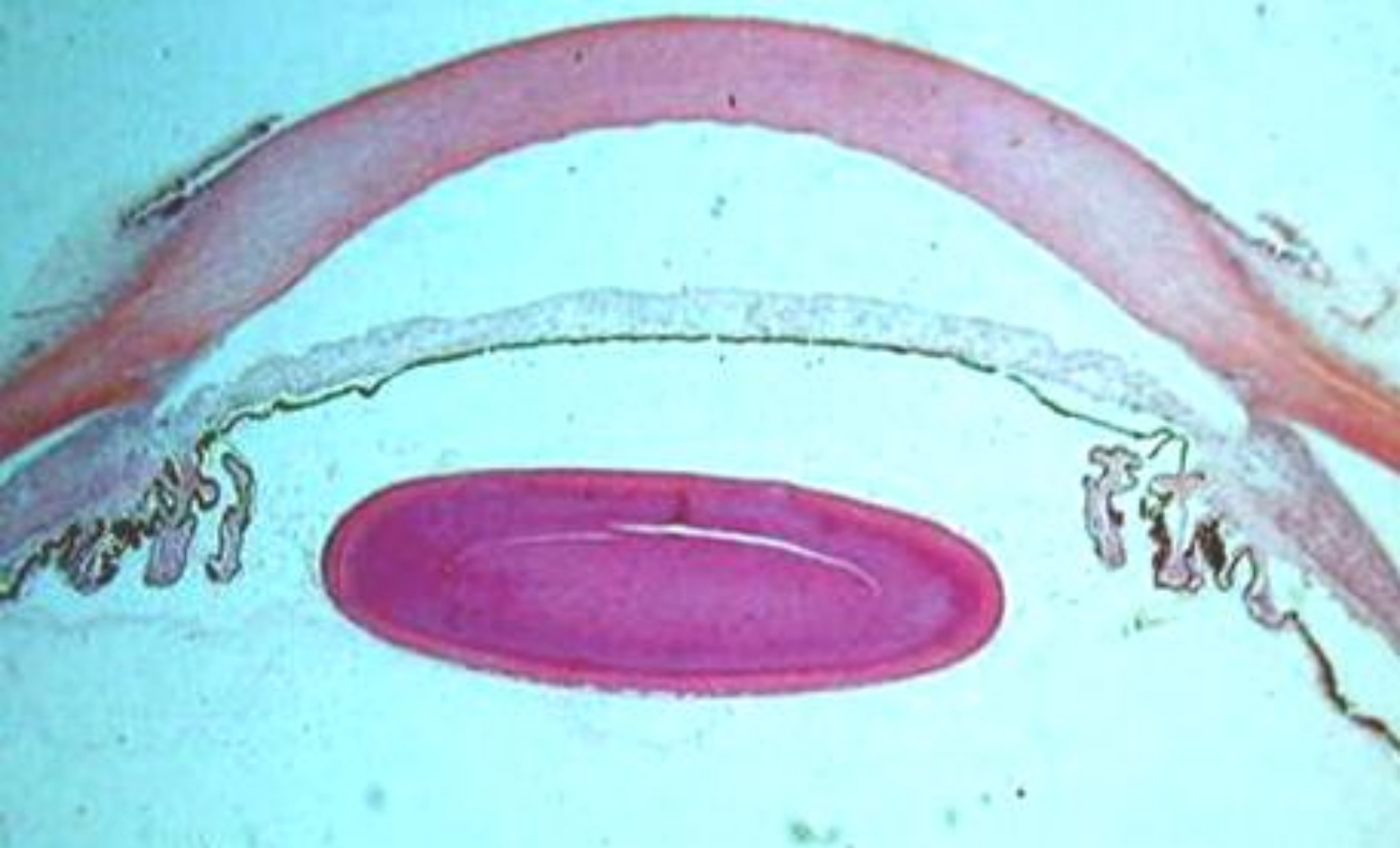
–Trachoma: 11.5%

–Corneal opacity: 7.8%

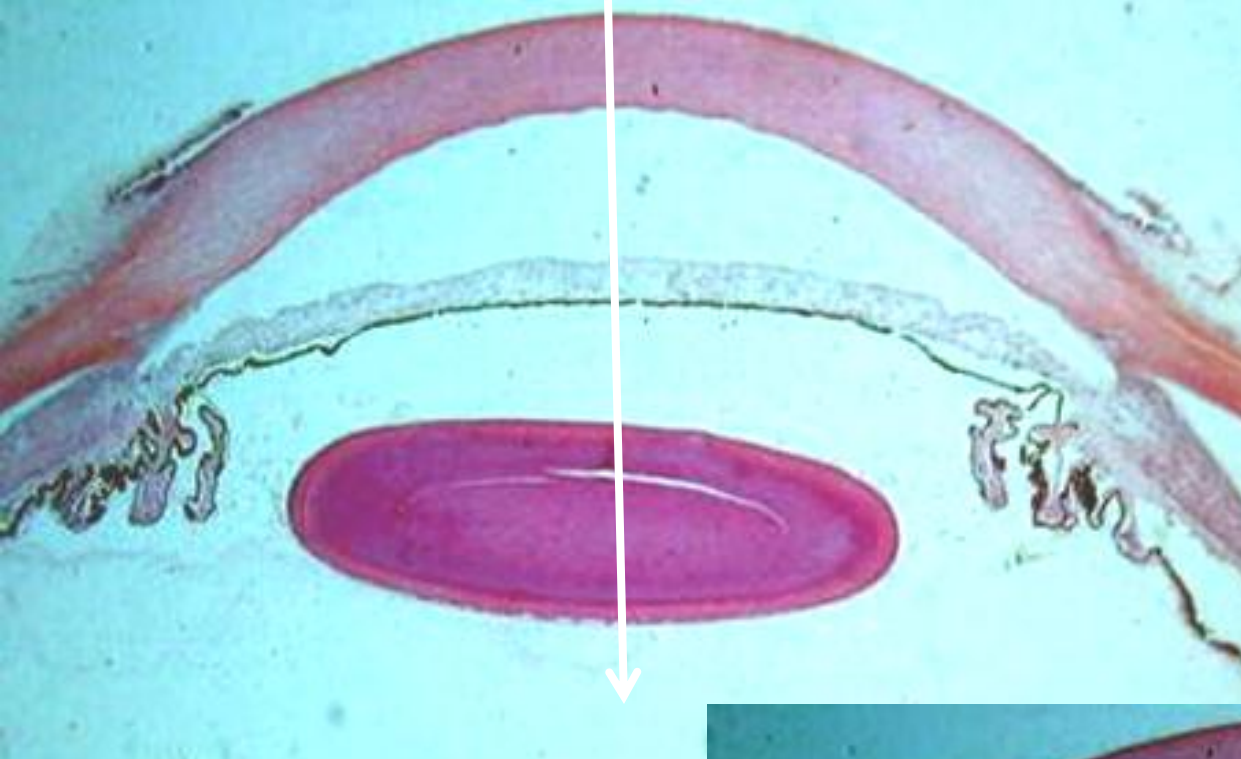


the lens

- Cataract, the leading cause of blindness in the world, also the leading cause of blindness in ET, accounting for 49.9%
- Trachoma: 11.5%
- Corneal opacity: 7.8%



Cataract



Cataract

leading cause

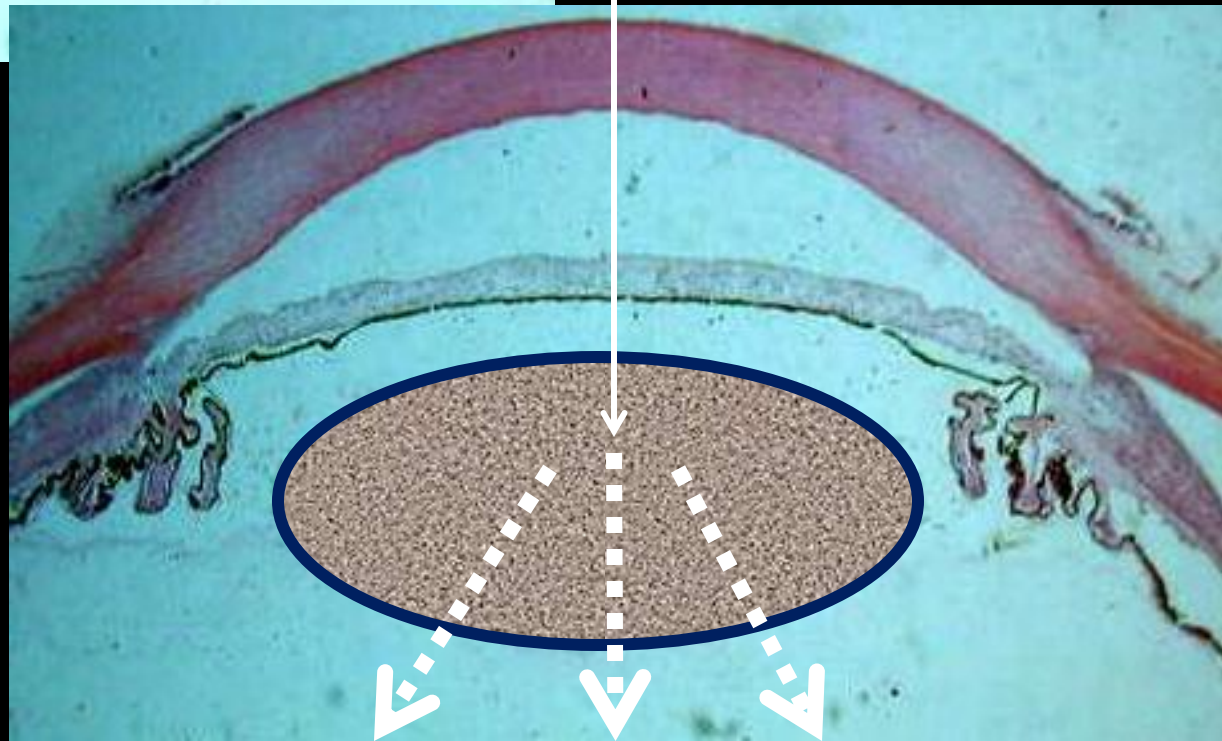
50%

Diabetes

Drugs

Irradiation

Age (= senile)



Cataract

- Opacification of the lens.
- Causes
 - Systemic Diseases (Diabetes Mellitus. Wilson Disease, atopic dermatitis)
 - Drugs-Corticosteroids
 - Radiation, Trauma
 - Age Related(Due to Sclerosis of the lens nucleus)

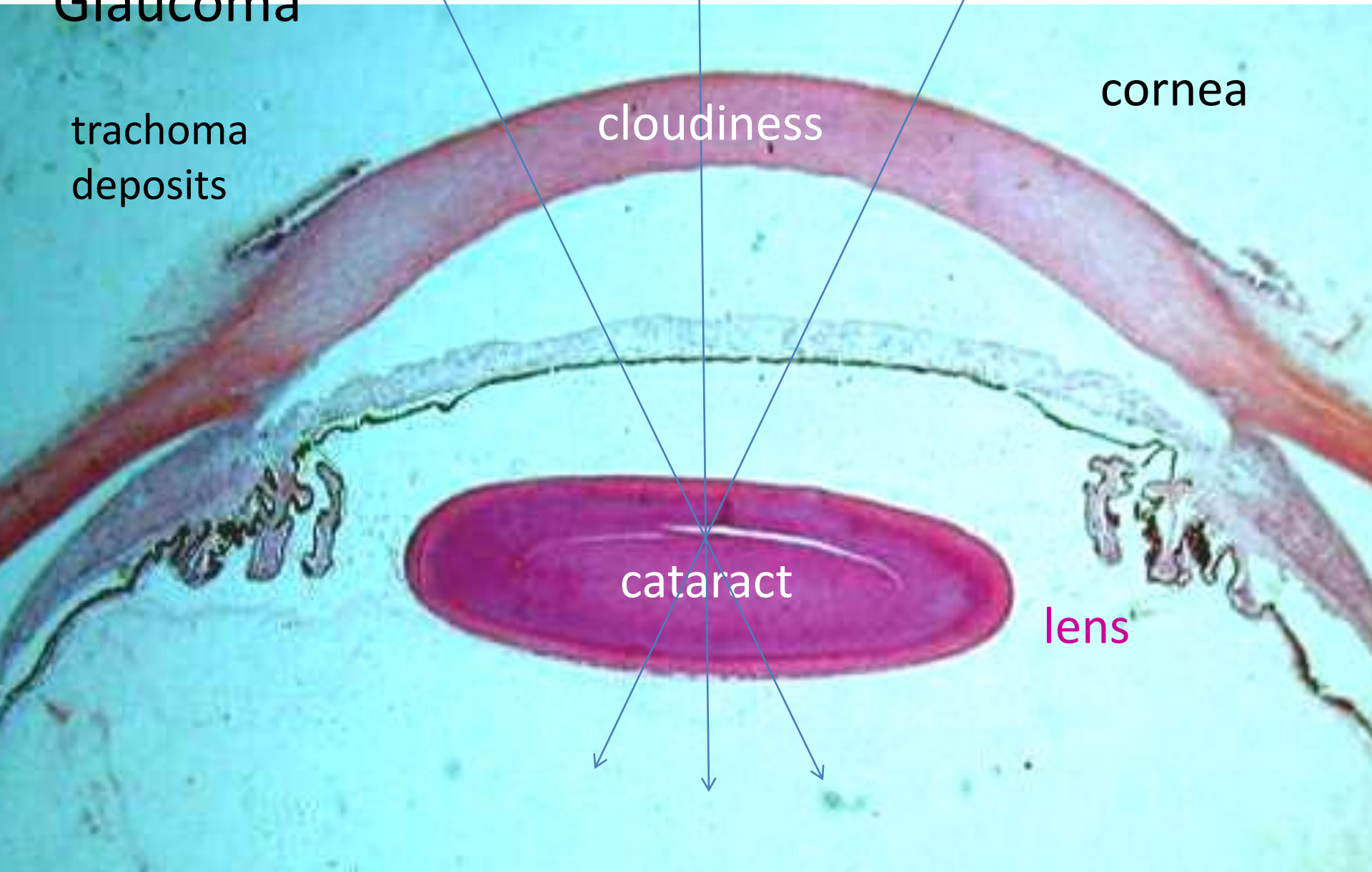
Mechanism

- Compaction and stiffening of the lens material
- Abnormal changes in lens proteins(crystallins)
- Pigmentation of lens proteins
- Changes in Ionic component of the lens

symptoms

- Blurred vision at distance or near
- Glare-halos or streaks around lights
- Difficulty in seeing bright lights or in poor lighting conditions.
- Loss of ability to discern colors

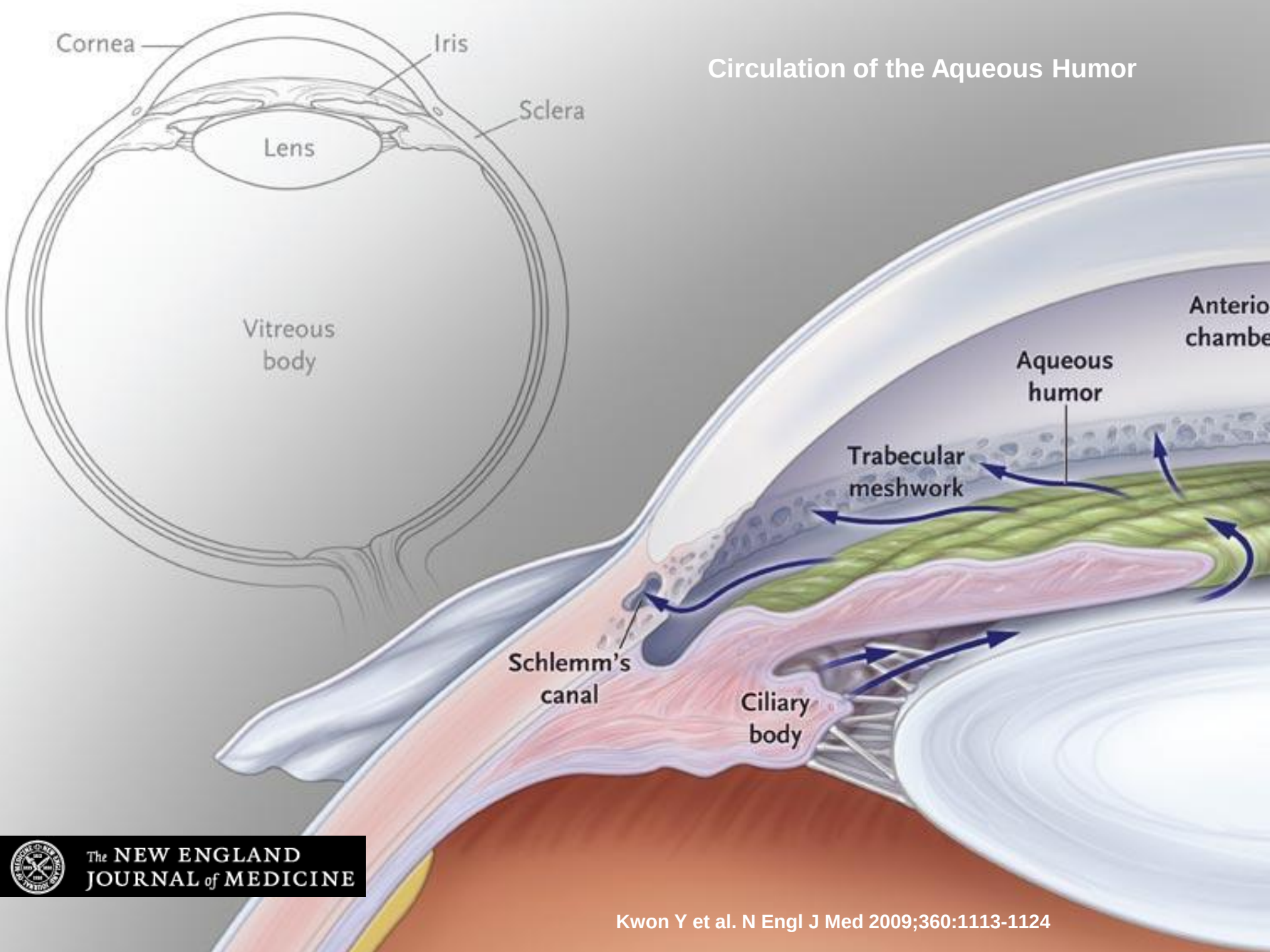
anterior segment and Glaucoma



Glaucoma

- Group of diseases with distinctive change in visual field and result in cupping of the optic nerve.
- Mostly associated with an increase in the intraocular pressure.
- Heavily impacted by flow aqueous humor.

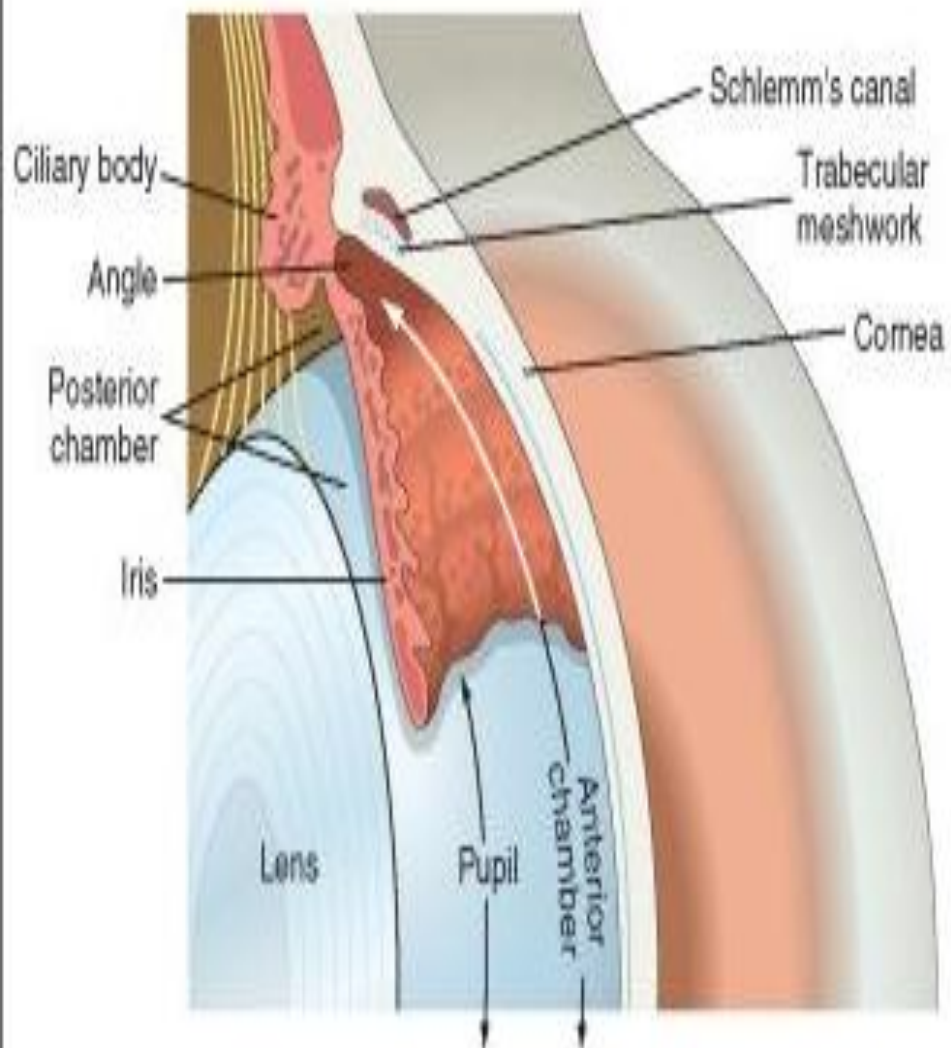
Circulation of the Aqueous Humor



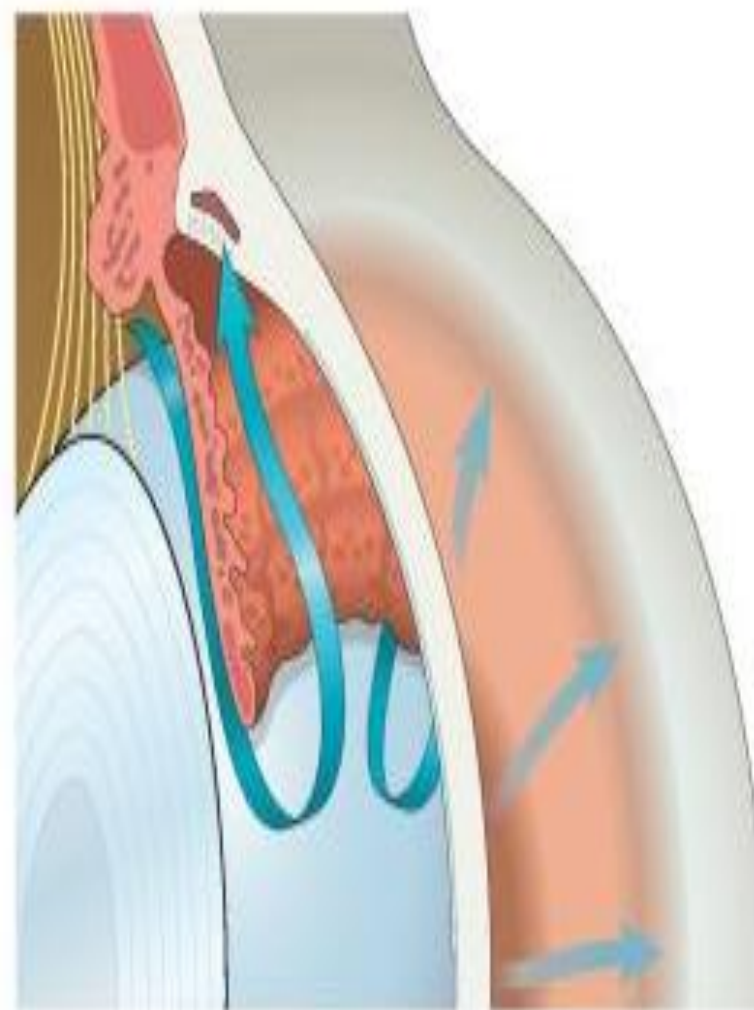
Glaucoma

- Classified into two
 - Open Angle Glaucoma
 - Angle Closure Glaucoma
- Angle- B/n the iris and the canal of schlemm

ANTERIOR AND POSTERIOR CHAMBERS

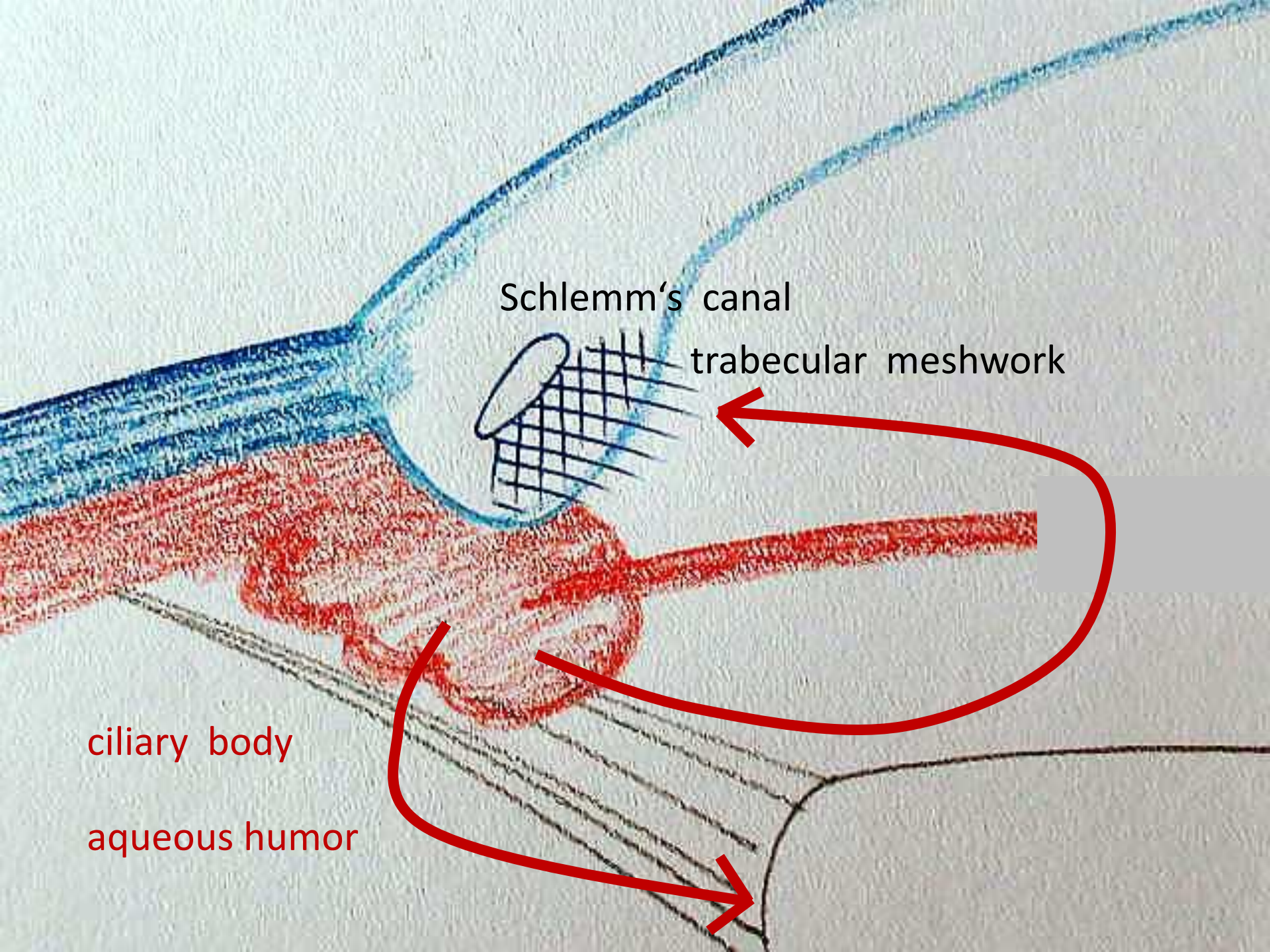


MAJOR AQUEOUS OUTFLOW PATHWAY



Angle Closure Glaucoma

- Primary
 - IN patients with shallow anterior chambers, hyperopic patients (small eyeball)
 - The iris transiently apposes to the lens.

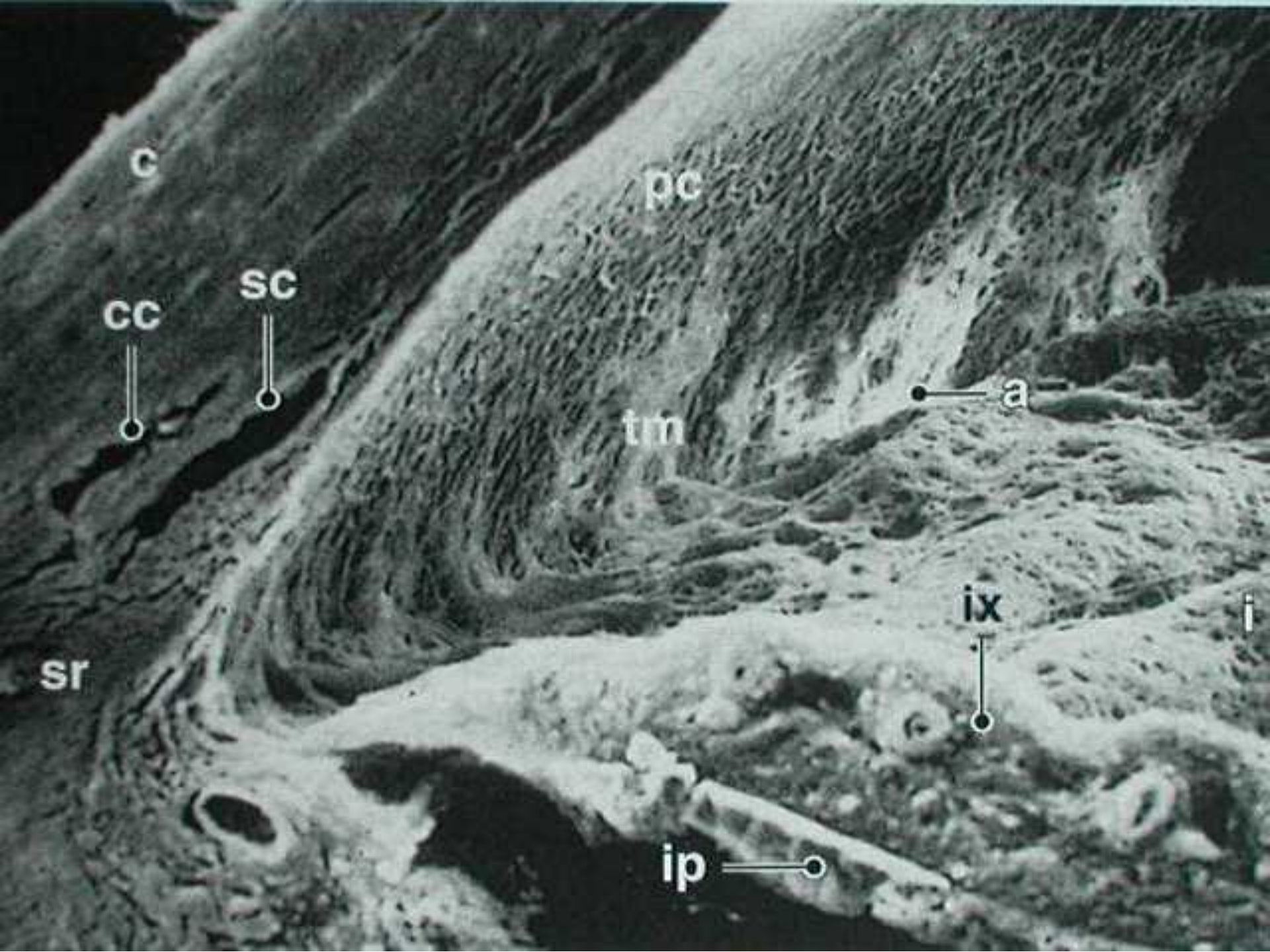


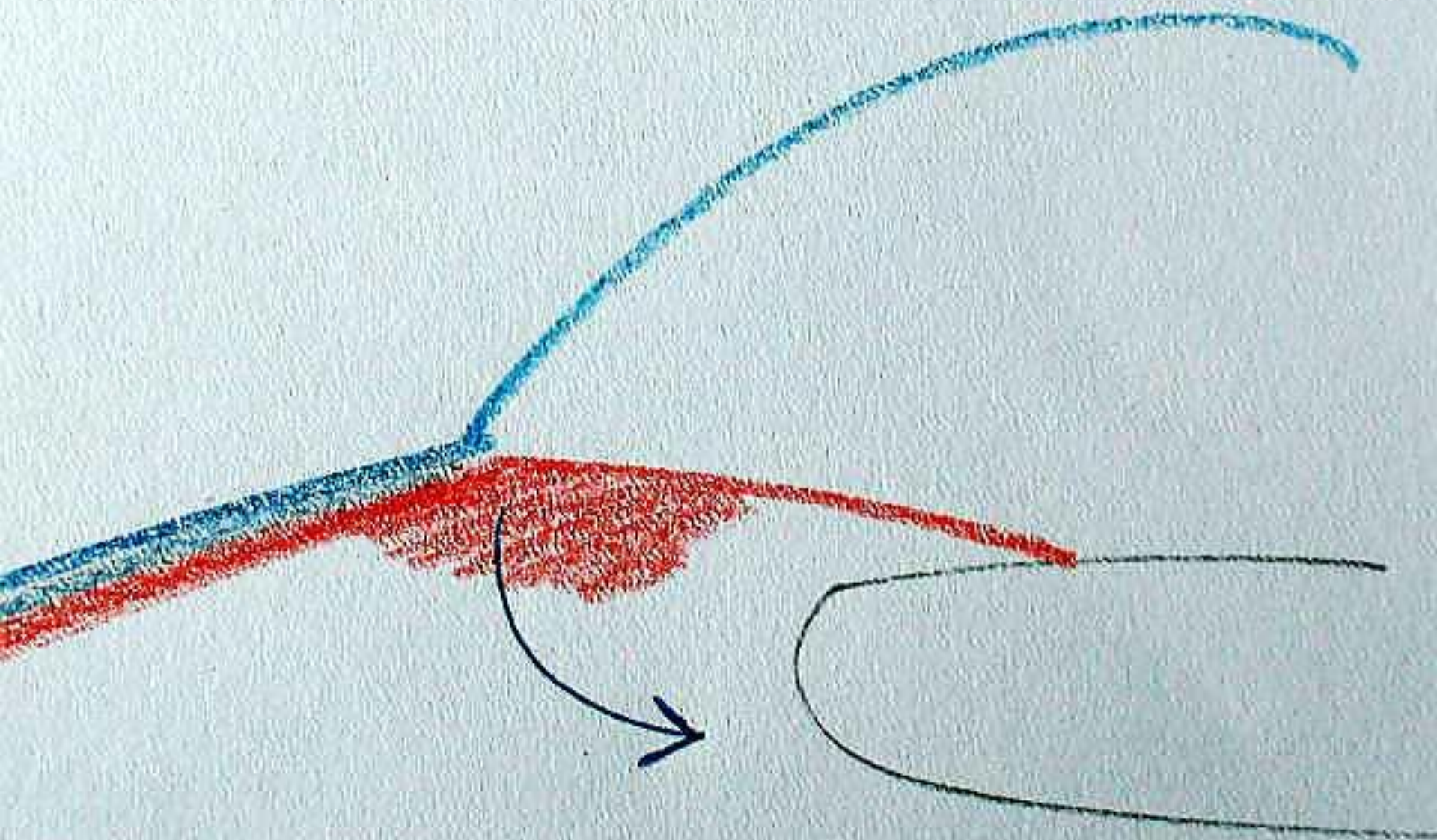
Schlemm's canal

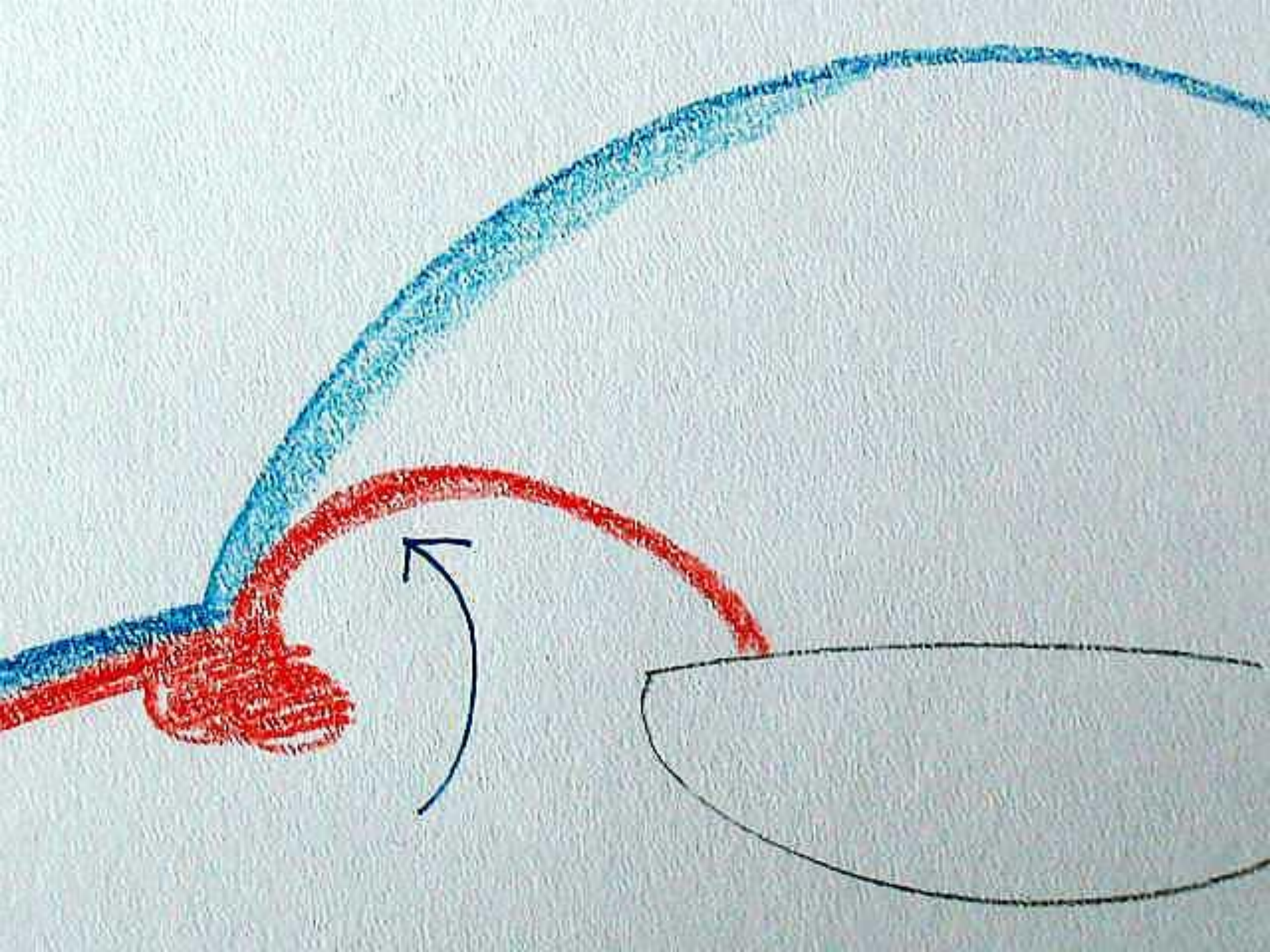
trabecular meshwork

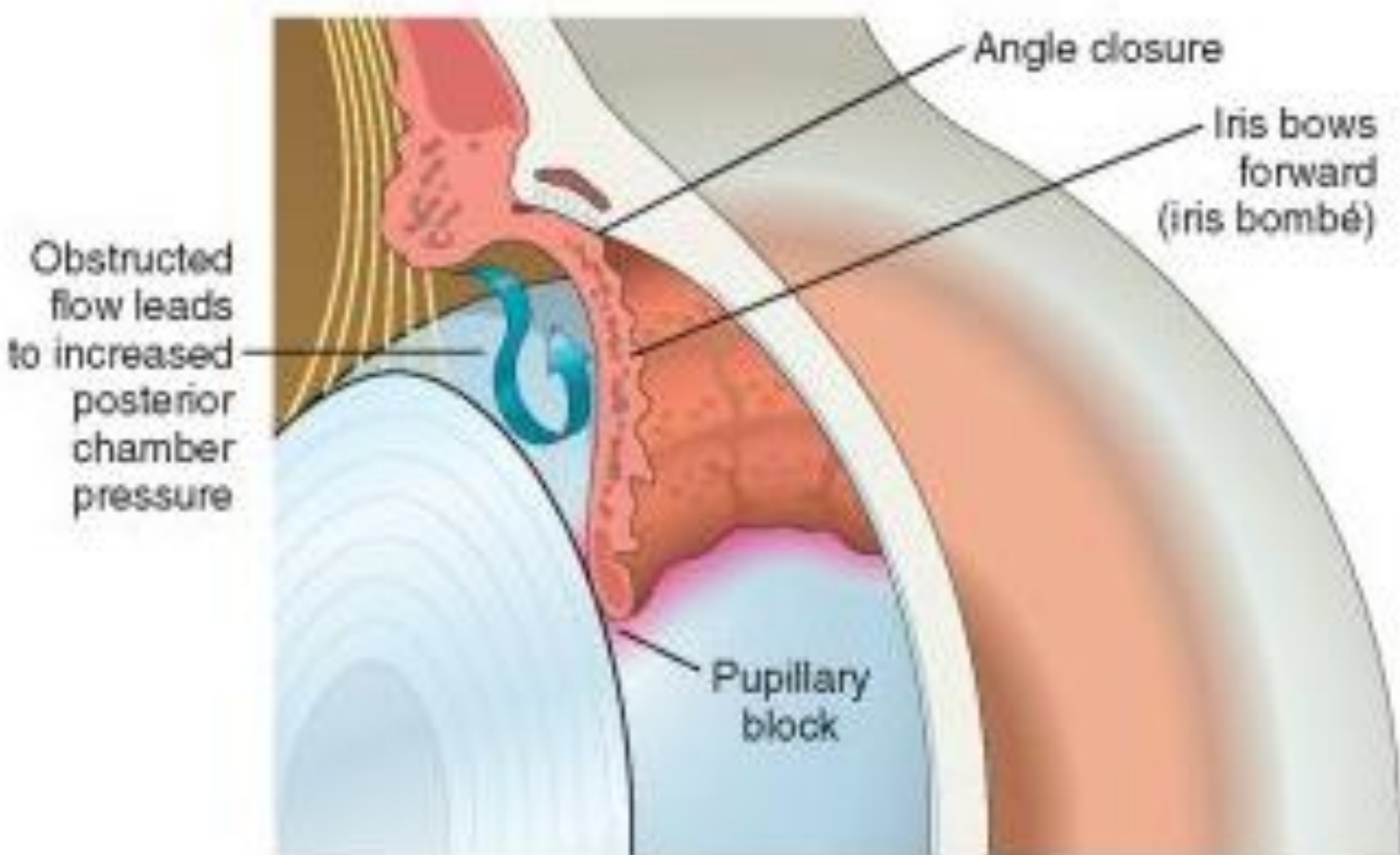
ciliary body

aqueous humor





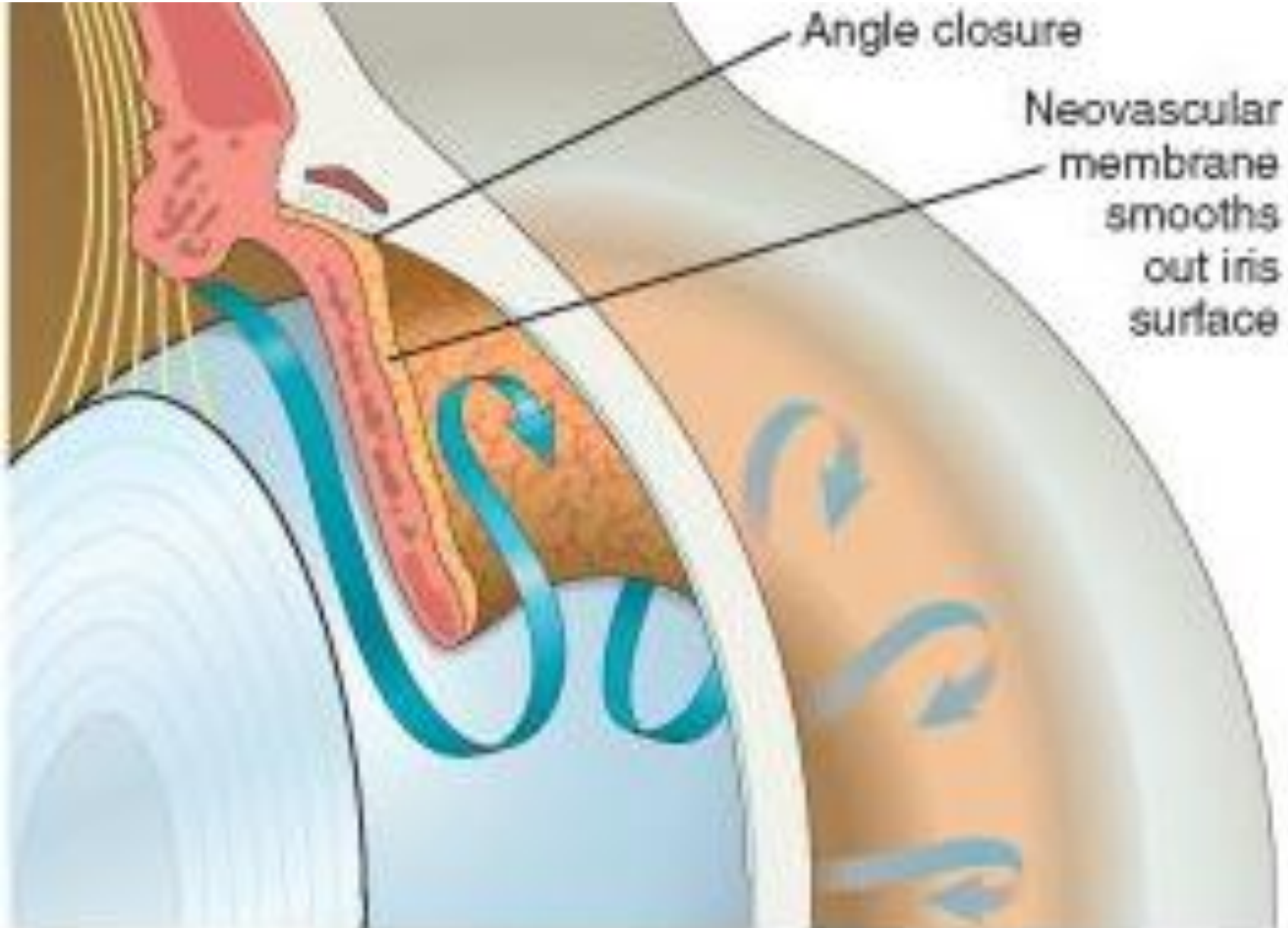




PRIMARY ANGLE-CLOSURE GLAUCOMA

Secondary Angle Closure Glaucoma

- Various pathologic membranes form over the iris. When they contract, it will draw the iris over the trabecular meshwork, blocking the way to the canal of schlemm.
- Eg.- Chronic retinal ischemia, retinoblastoma => overexpression of VEGF => formation of vascular membrane over iris. (Neovascular Glaucoma)
- Tumors on the ciliary body



NEOVASCULAR GLAUCOMA

Open Angle Glaucoma

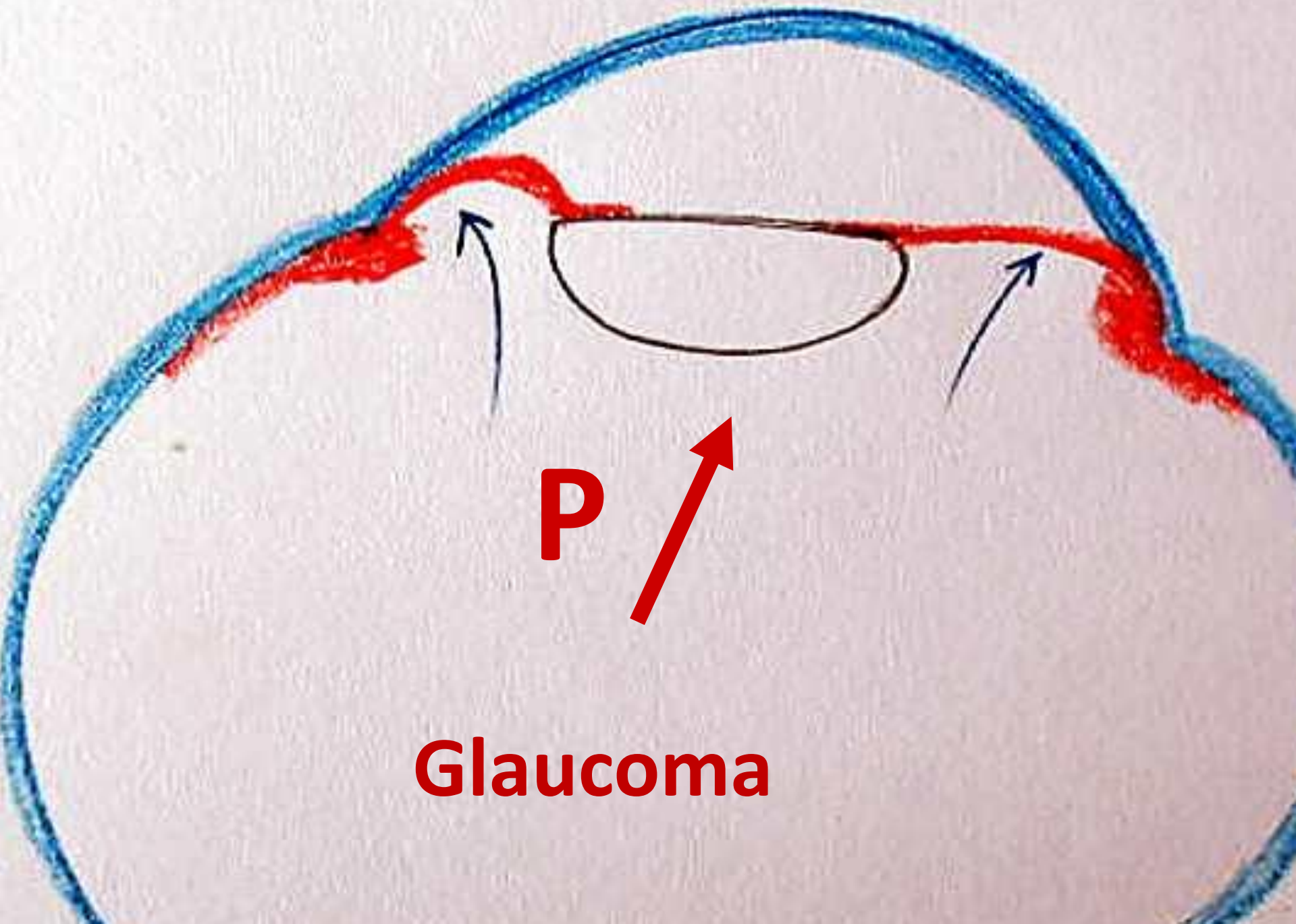
Primary- Idiopathic pathogenesis

- mutation in MYOC gene encoding for a protein called myocilin found in the trabecular meshwork.

- Function not identified.

Open Angle Glaucoma

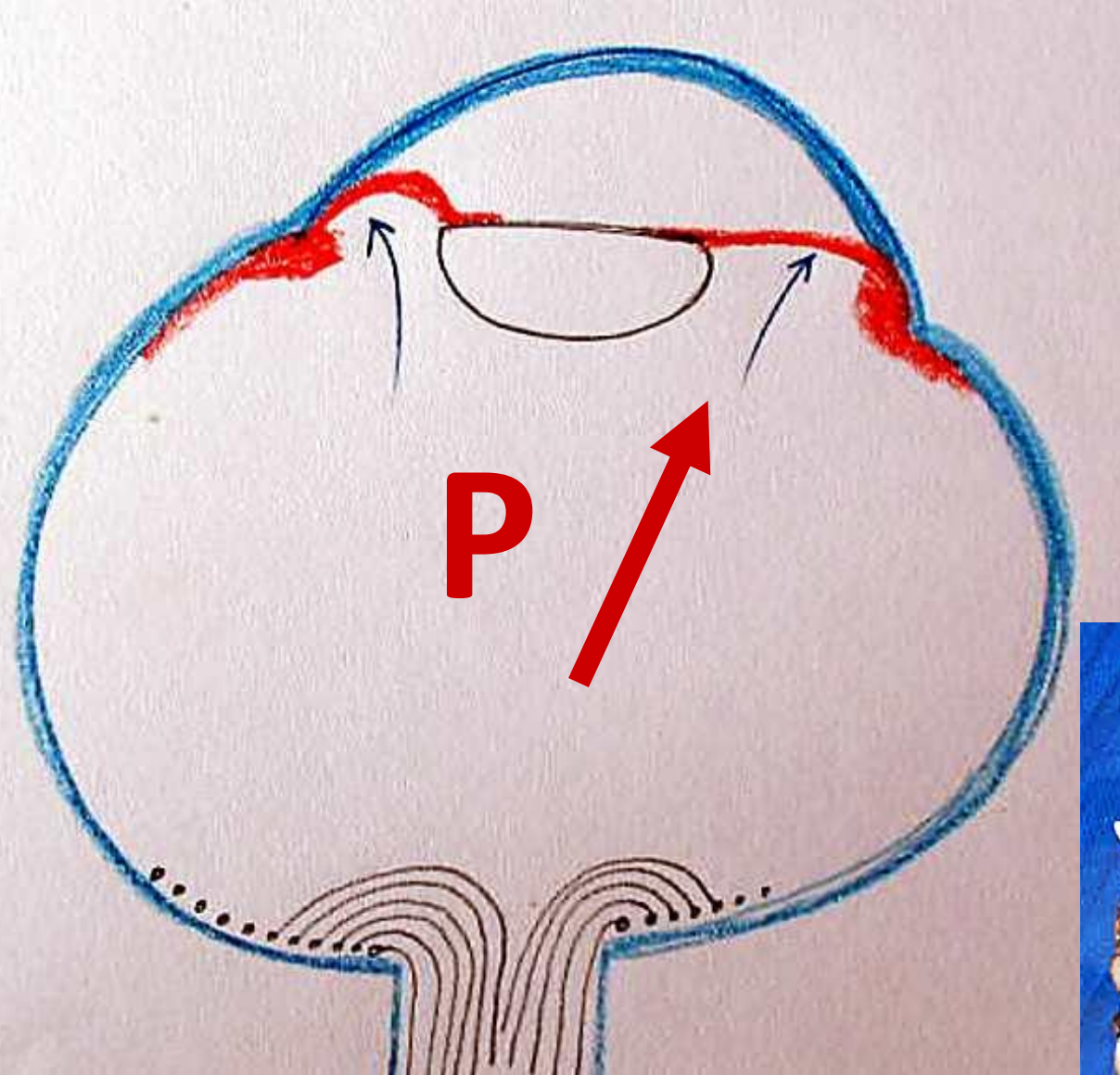
- Secondary
 - Particulate matter clogging the trabecular meshwork.
 - Protein from lens liquefaction
 - Iris epithelial pigment granules
 - Necrotic tumors

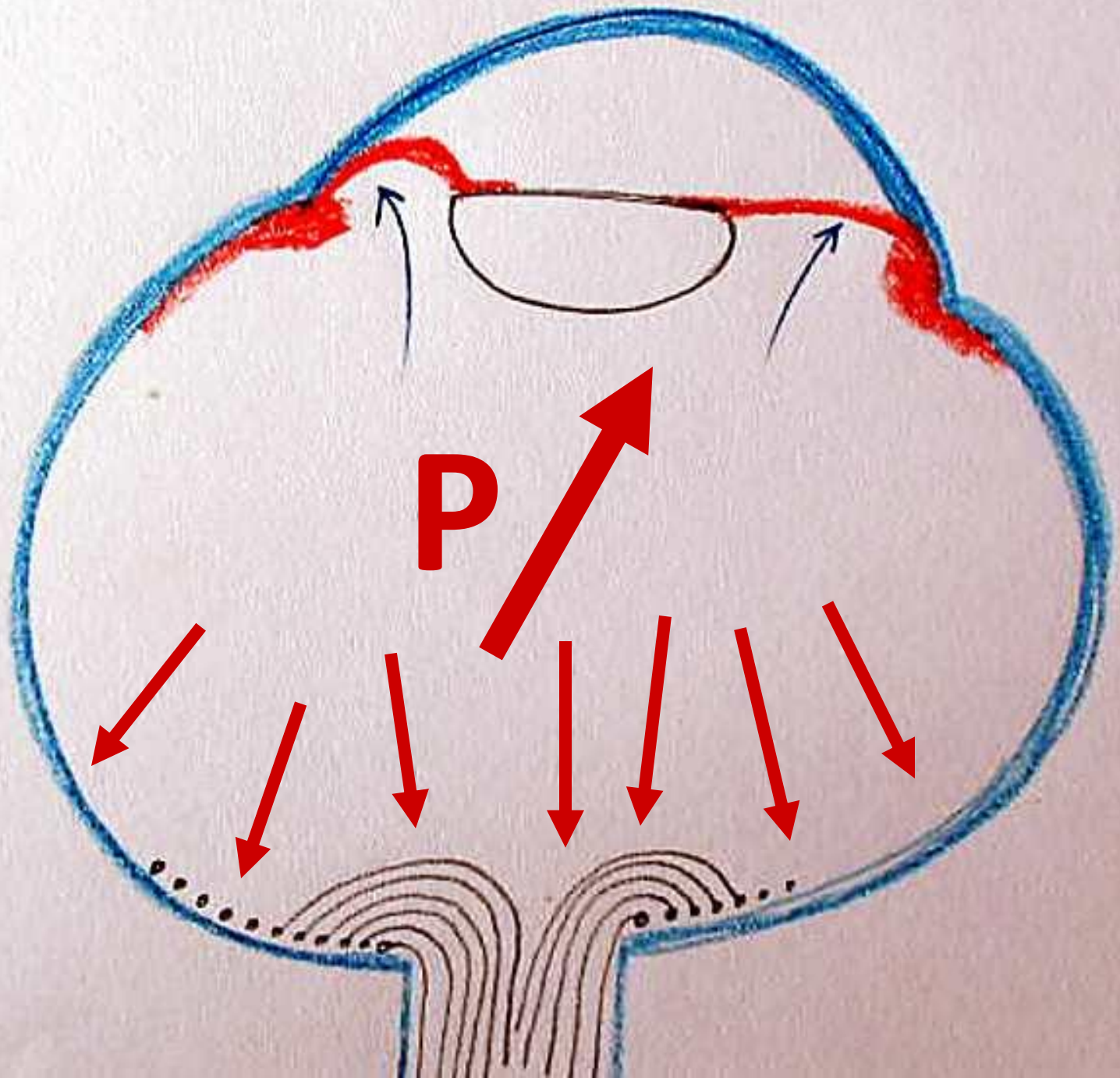


P

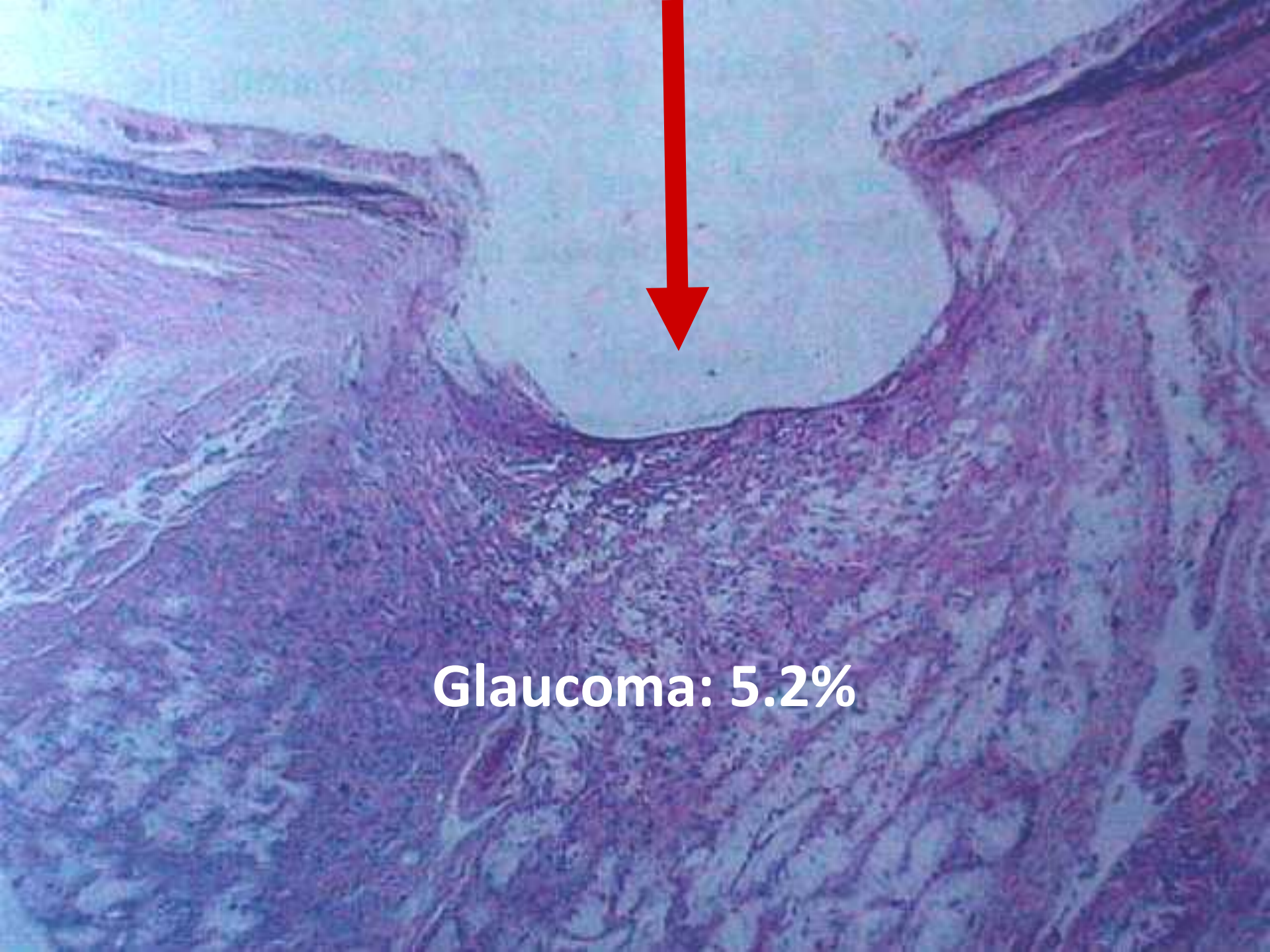


Glaucoma









Glaucoma: 5.2%



A hand-drawn diagram of a human eye in profile, facing left. The eye is outlined in blue. The iris is shaded with grey and has a red highlight. The pupil is a black circle. The sclera is white. The eyelid is drawn with black lines. The eye is positioned on the left side of the frame, with the rest of the face and head area being blank white paper.

- **Major causes:**

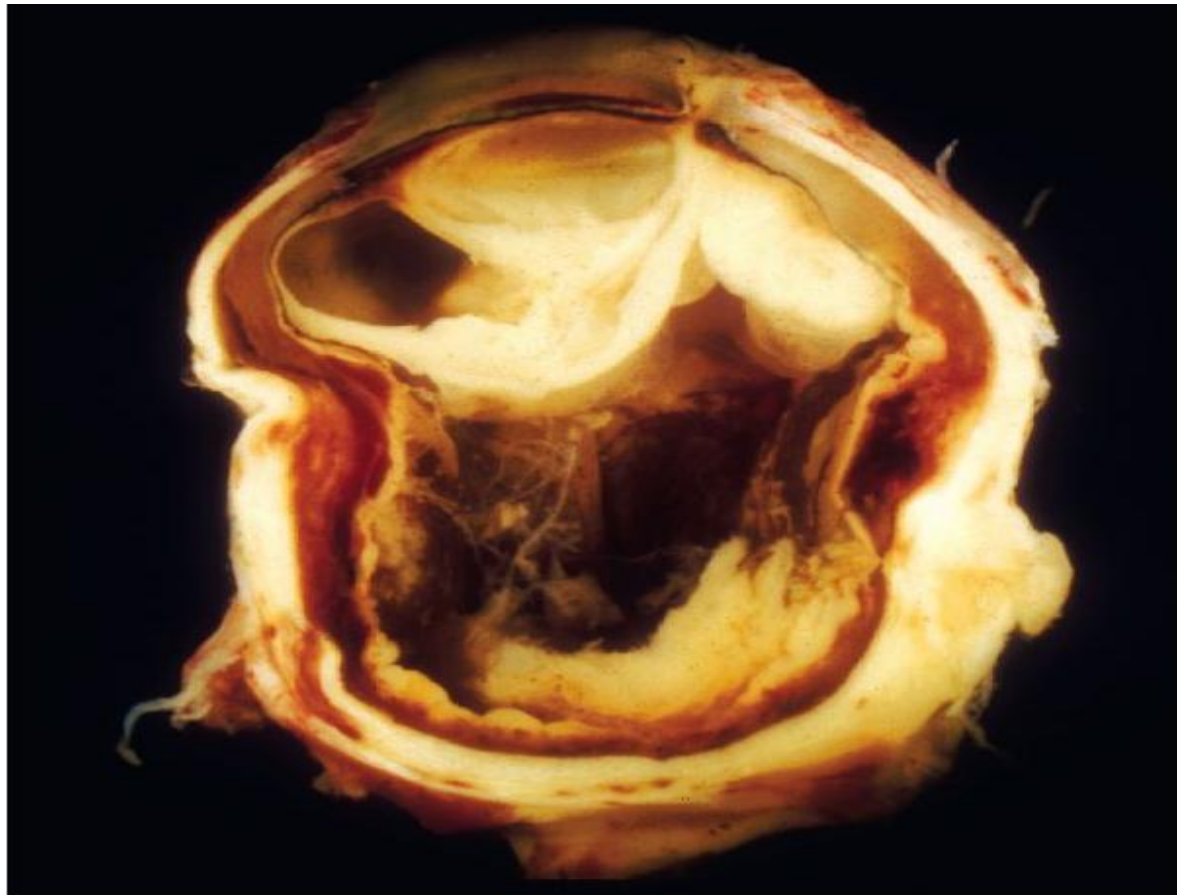
- Cataract, the leading cause of blindness in the world, also the leading cause of blindness in ET, accounting for 49.9%
- Trachoma: 11.5%
- Corneal opacity: 7.8%
- **Glaucoma: 5.2%**

Endophthalmitis and Panophthalmitis

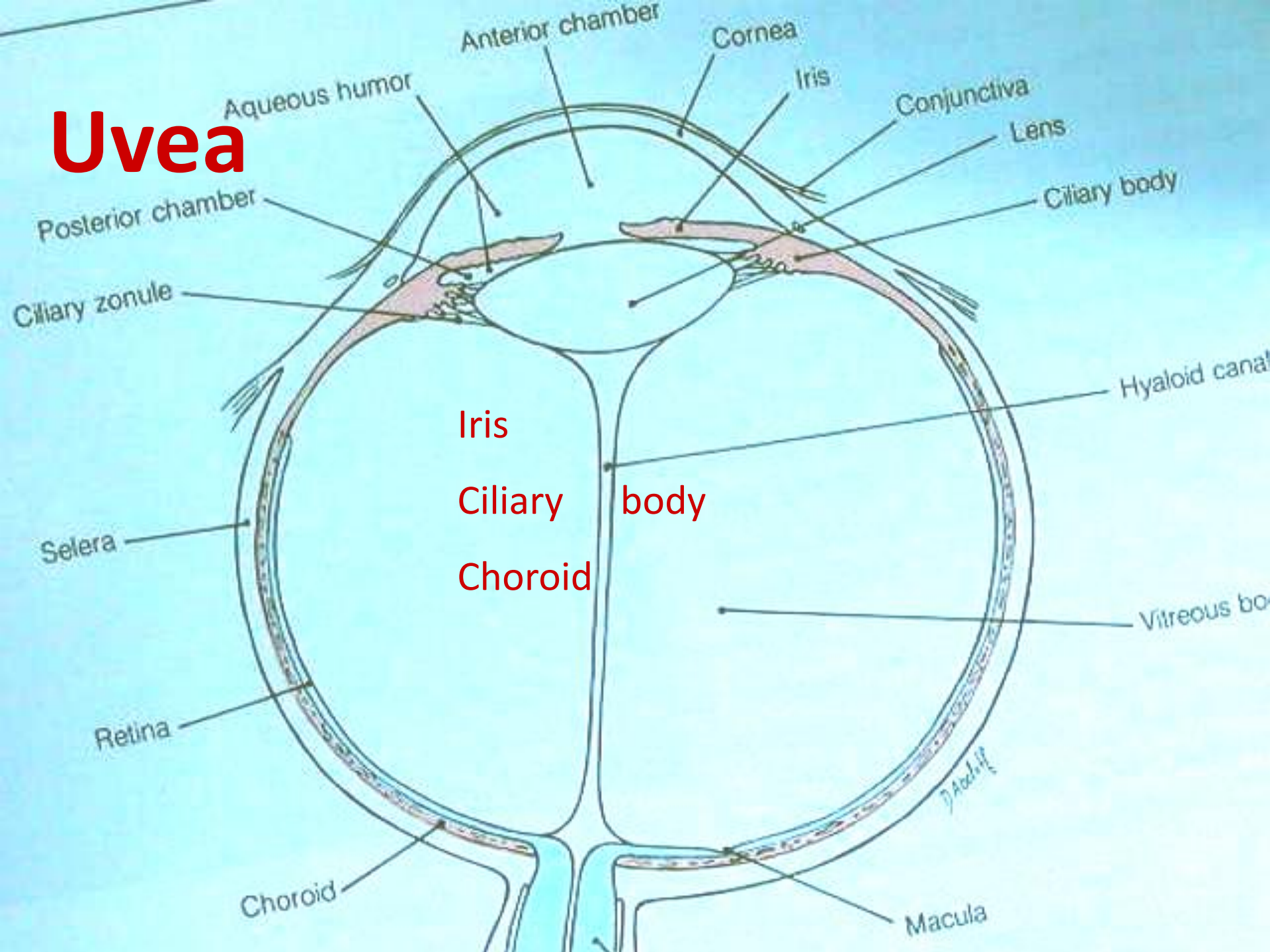
- Endophthalmitis- Intraocular inflammation extending to the vitreous humor
 - Endogenous- from hematogenous source
 - Exogenous- from the environment getting access through the eye.
- May cause adhesion of iris to trabecular meshwork(ant. Synechiae) or to the lens(post. Synechiae)

Panophthalmitis

- Involving the retina, choroid and sclera even extending into the orbit.



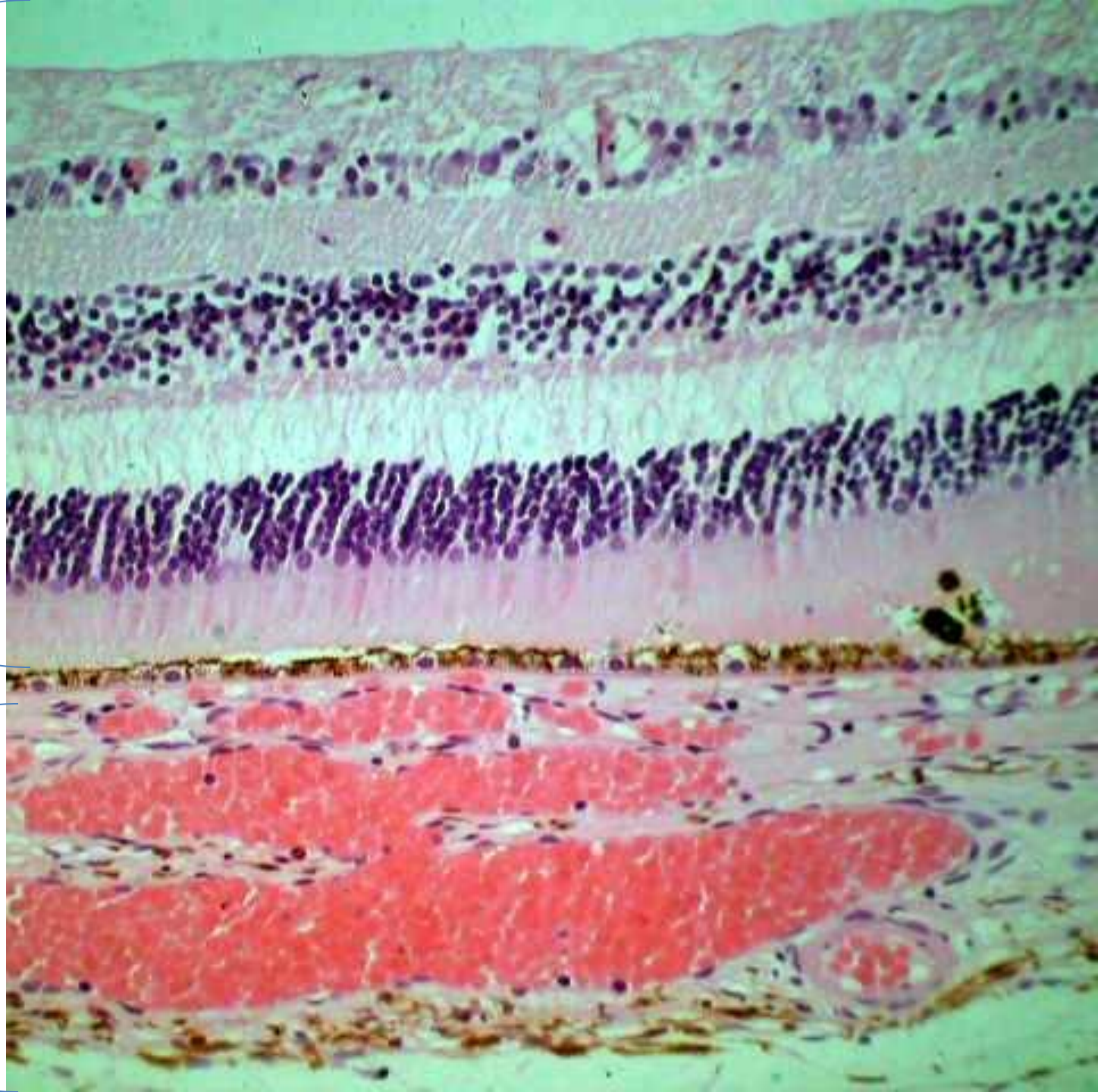
Uvea

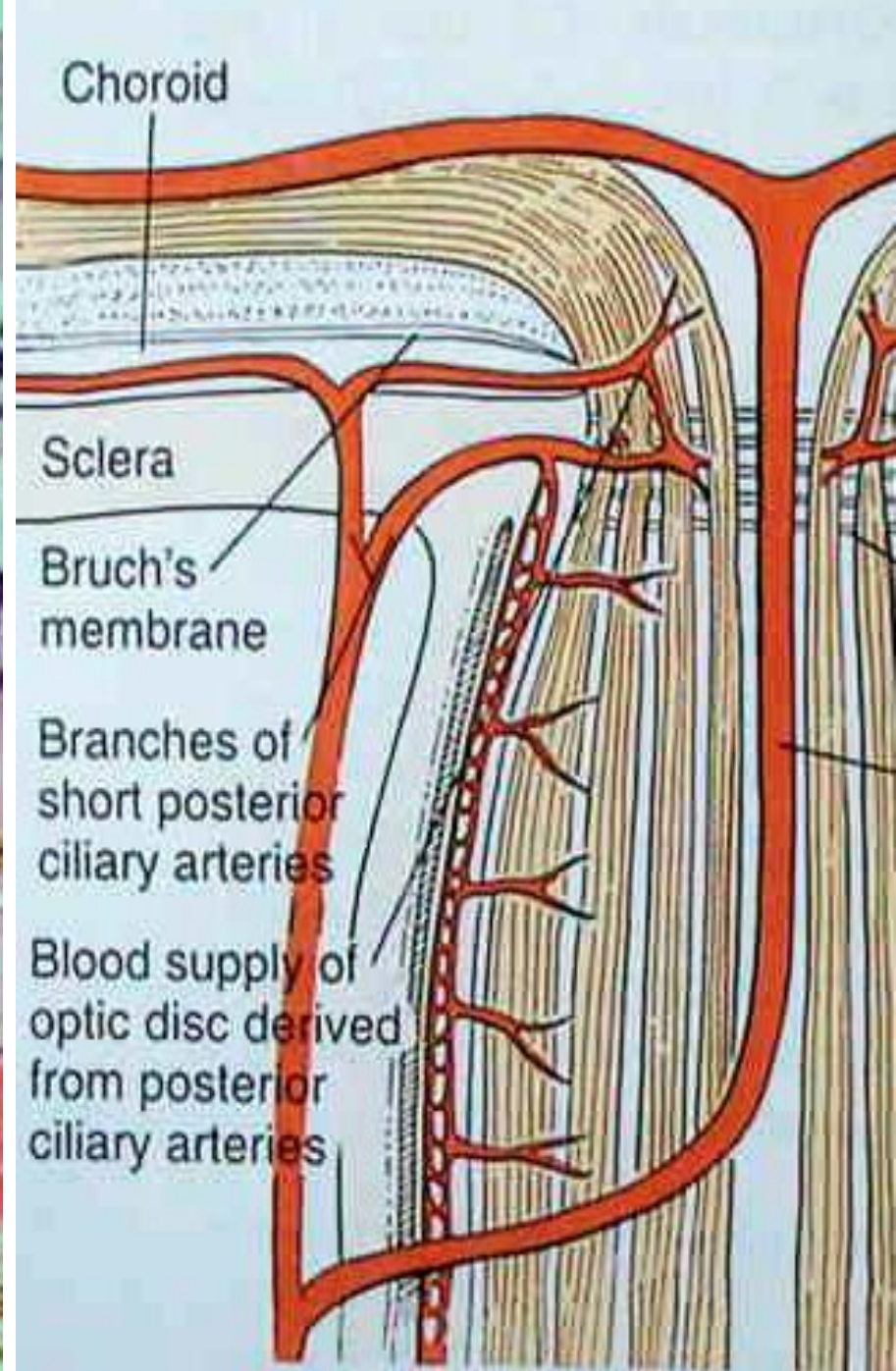
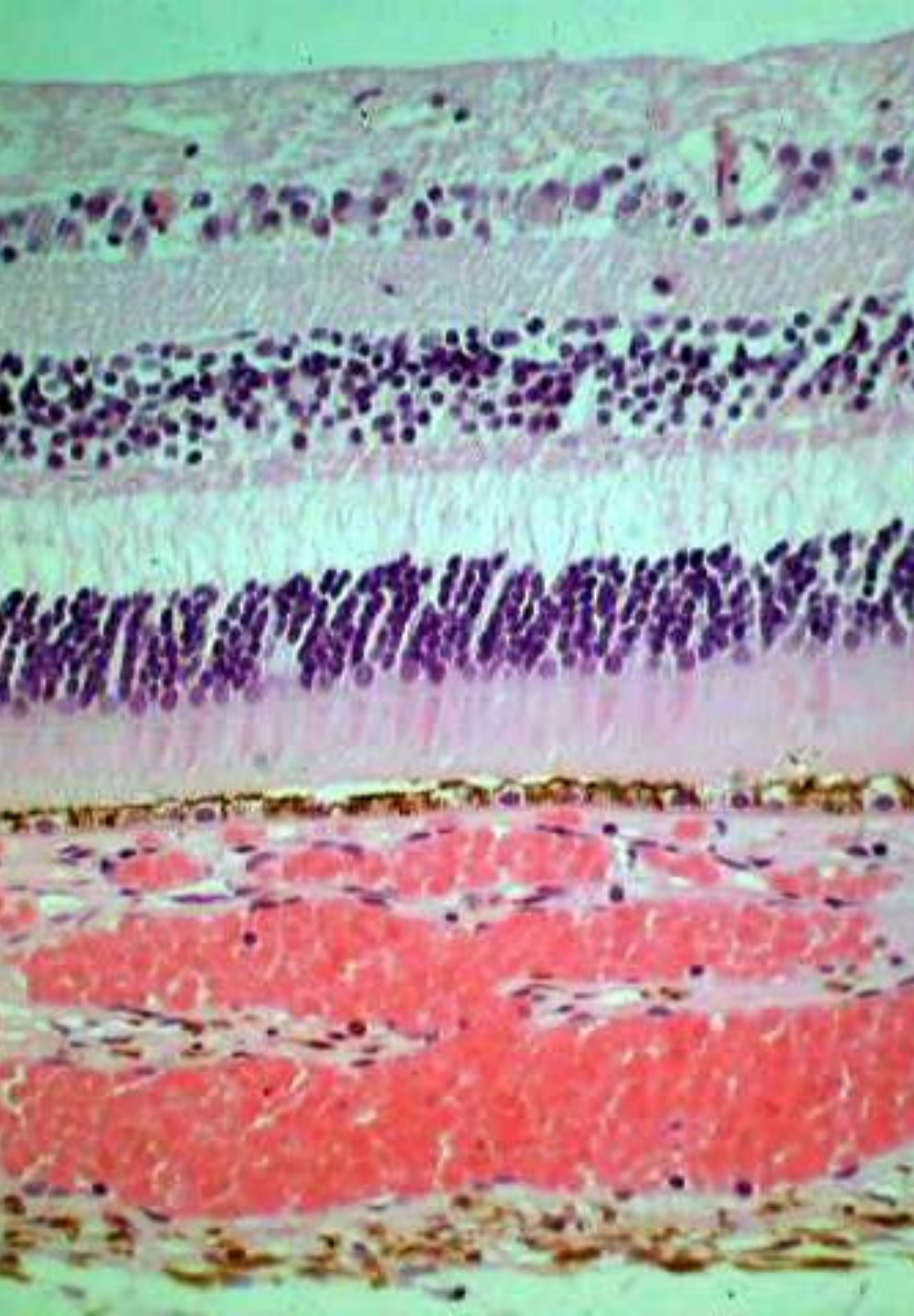


retina

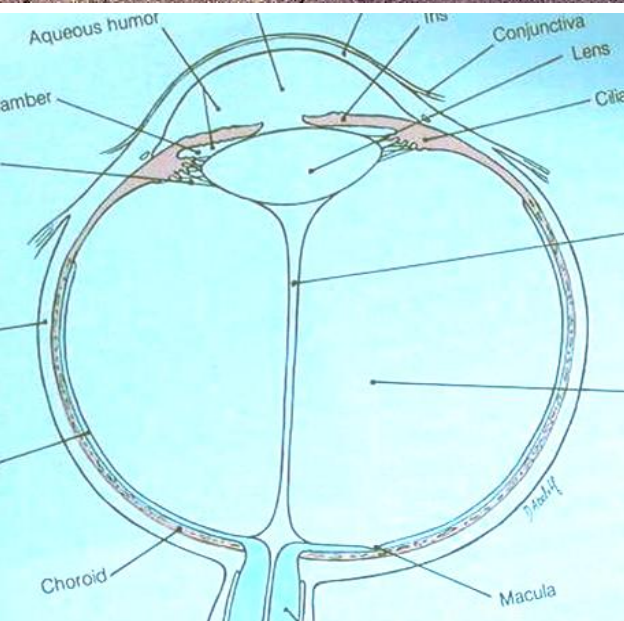
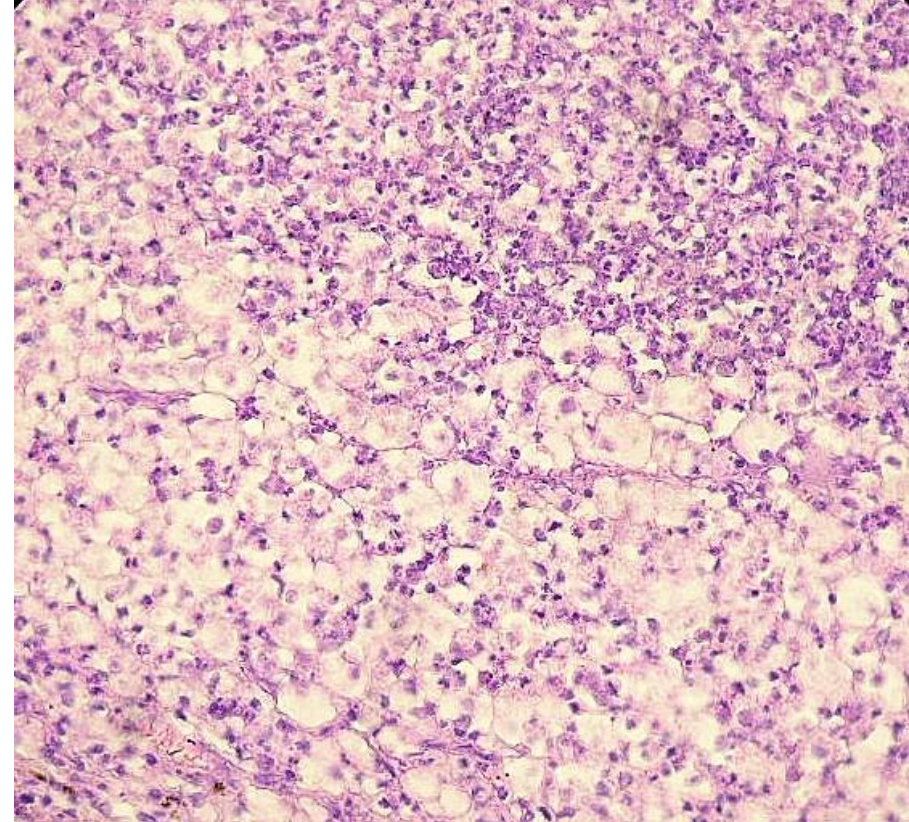
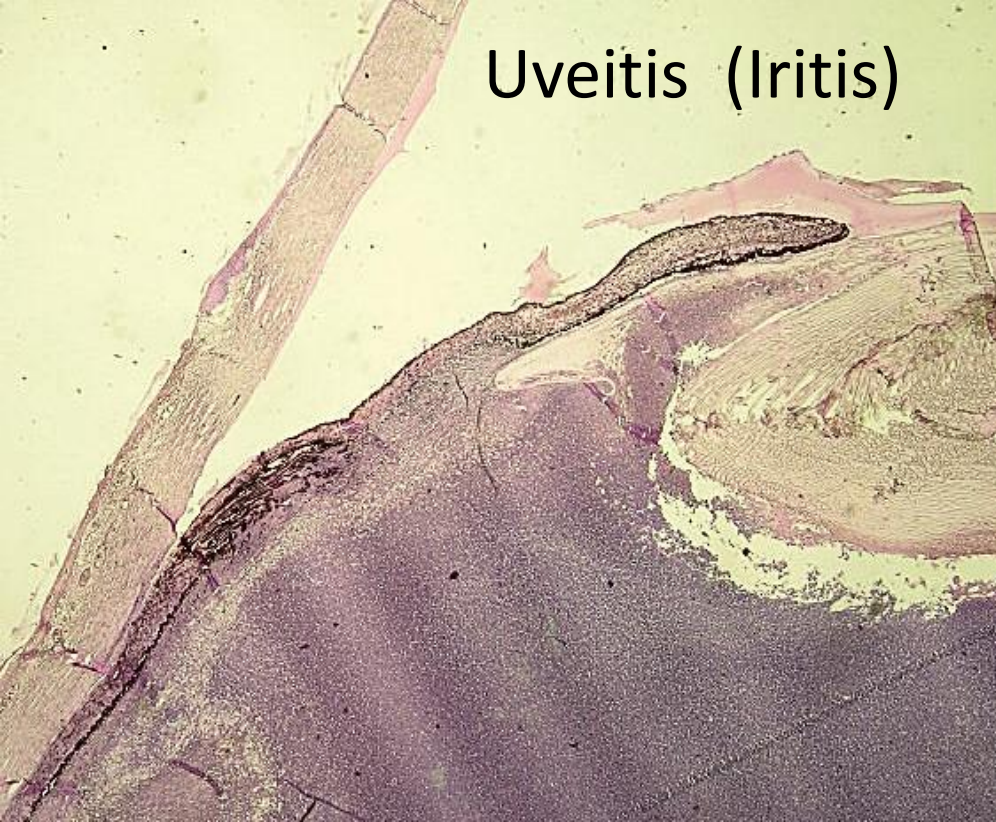
pigmented epithelium

choroid





Uveitis (Iritis)



HIV / AIDS

Pneumocystis carinii

Toxoplasma Gondii

Cytomegalovirus

Fungal

Armed Force Hospital

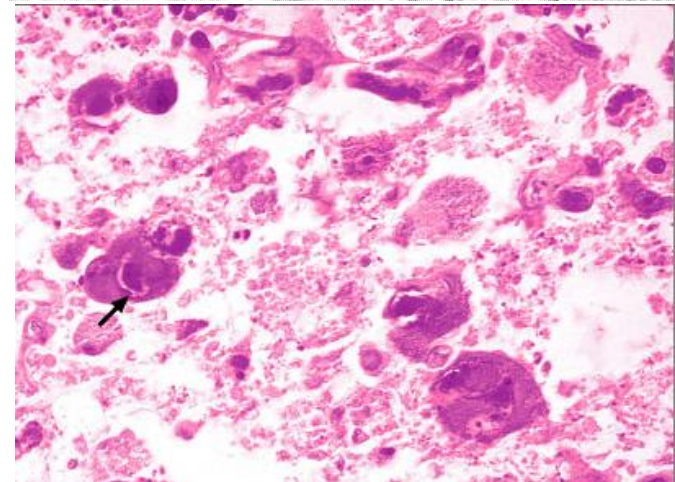
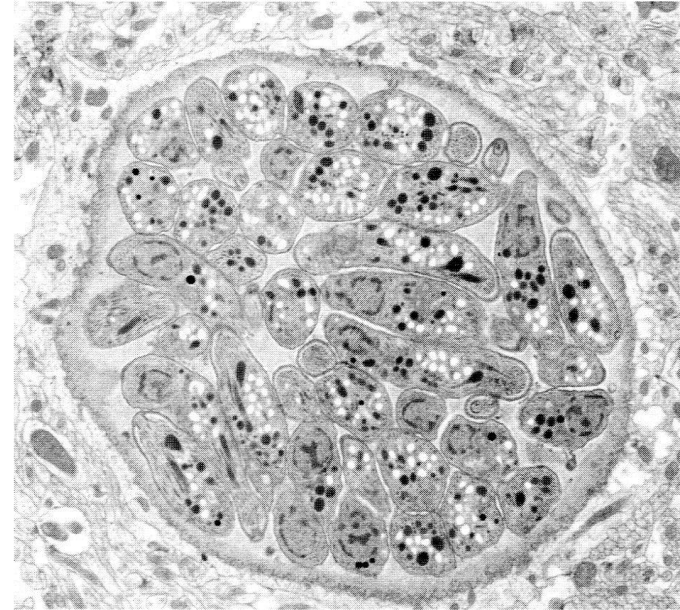
Abeba T/Giorgis et al

- 65 patients with HIV / AIDS

Uveitis: anterior segment: 6%

toxoplasma: 3%

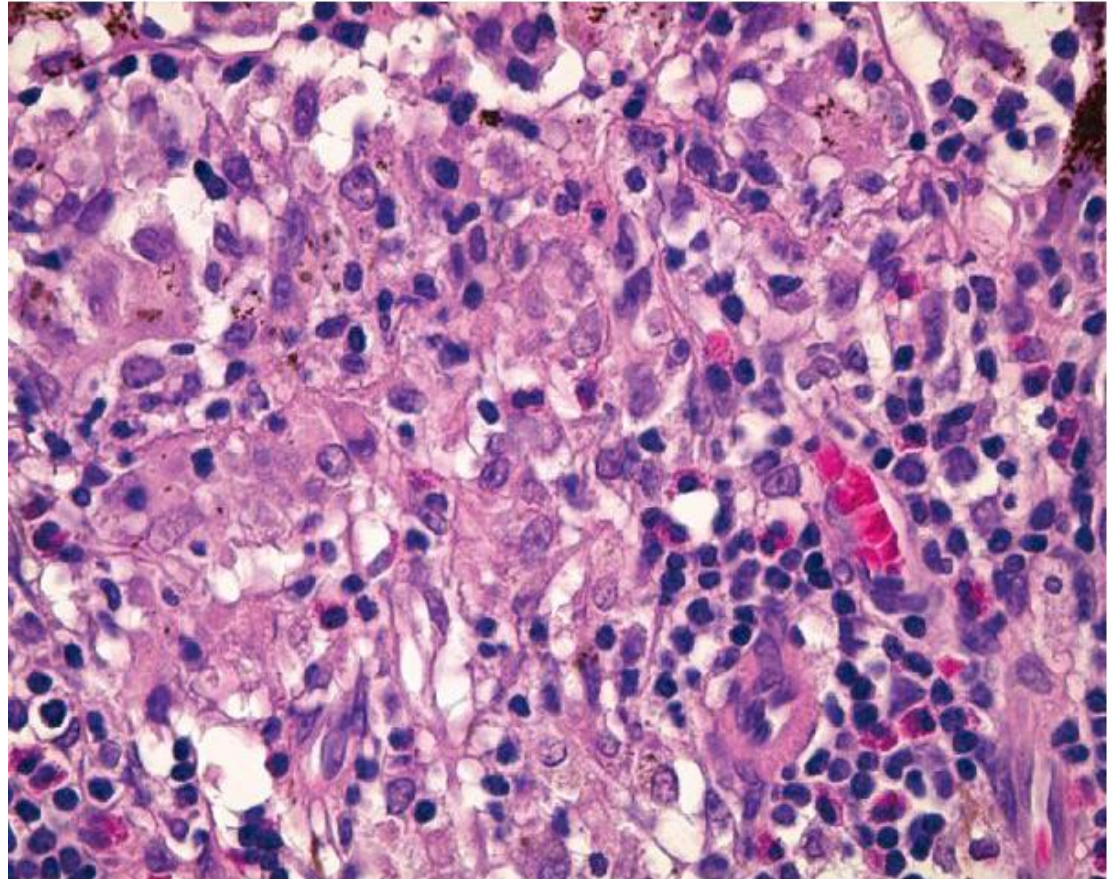
Retinitis: CMV: 6%



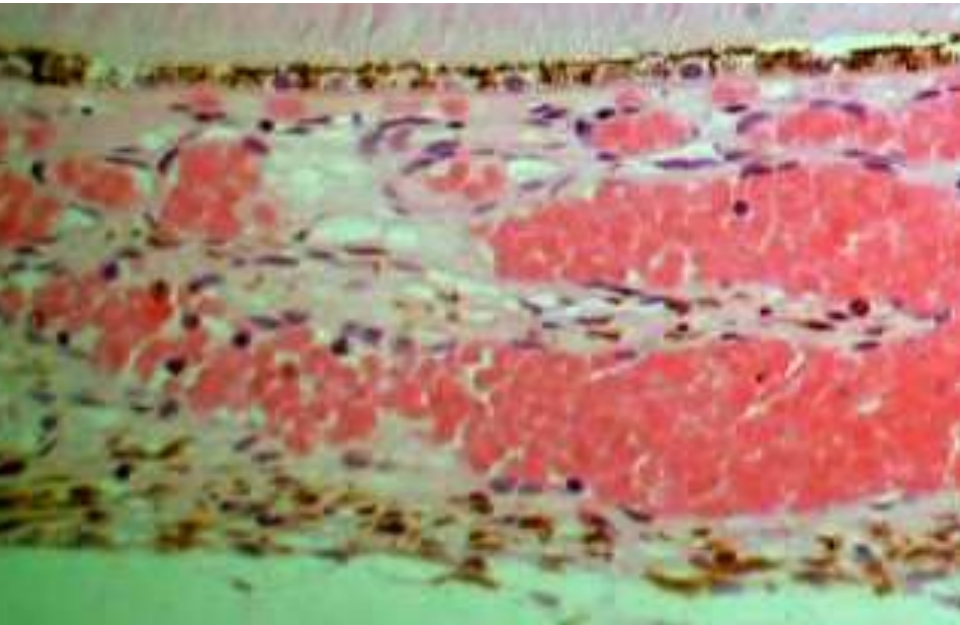
Sympathetic Ophthalmia

- Autoimmune reaction to uninjured eye following a penetrating trauma to the contralateral eye.
- Penetrating Trauma=>Retinal antigens exposed to the immune system=> Immune system sensitized against antigens=>Immune system acts against both eyes.

- Diffuse granulomatous inflammation of the uvea(iris, choroid and ciliary body)

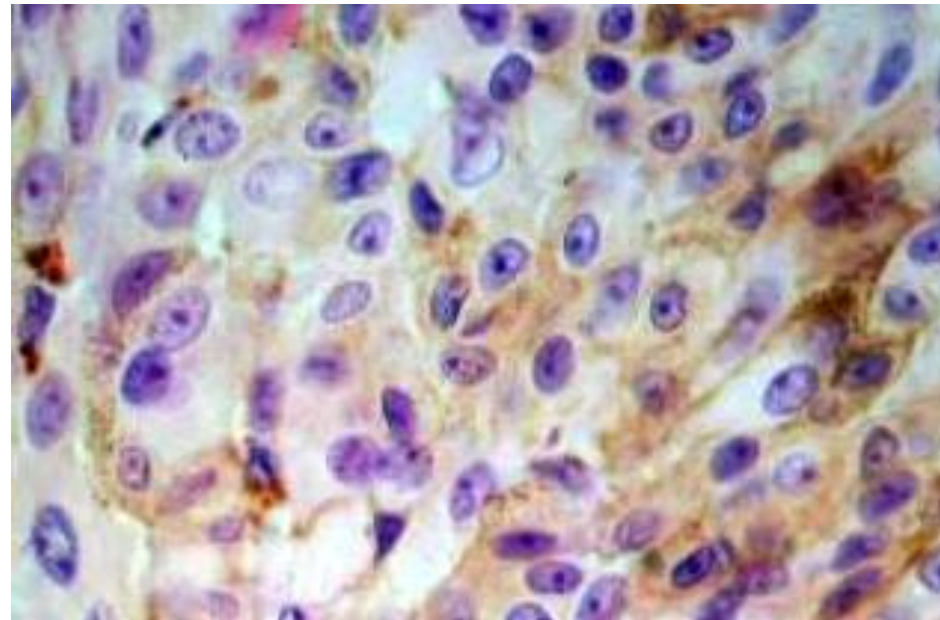


[Uvea] Choroid



metastasis

= most common intraocular malignancy of adults



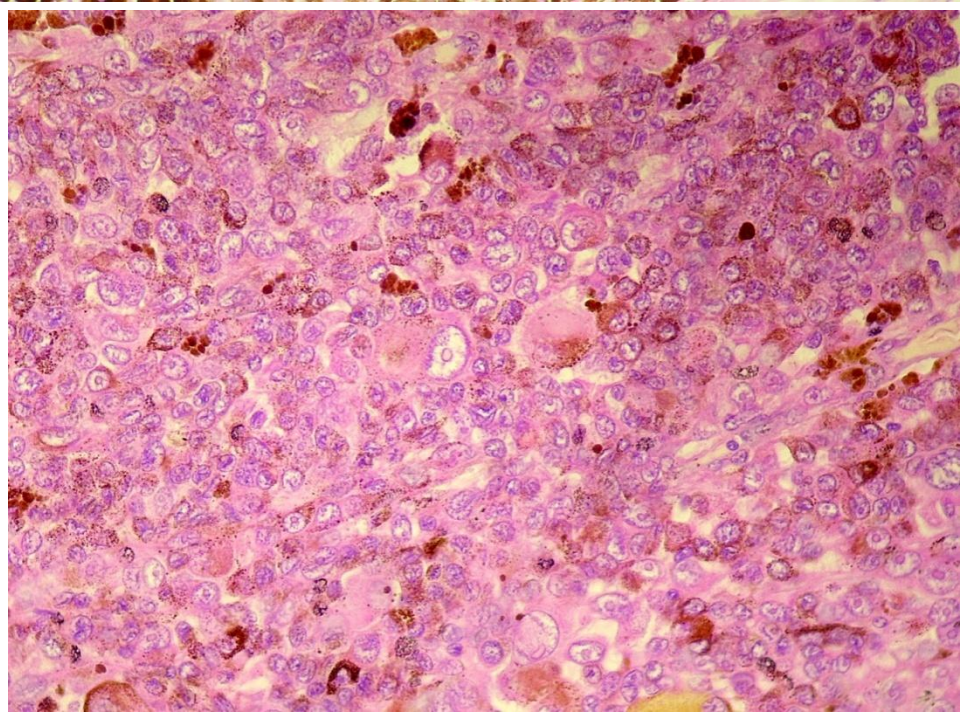
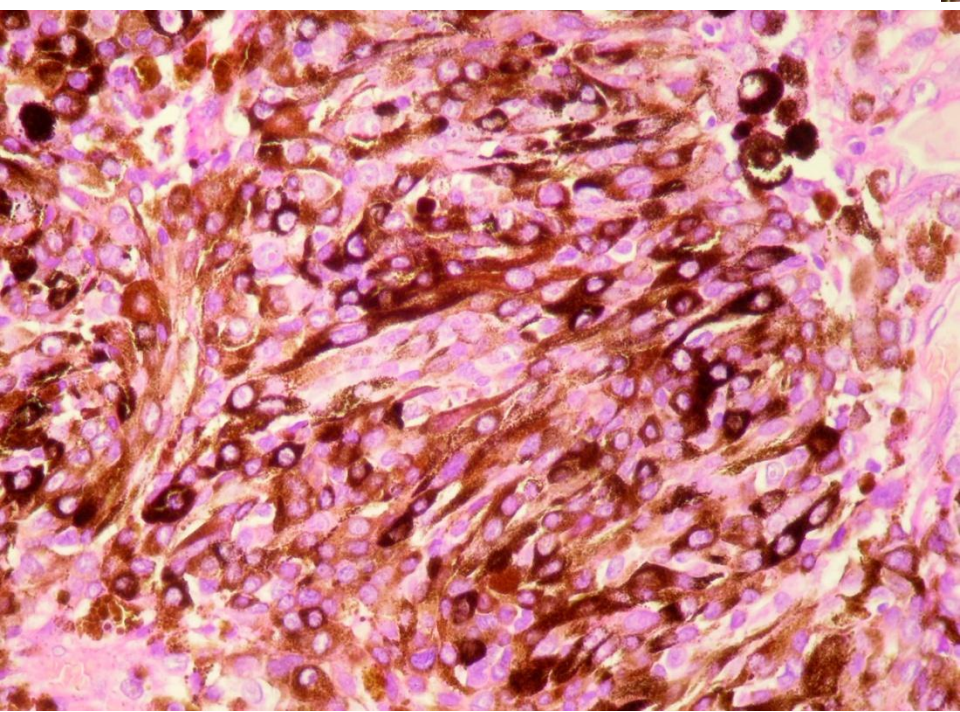
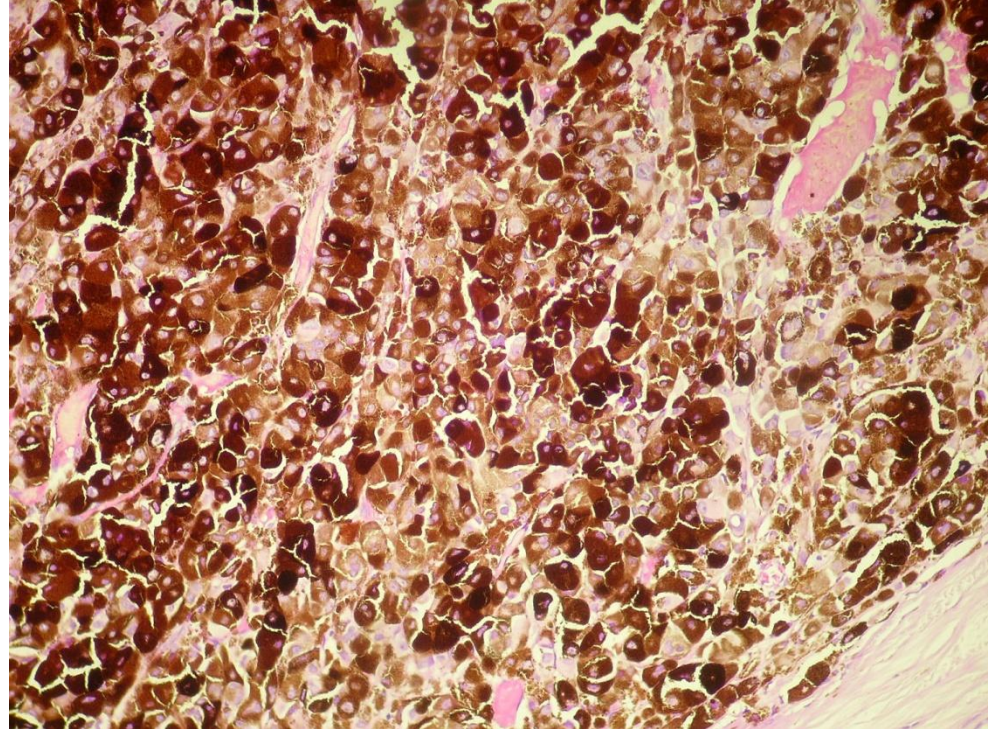
melanoma

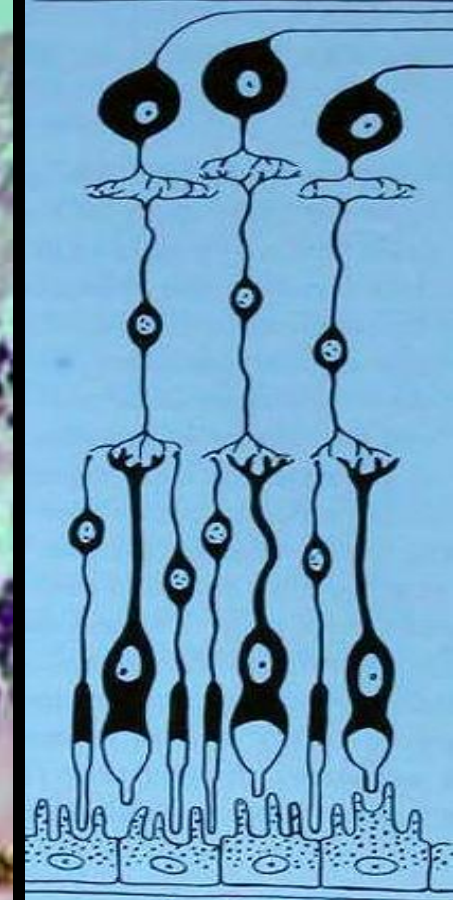
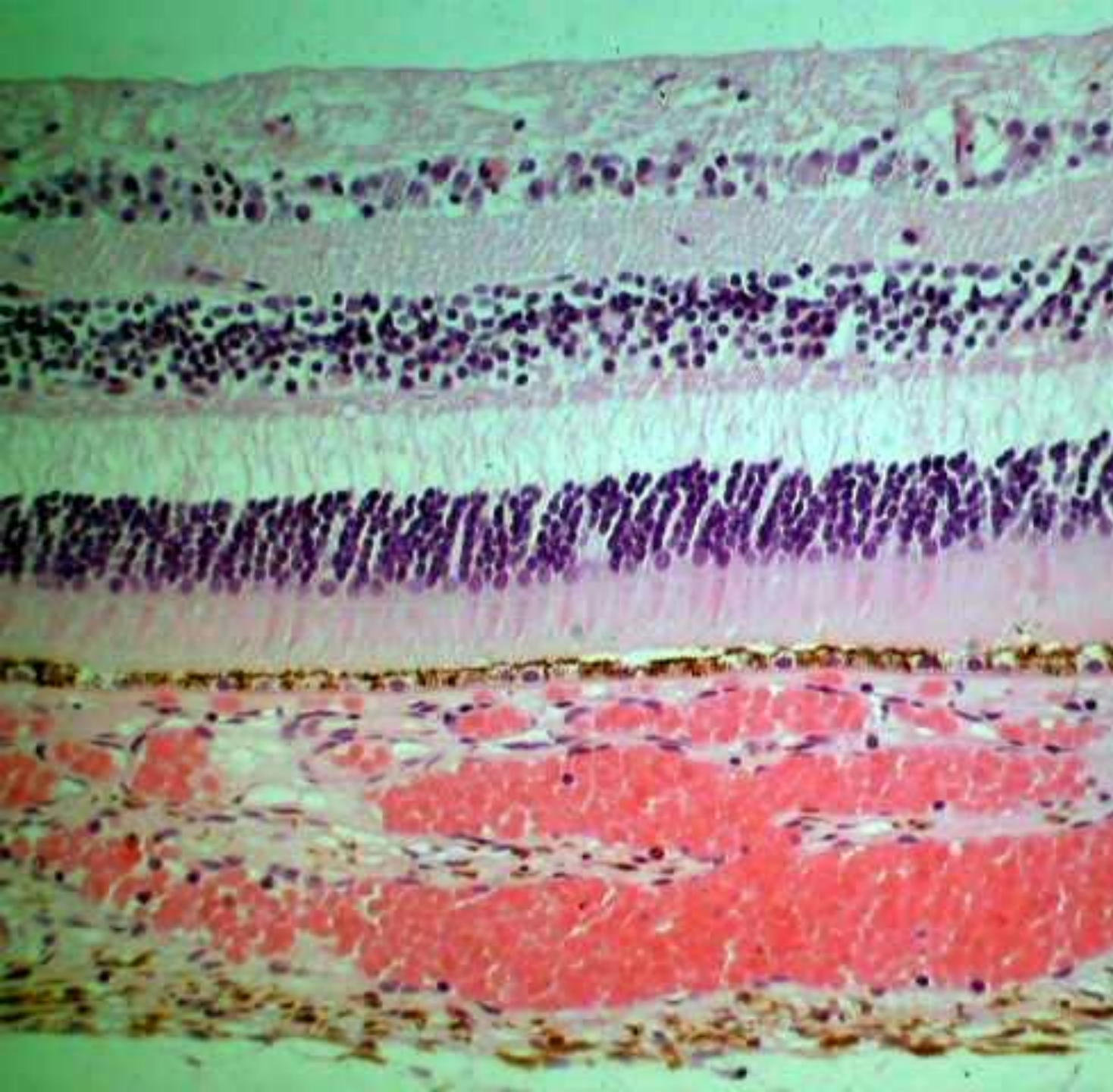
= the most common primary intraocular malignancy of adults

09.1055



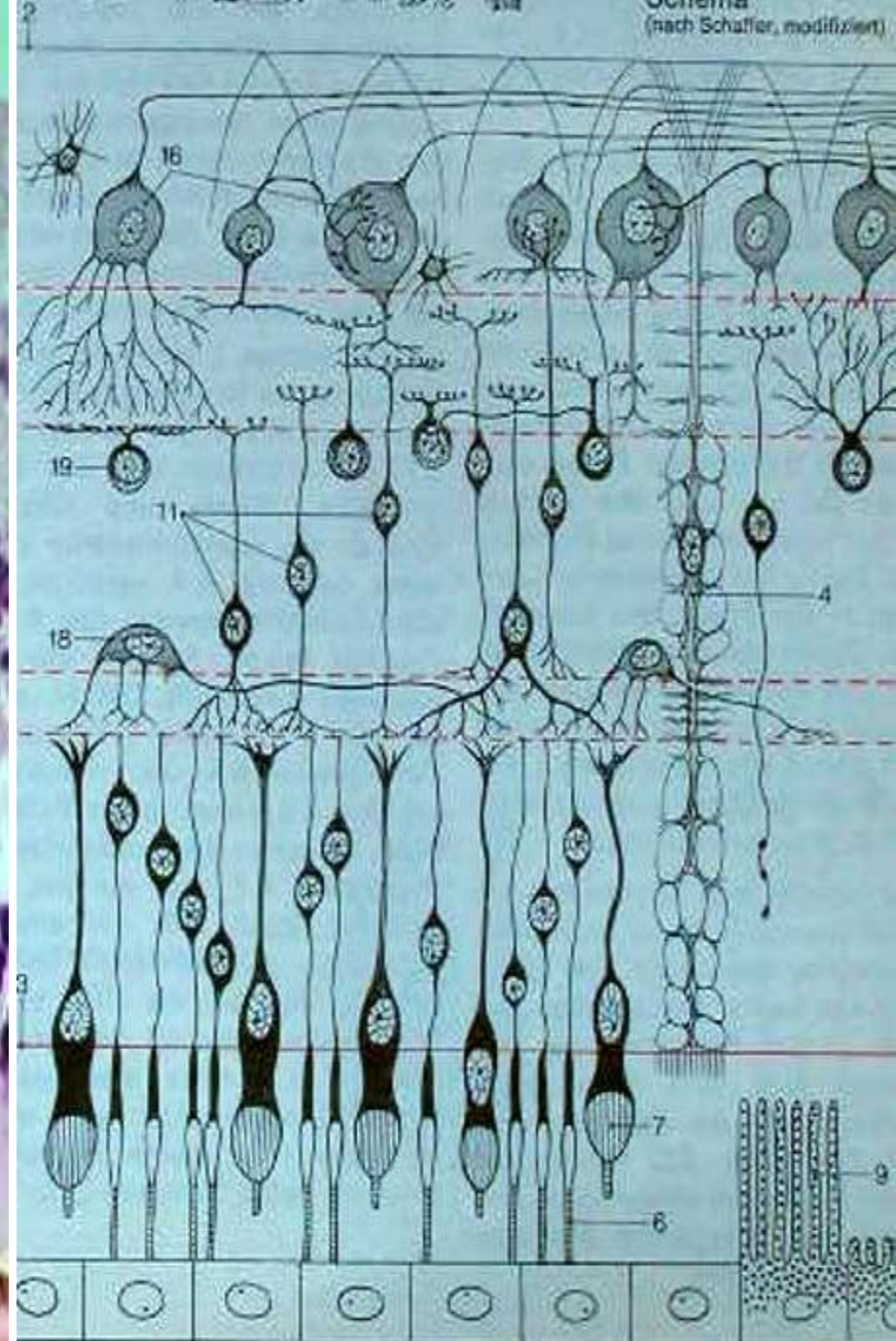
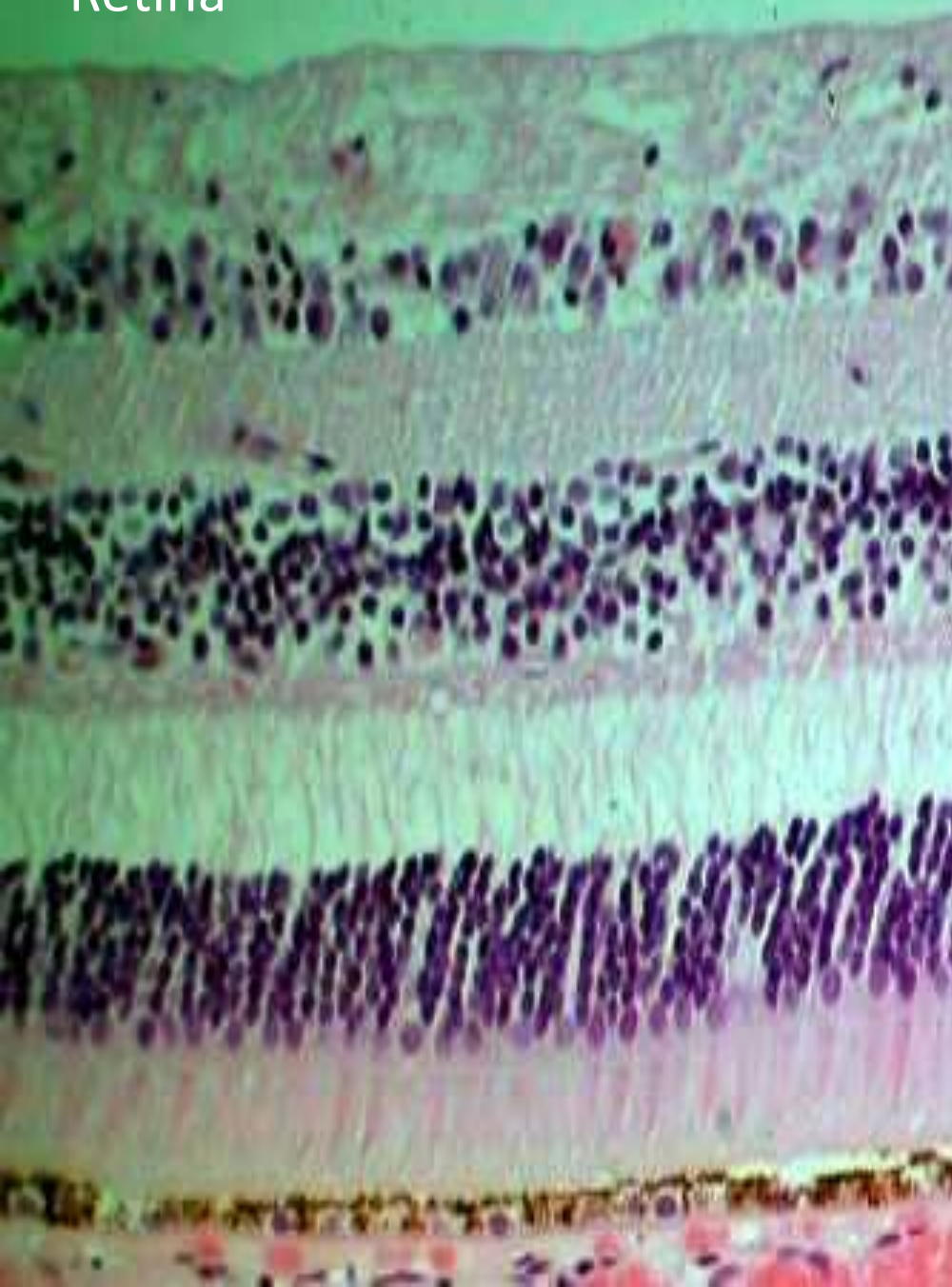
M 75

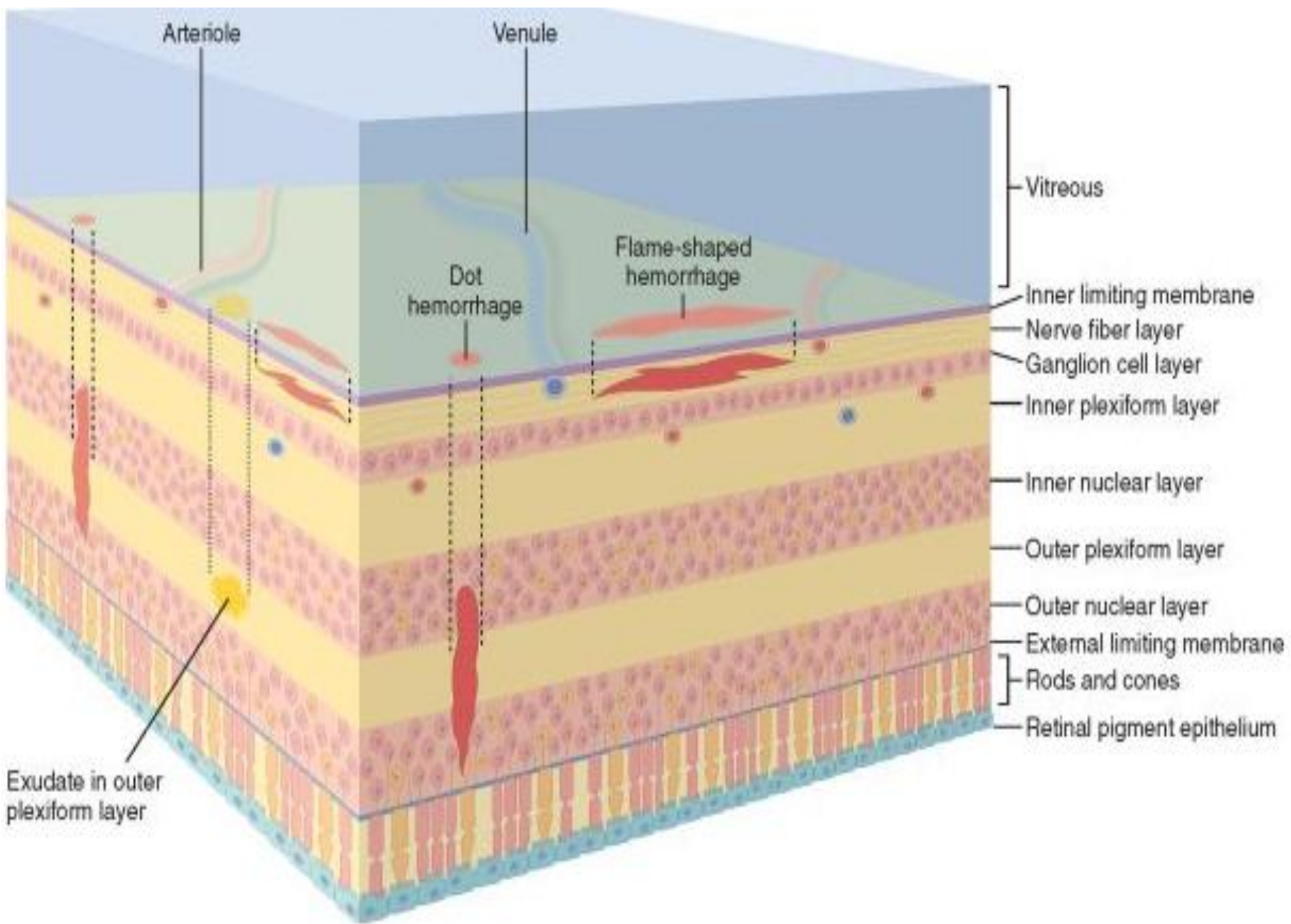


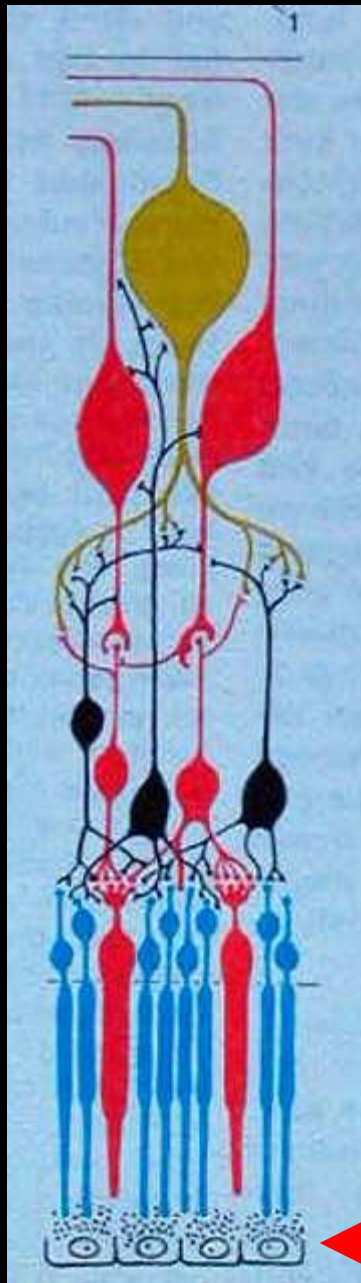


Choroid

Retina

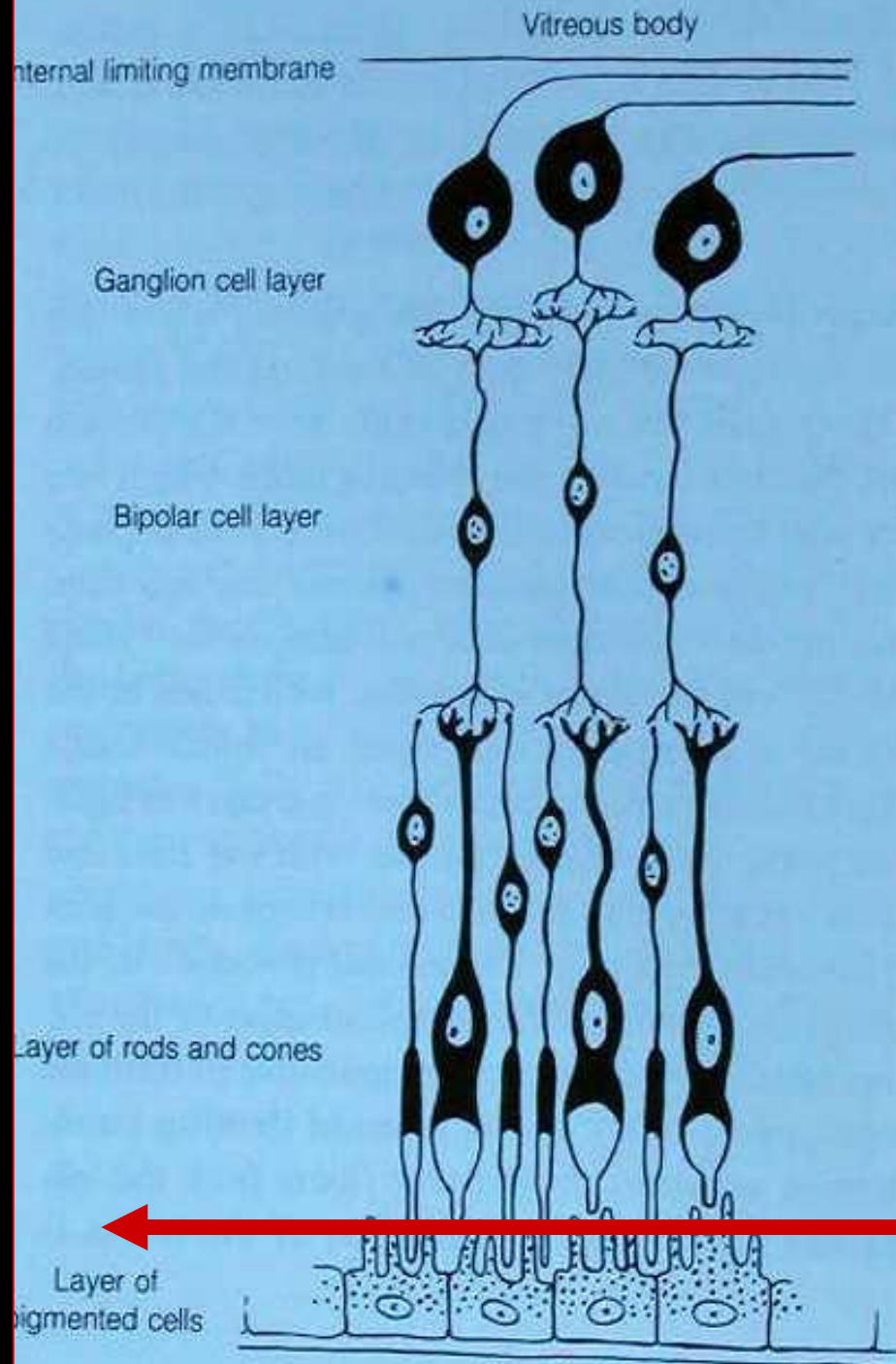






pigmented epithelium

Detachment



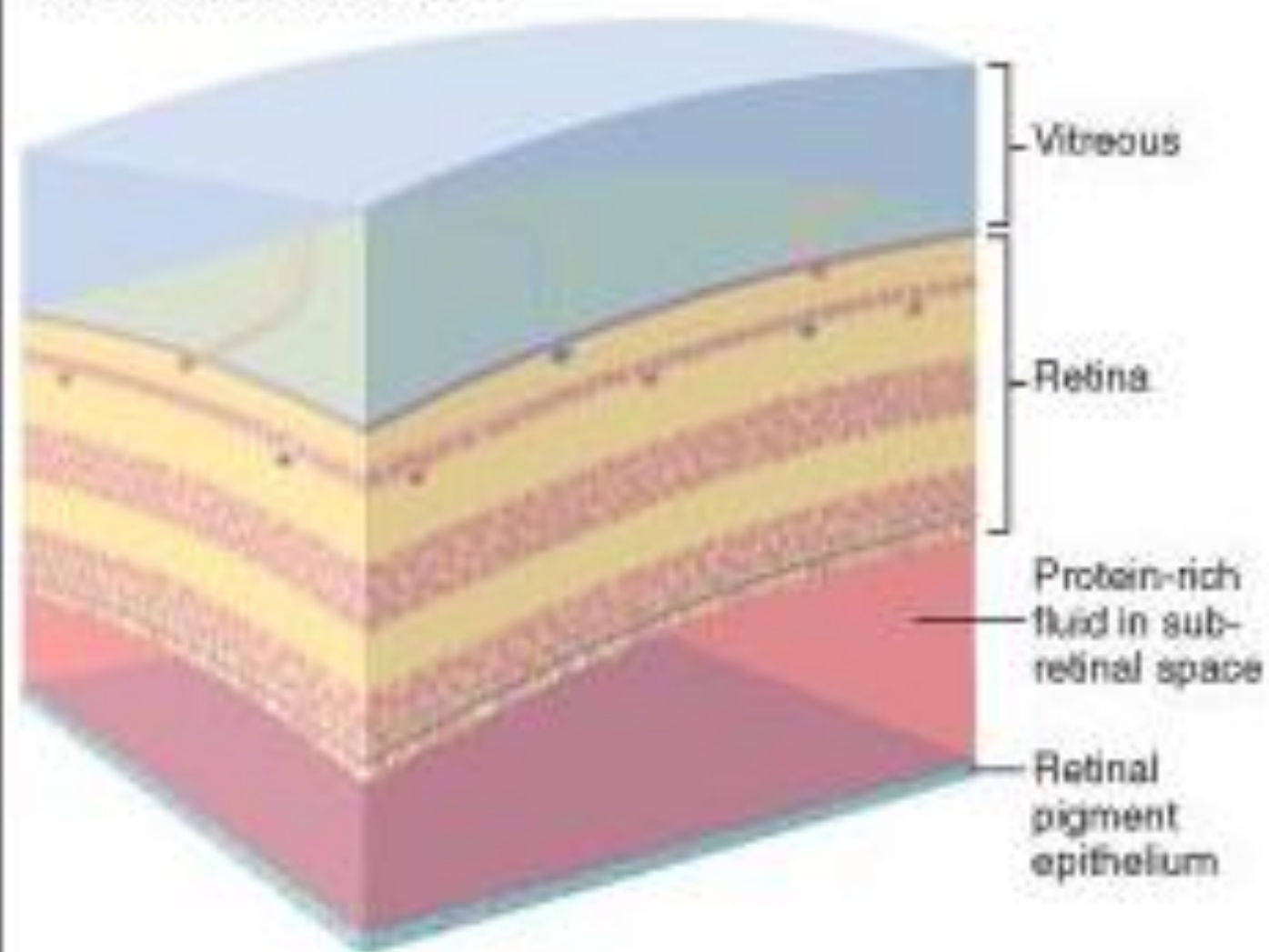
Detachment



Two Types

- Non Rhegmatogenous
 - With out a break in the retina
 - Due to accumulation of protein rich fluid
 - Hypertension
 - Choroidal Tumors

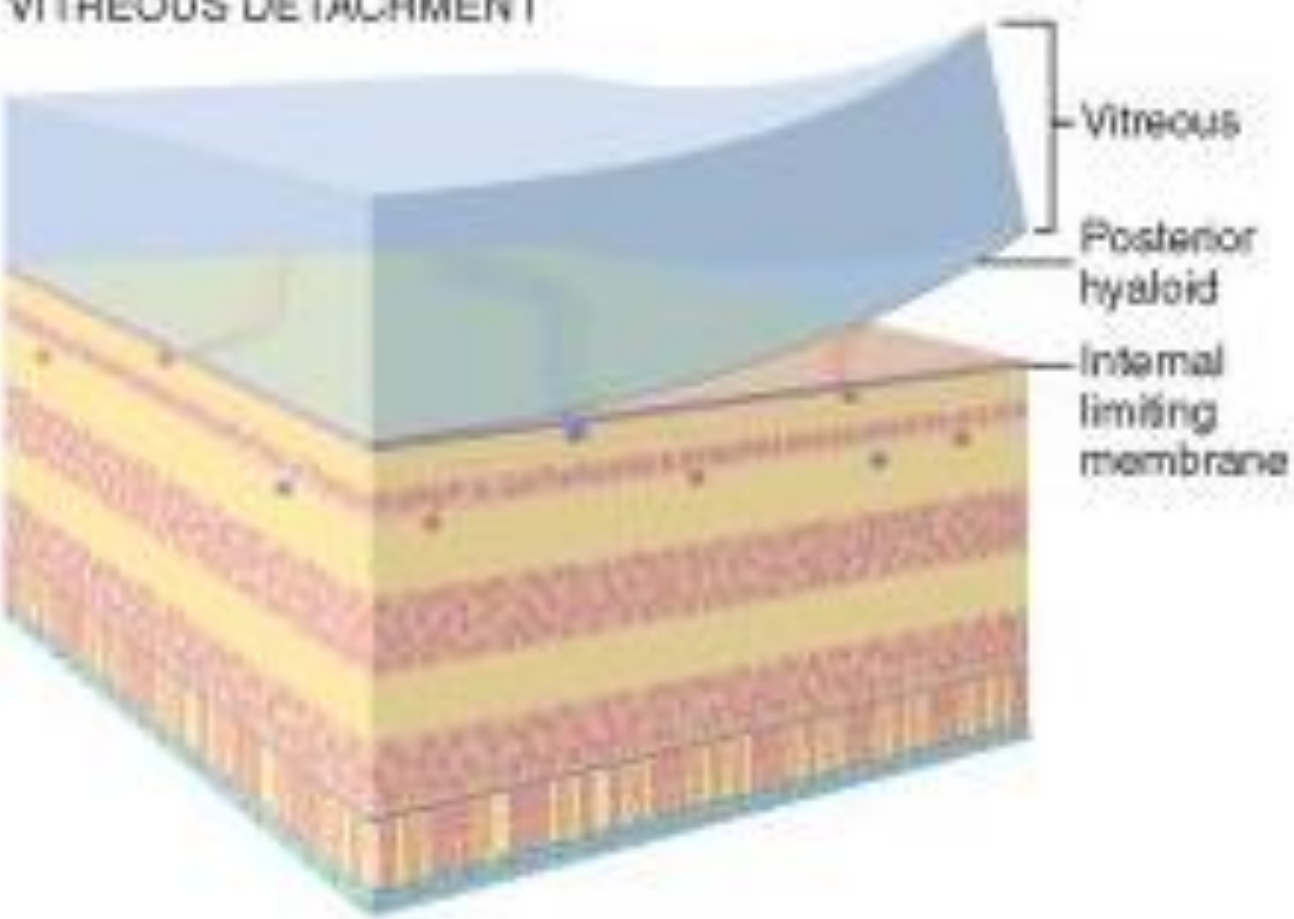
NON-RHEGMATOGENOUS RETINAL DETACHMENT



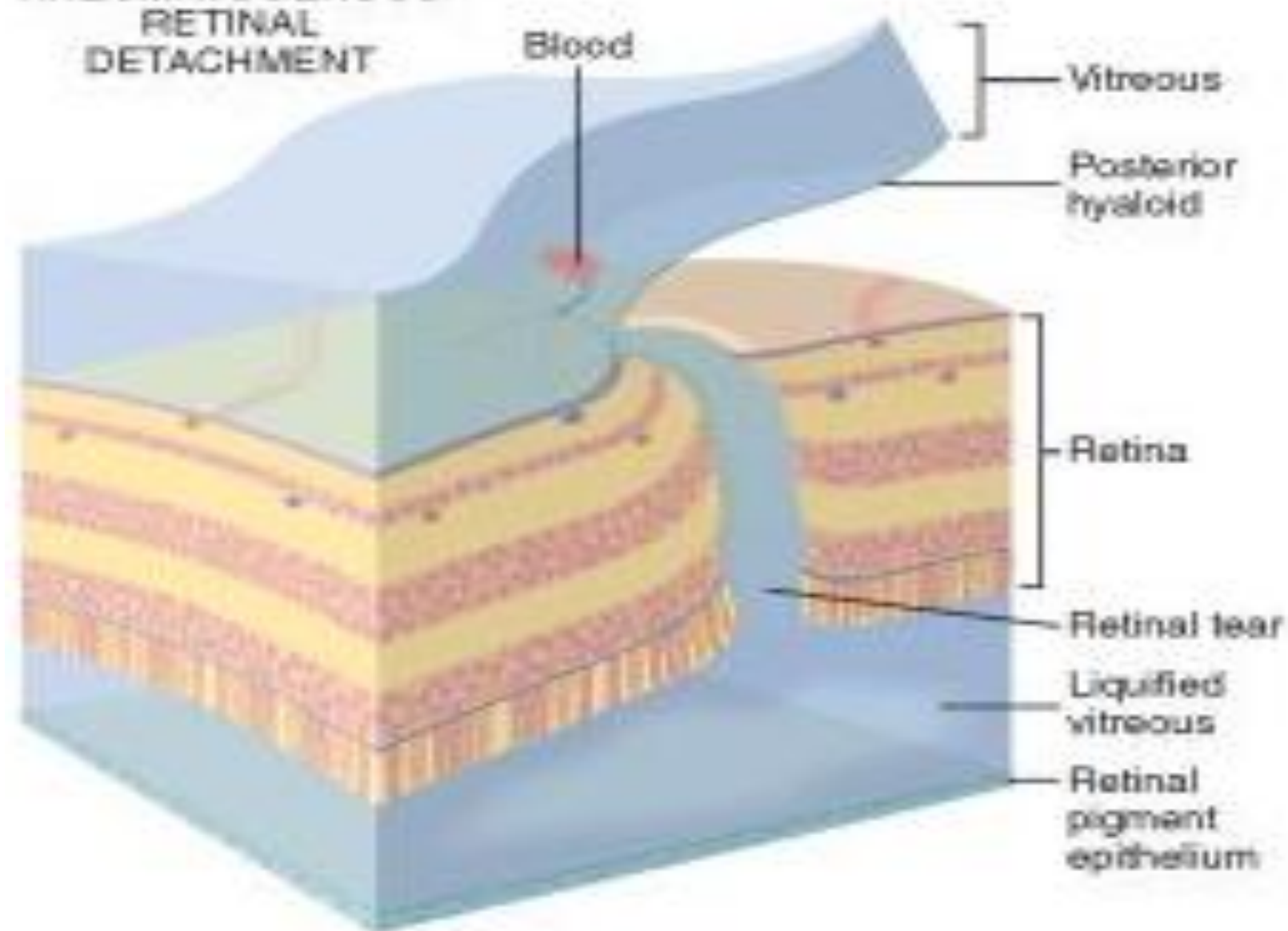
Rhegmatogenous

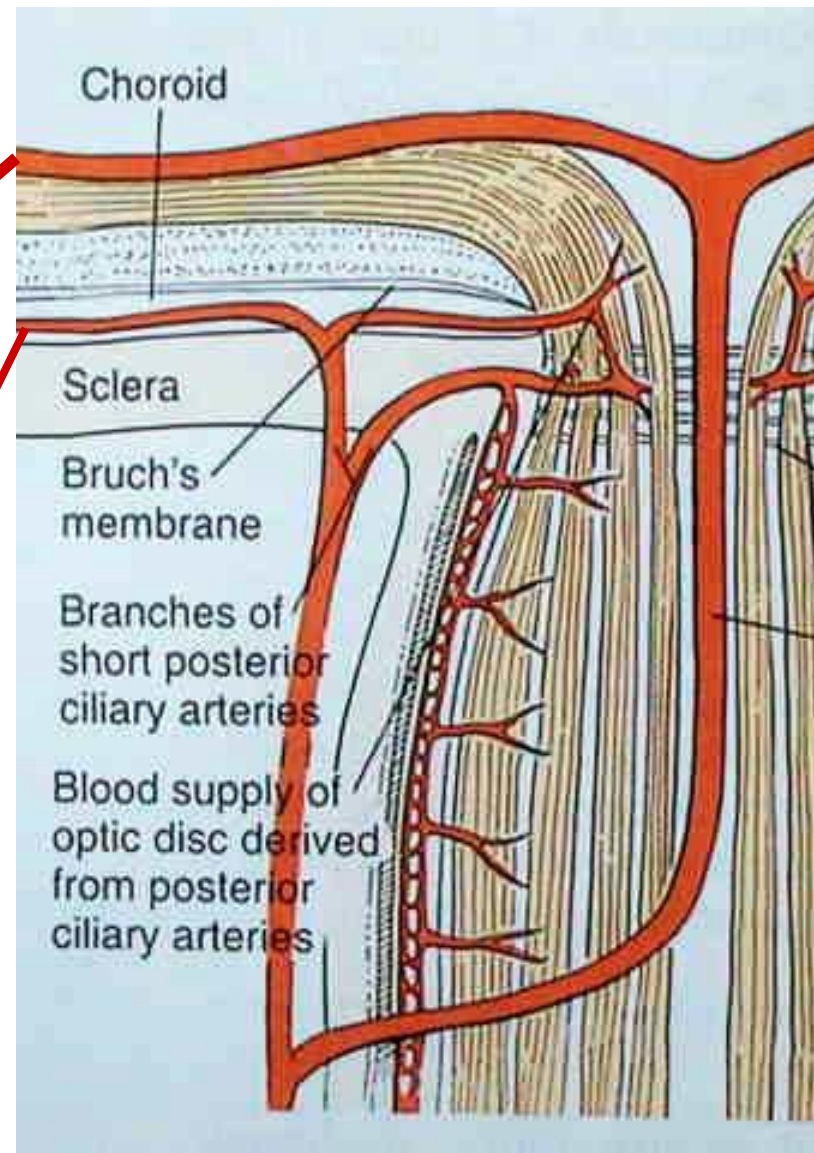
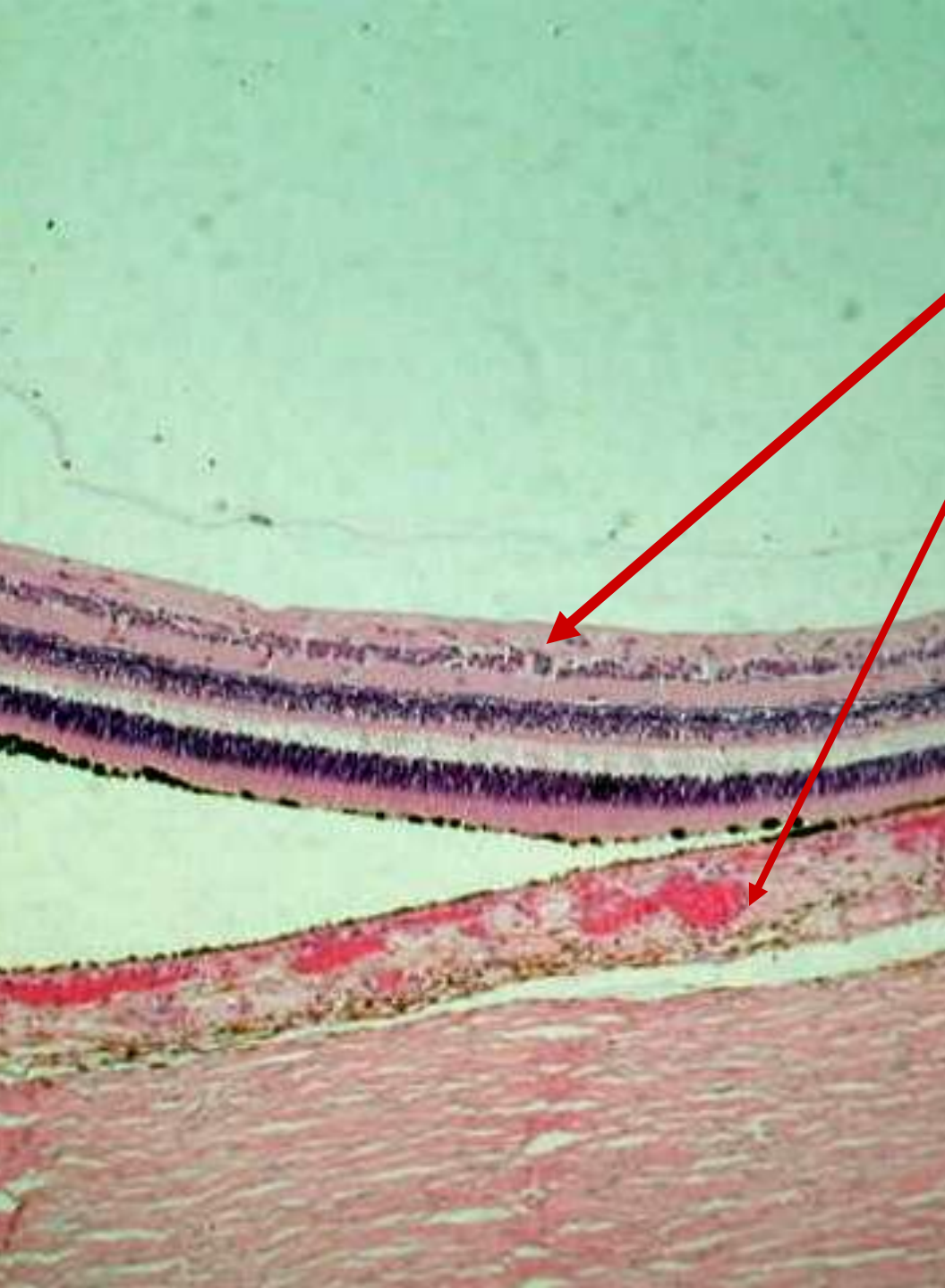
- Associated with a defect in the retinal wall
- The posterior hyaloid(link b/n vitreous humour and retina) fails to cleanly separate from the retina.

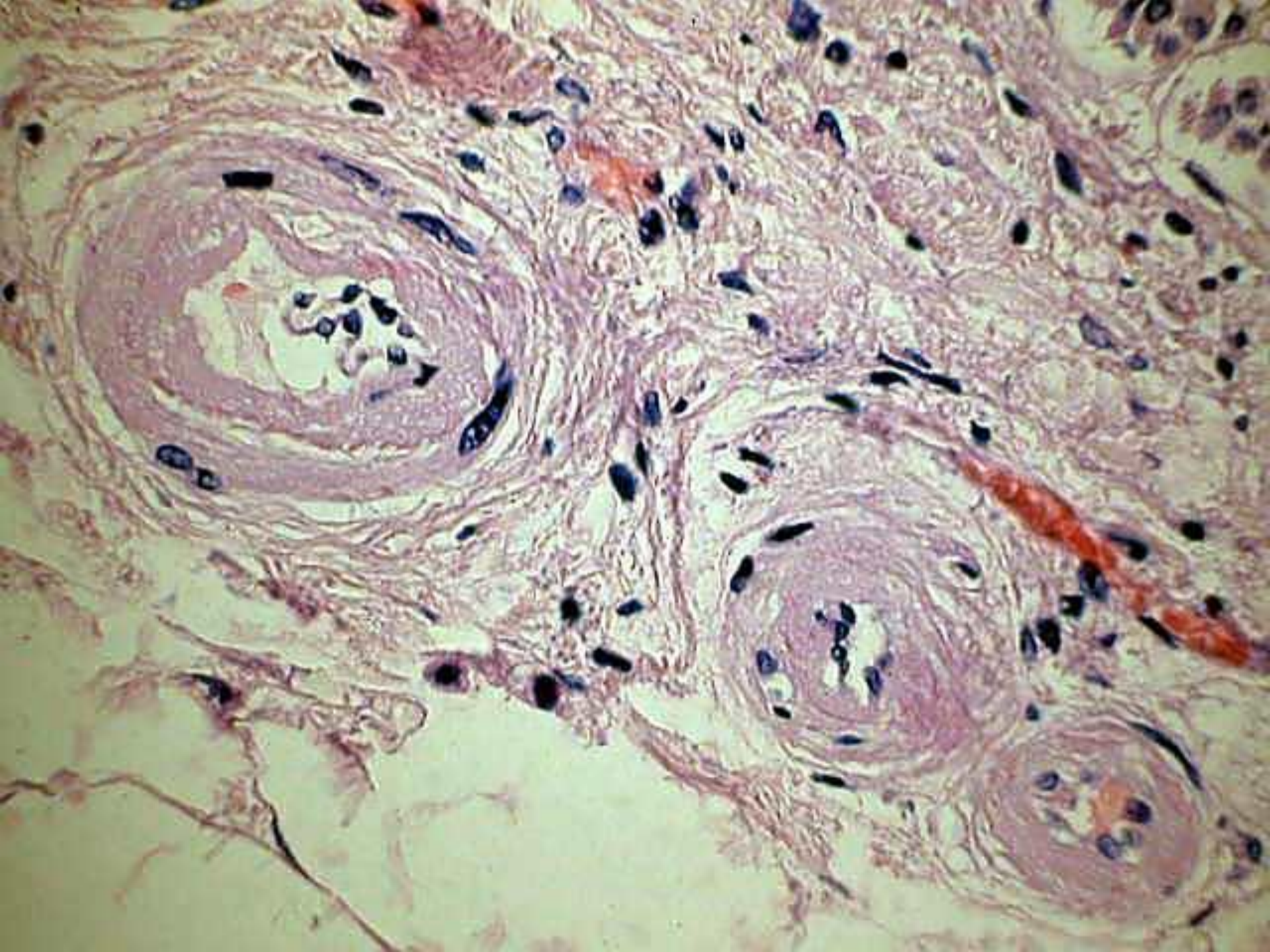
VITREOUS DETACHMENT



**RHEGMATOGENOUS
RETINAL
DETACHMENT**

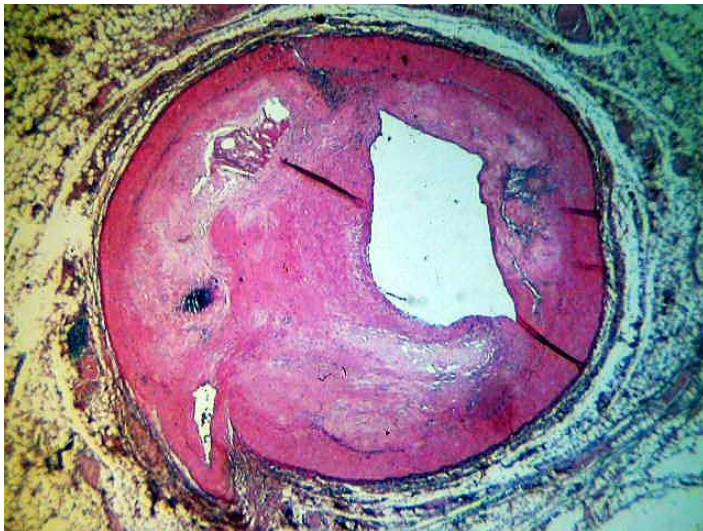




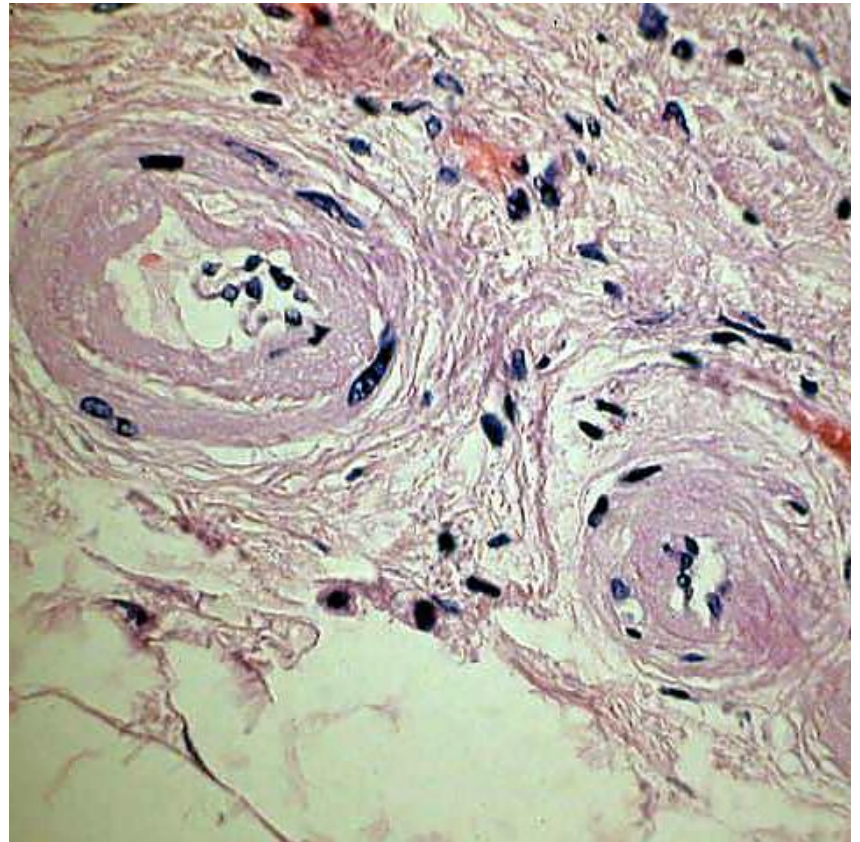


Arteriosclerosis

- Atherosclerosis

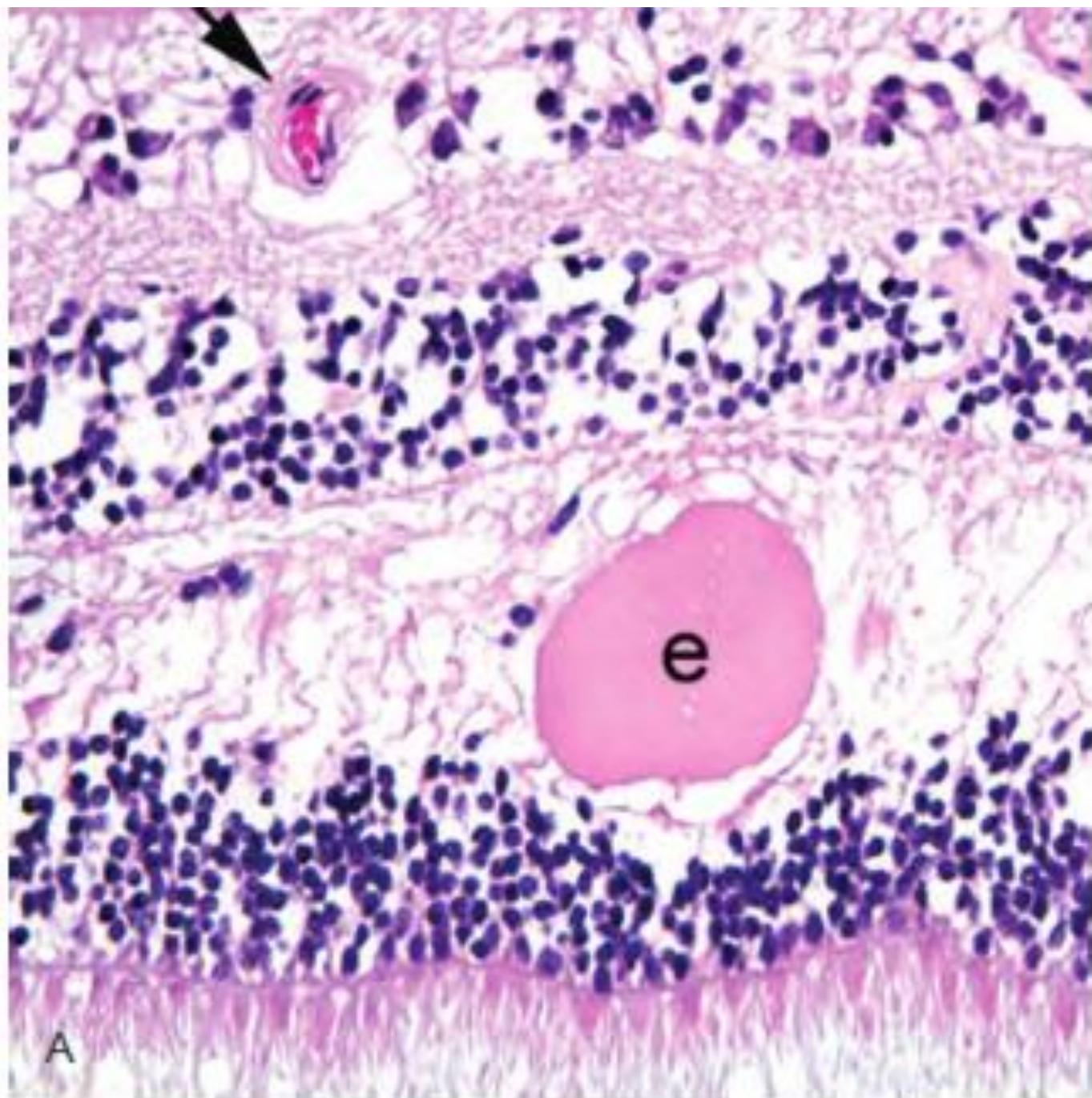


- Arteri~~o~~losclerosis

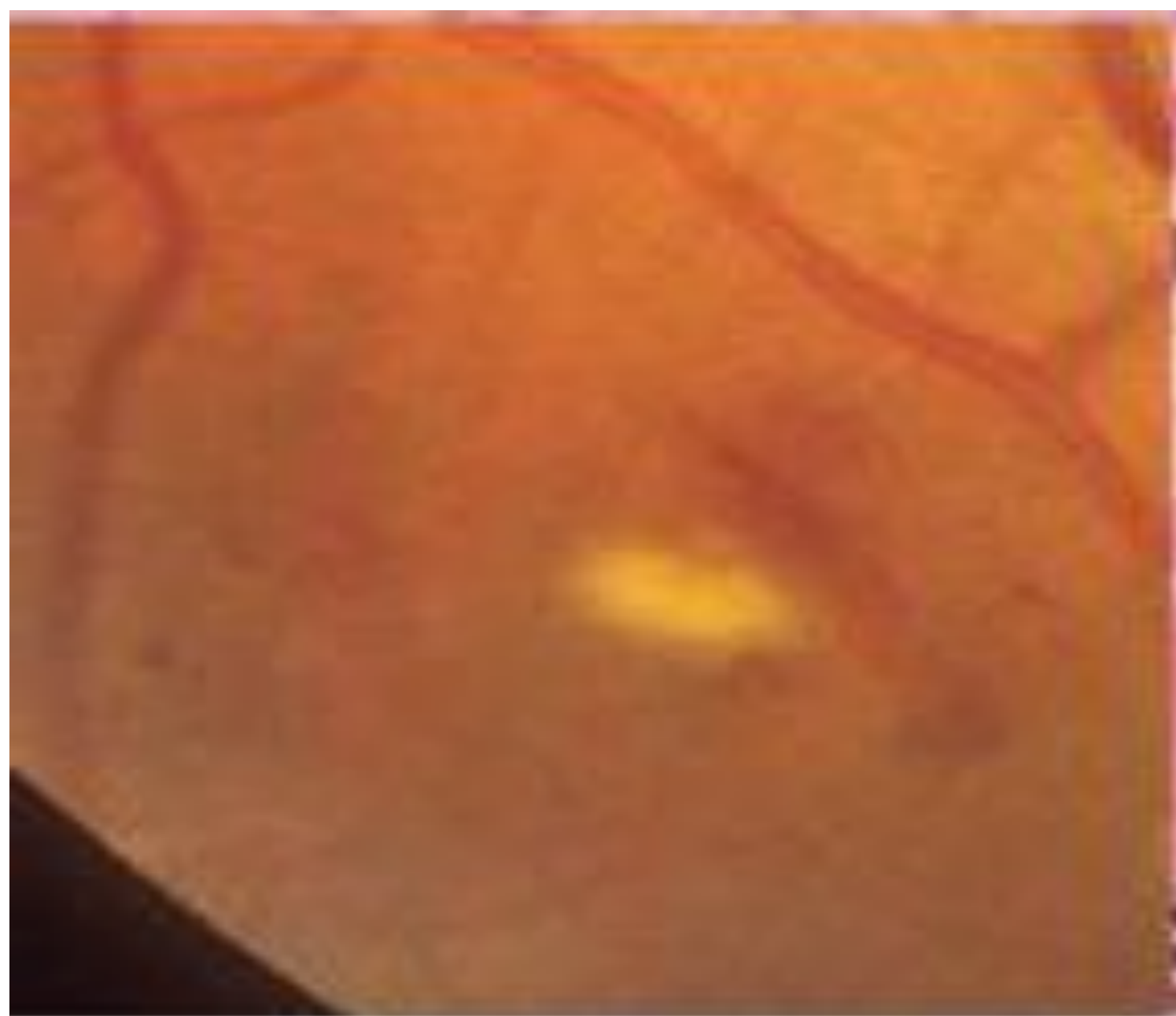


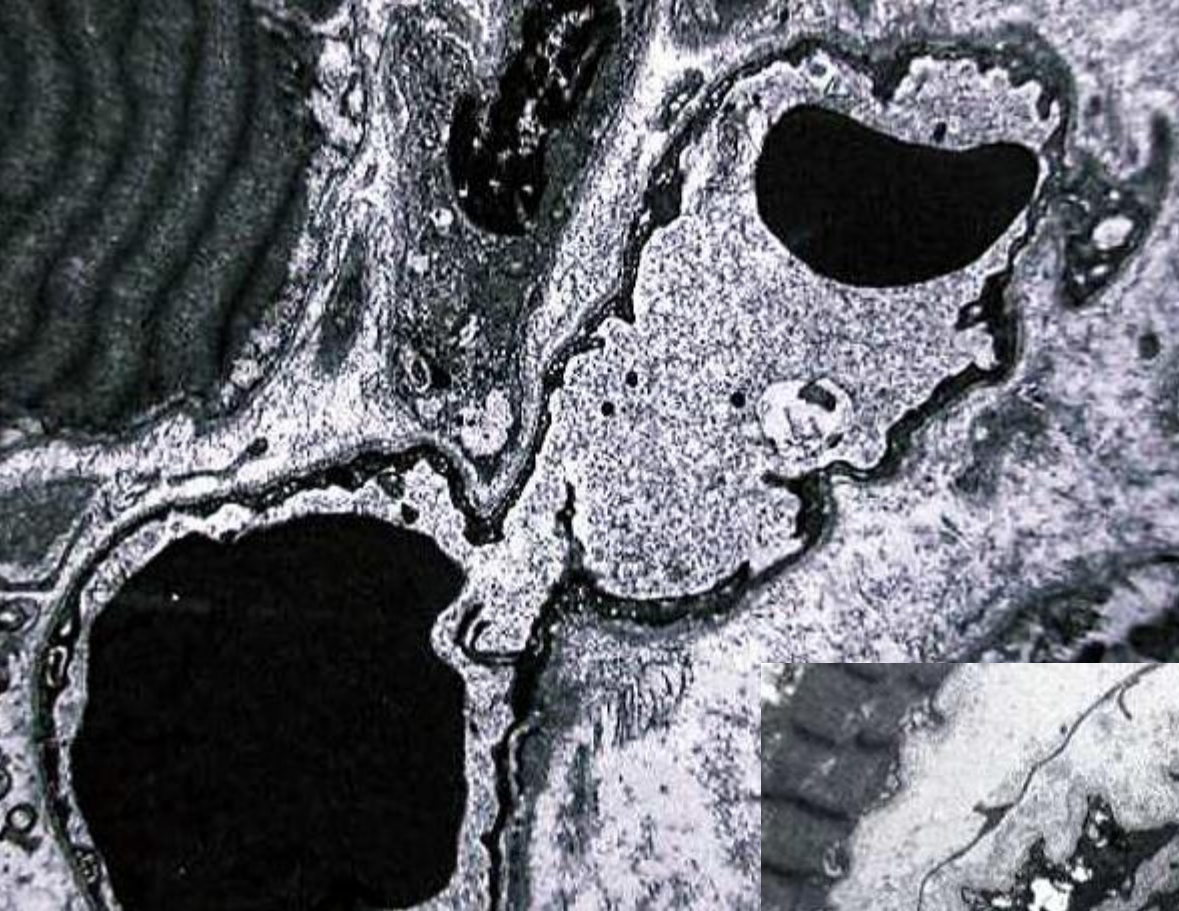
Choroidal arteriolosclerosis

- Seen in hypertension
- Focal Choroidal Infarct
- Accumulation of exudative fluid(if capillaries are damaged), leading to detachment of pigment epithelium
- Retinal Artery involved- Focal retinal infarcts(Cotton-Wool) spots
- Narrowing of retinal arteries
- Arteries may compress veins.
- ON ophthalmoscope, narrower and pallor arteries
 - Copper wiring and silver wiring.





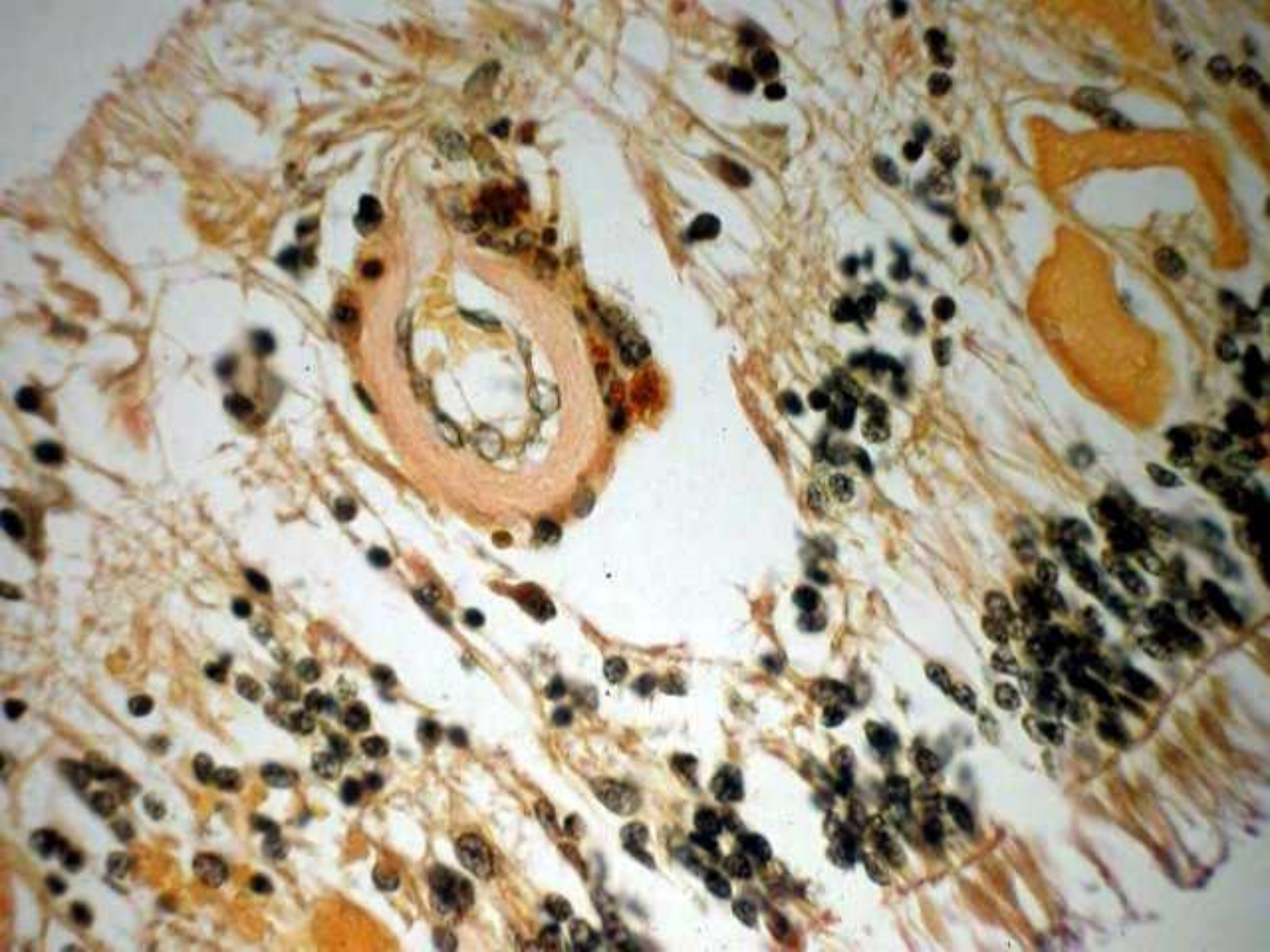




Diabetes



Microangiopathy



Mechanism

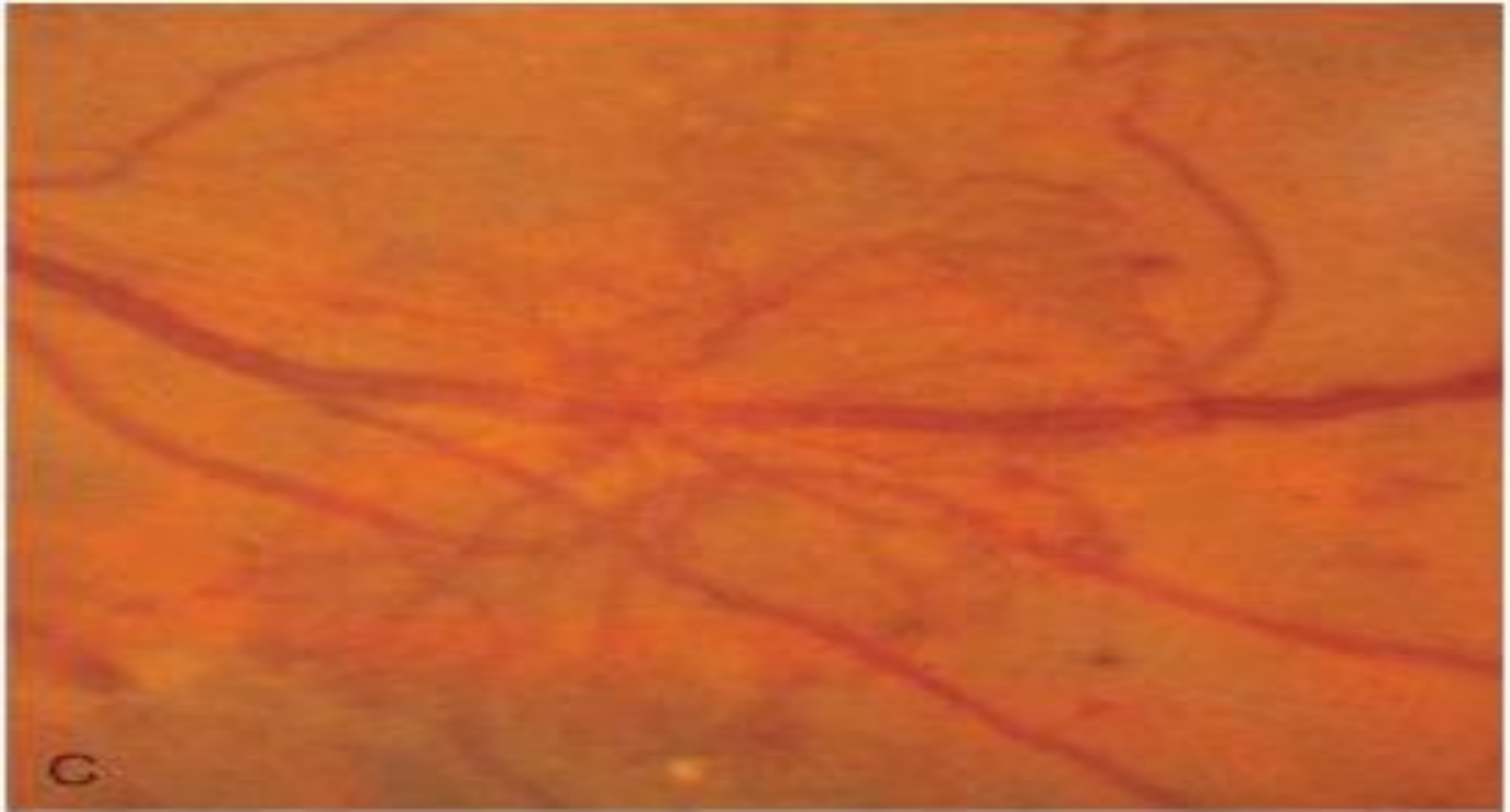
- Thickening of basement membrane of blood vessels
- Endothelial Dysfunction
- (Both associated with non-enzymatic glycation)
- Preproliferative and Proliferative

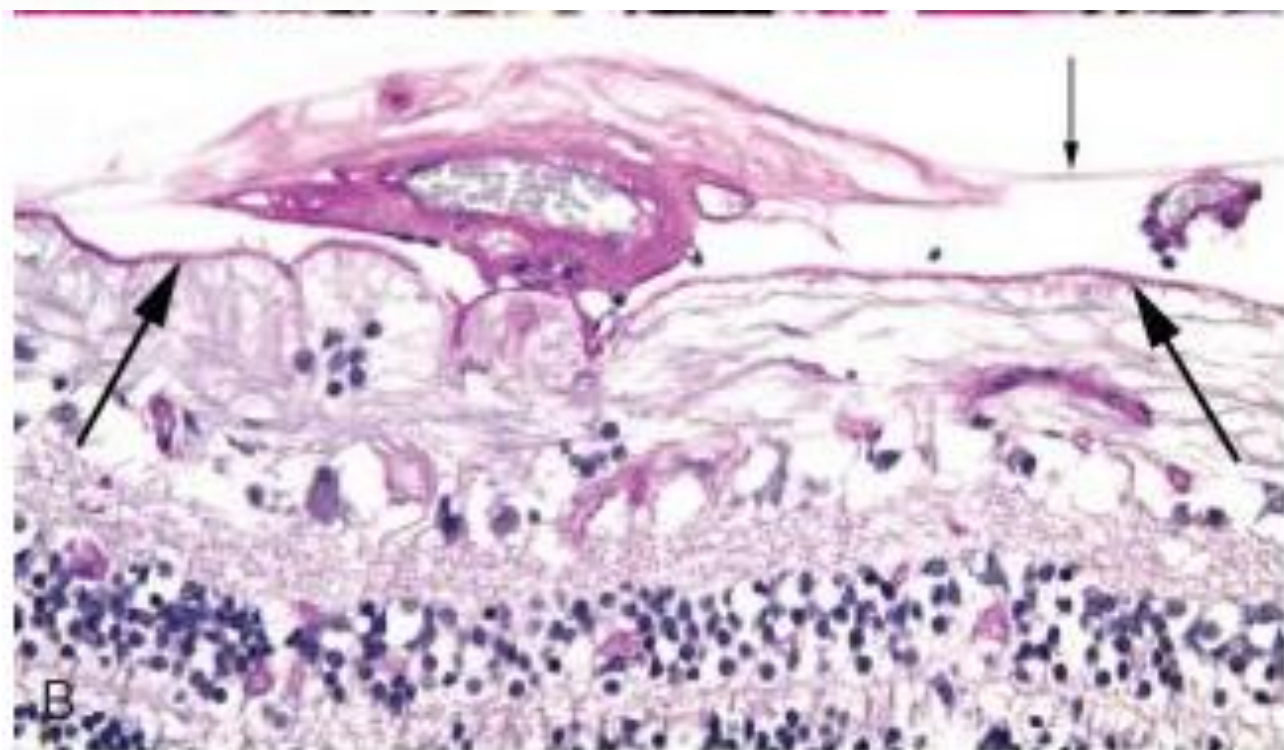
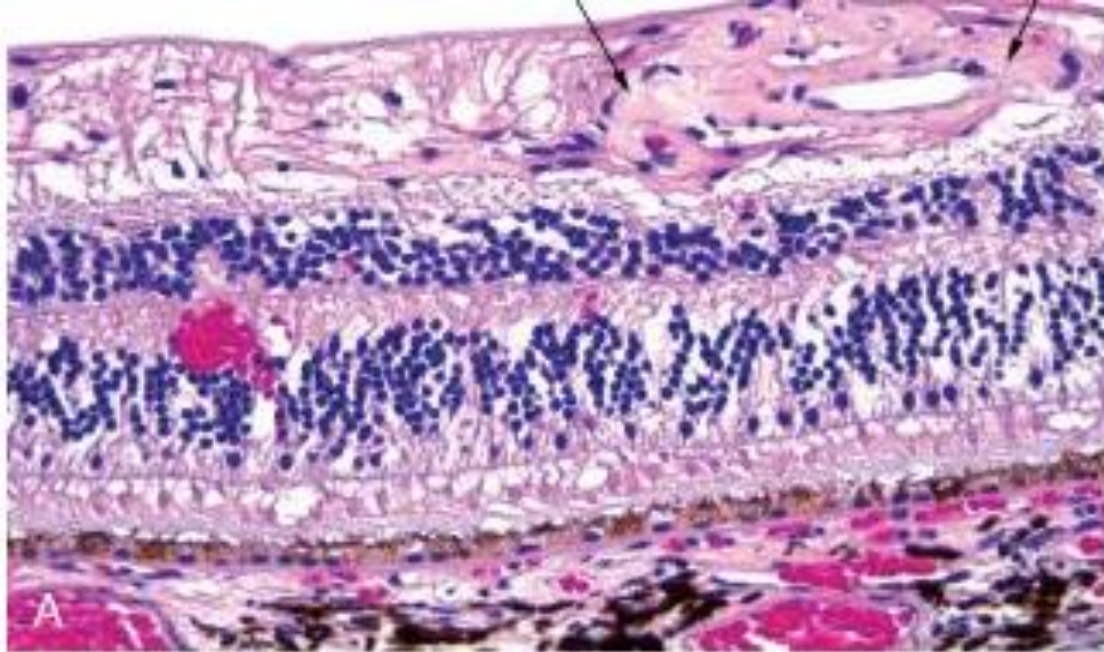
Preproliferative

- Microaneurysms
- Macular edema-due to leaky capillaries(endothelial dysfunction)
- Exudates-on outer plexiform layer.

Proliferative

- Chronic non-perfusion=>secretion of VEGF=>Retinal neoangiogenesis





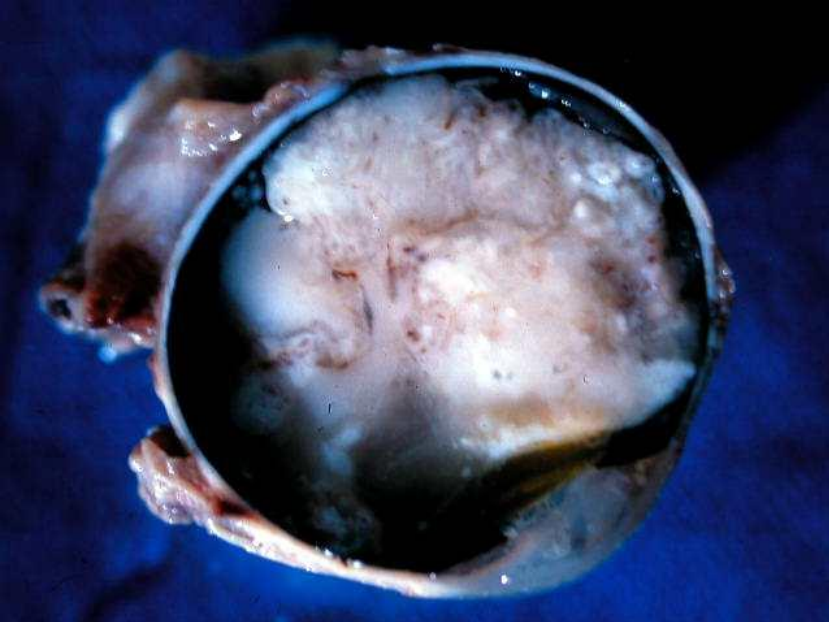
Complication

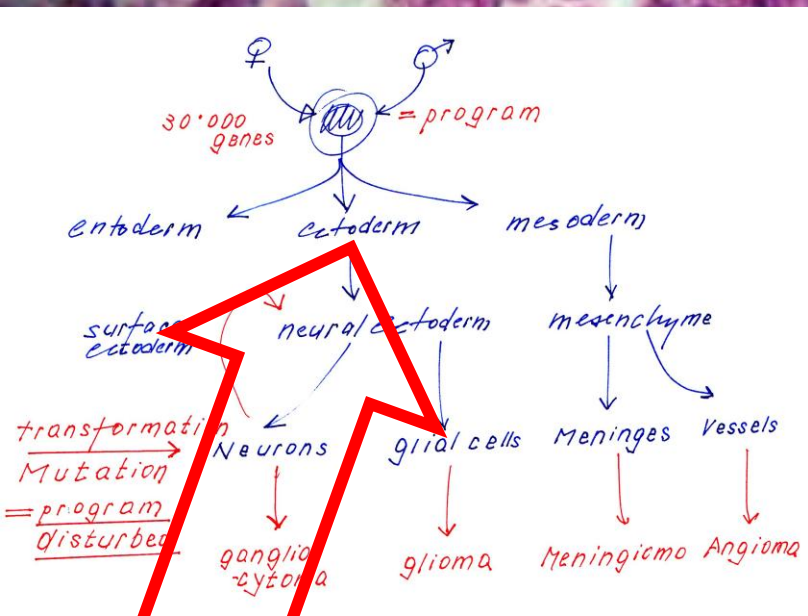
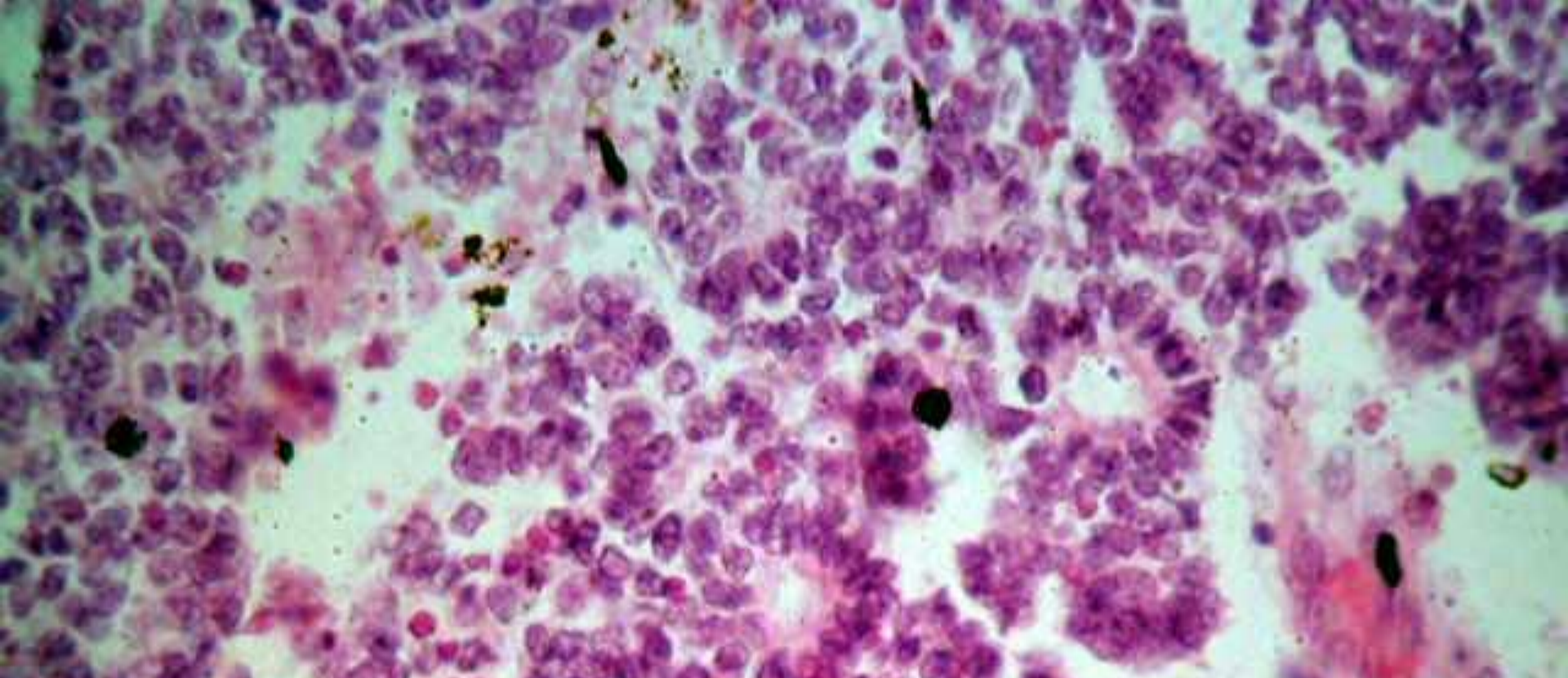
- Obstructs vision
- Large incidence of hemorrhage incase of vitreous separation

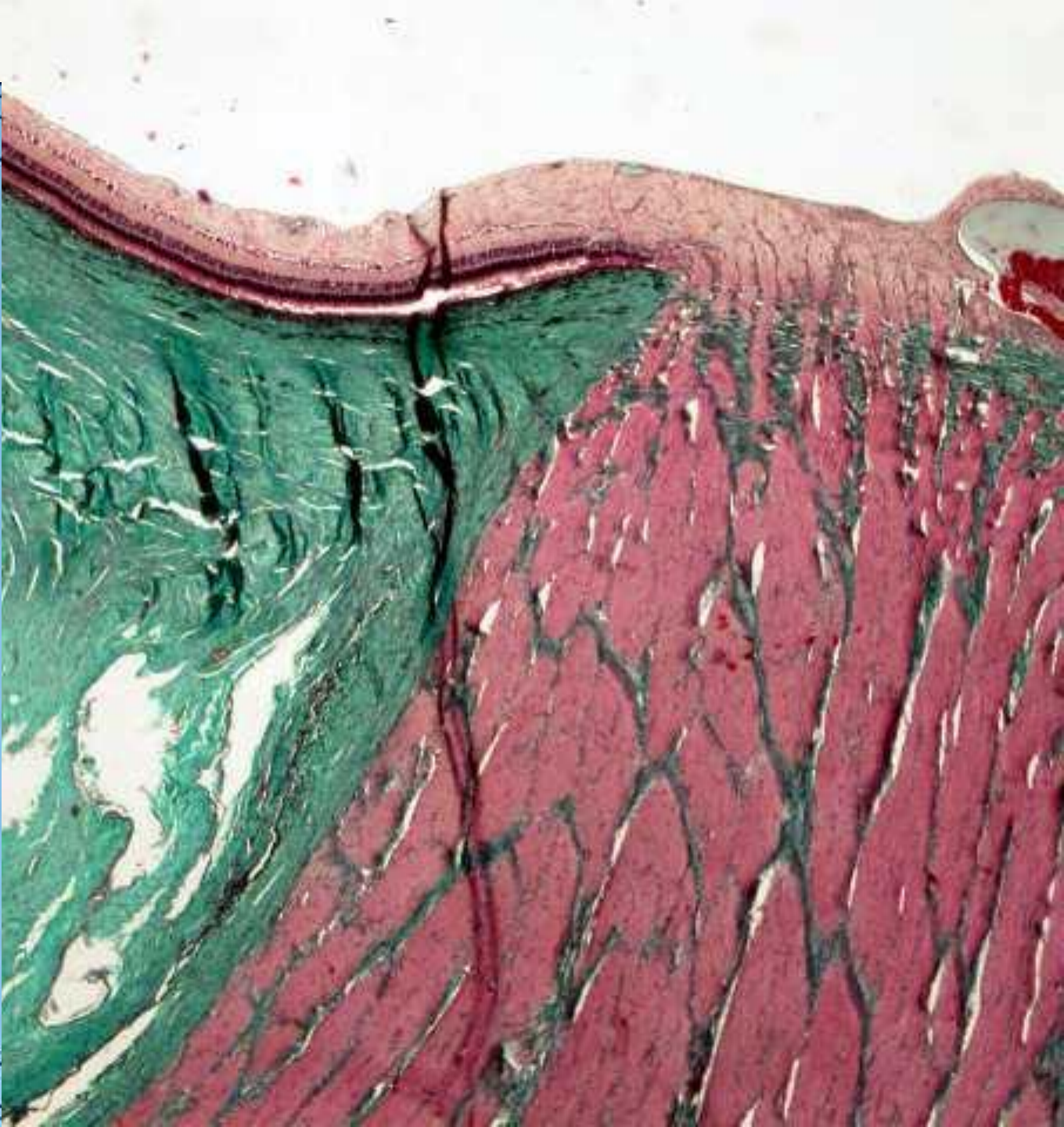
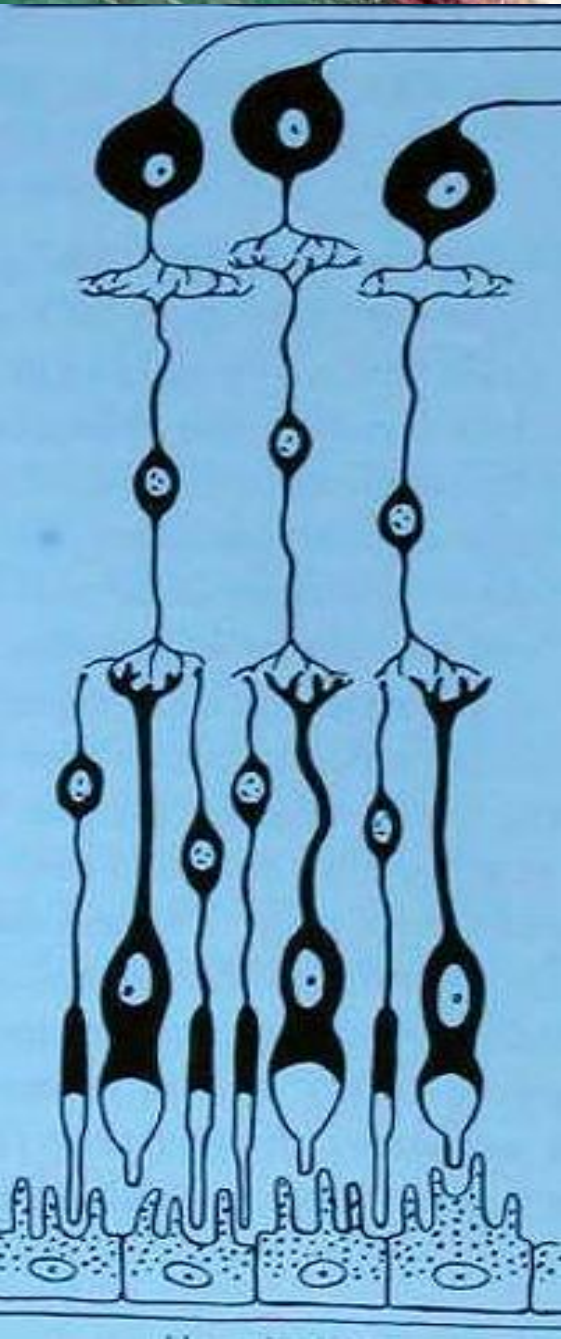


retinoblastoma



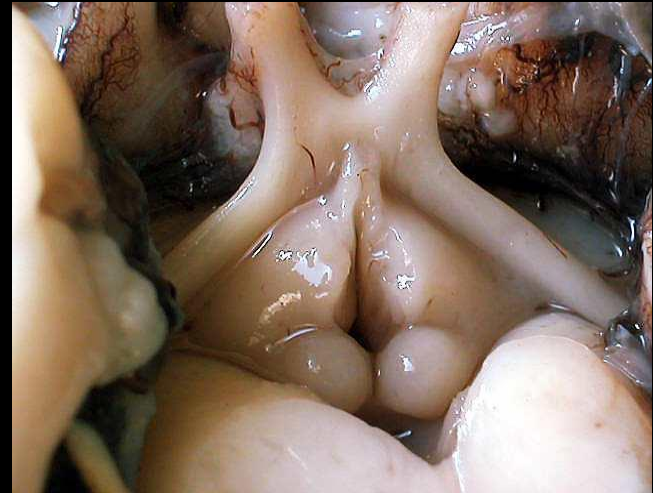




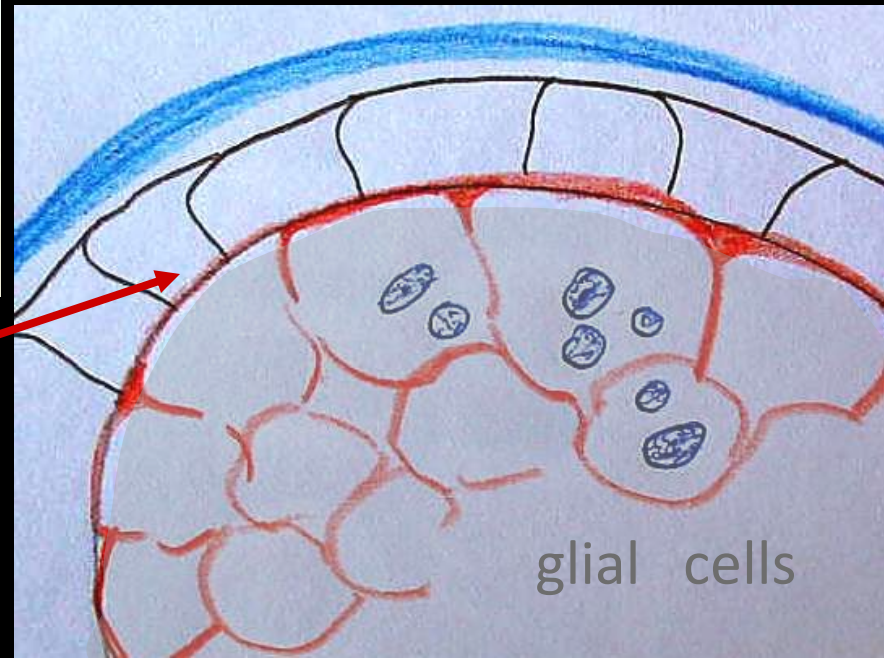




optic nerve
= white matter



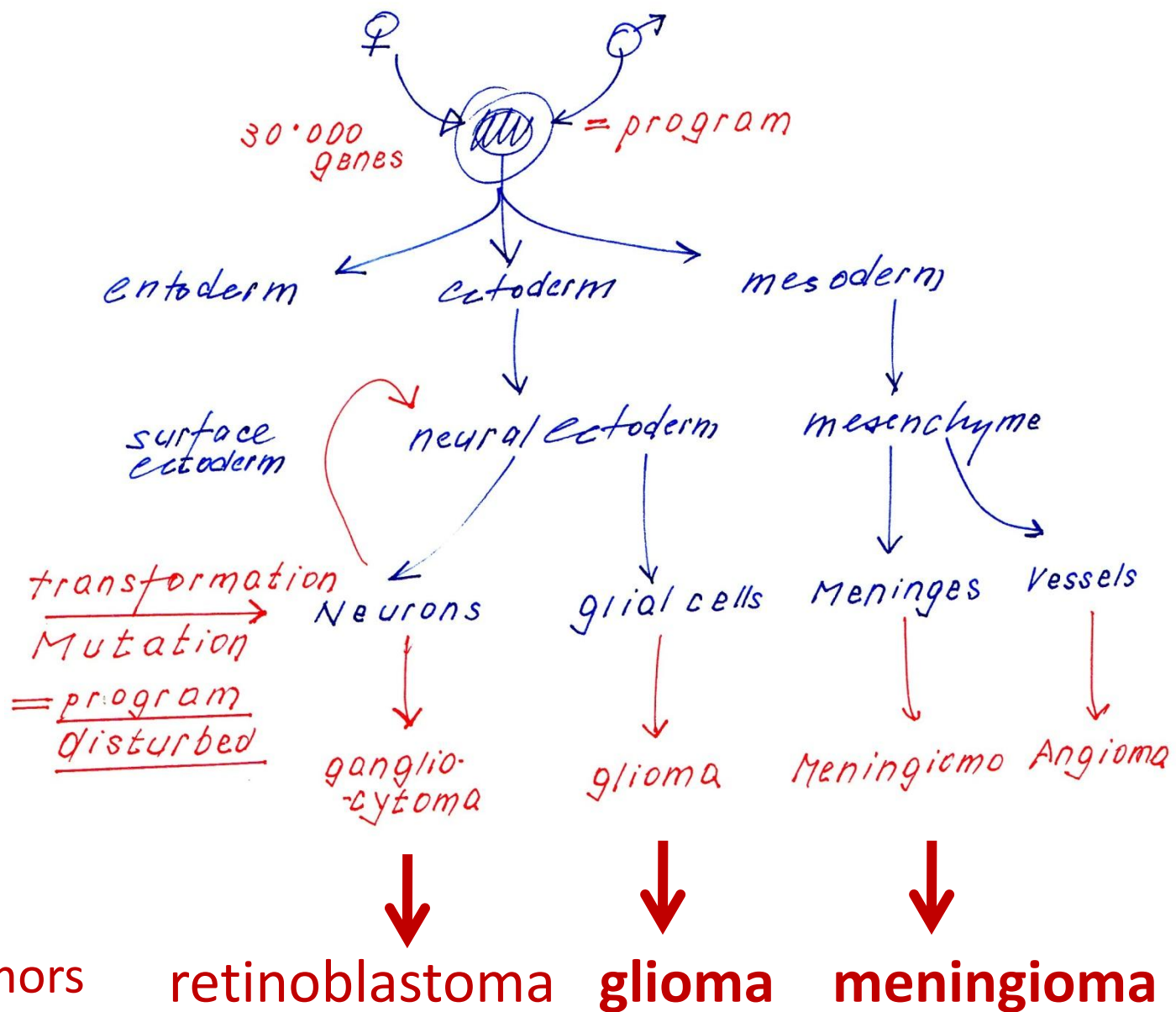
subarachnoid space



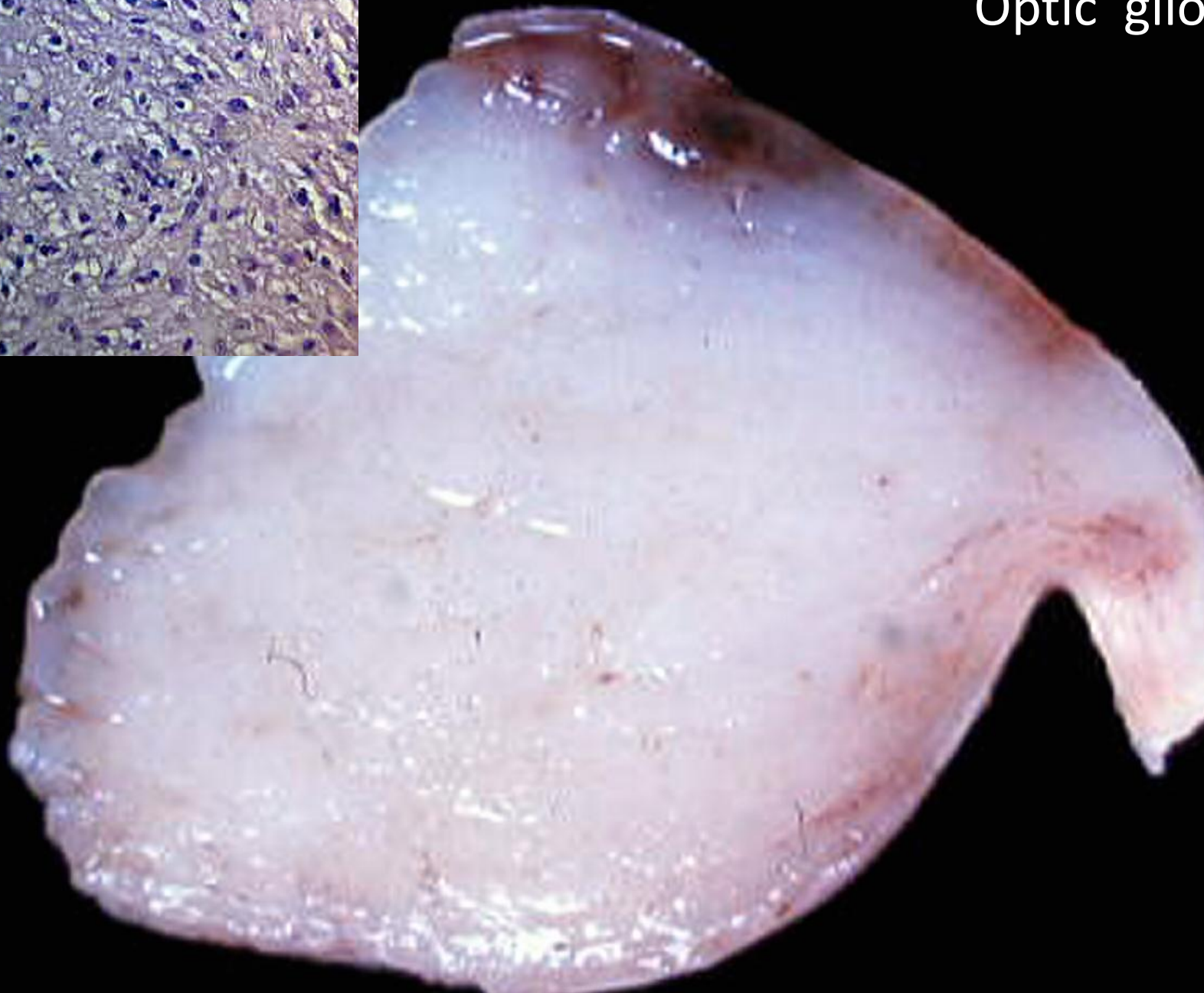
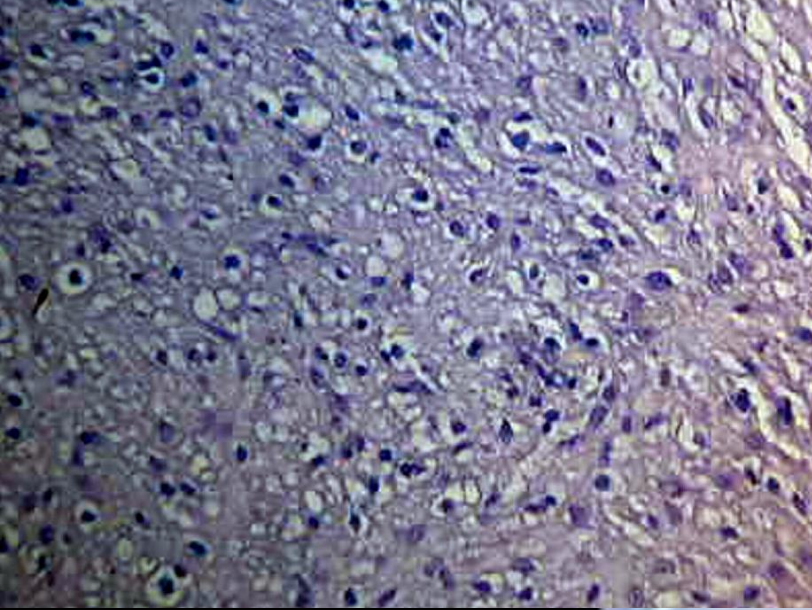
glial cells

Retina
optic "nerve"

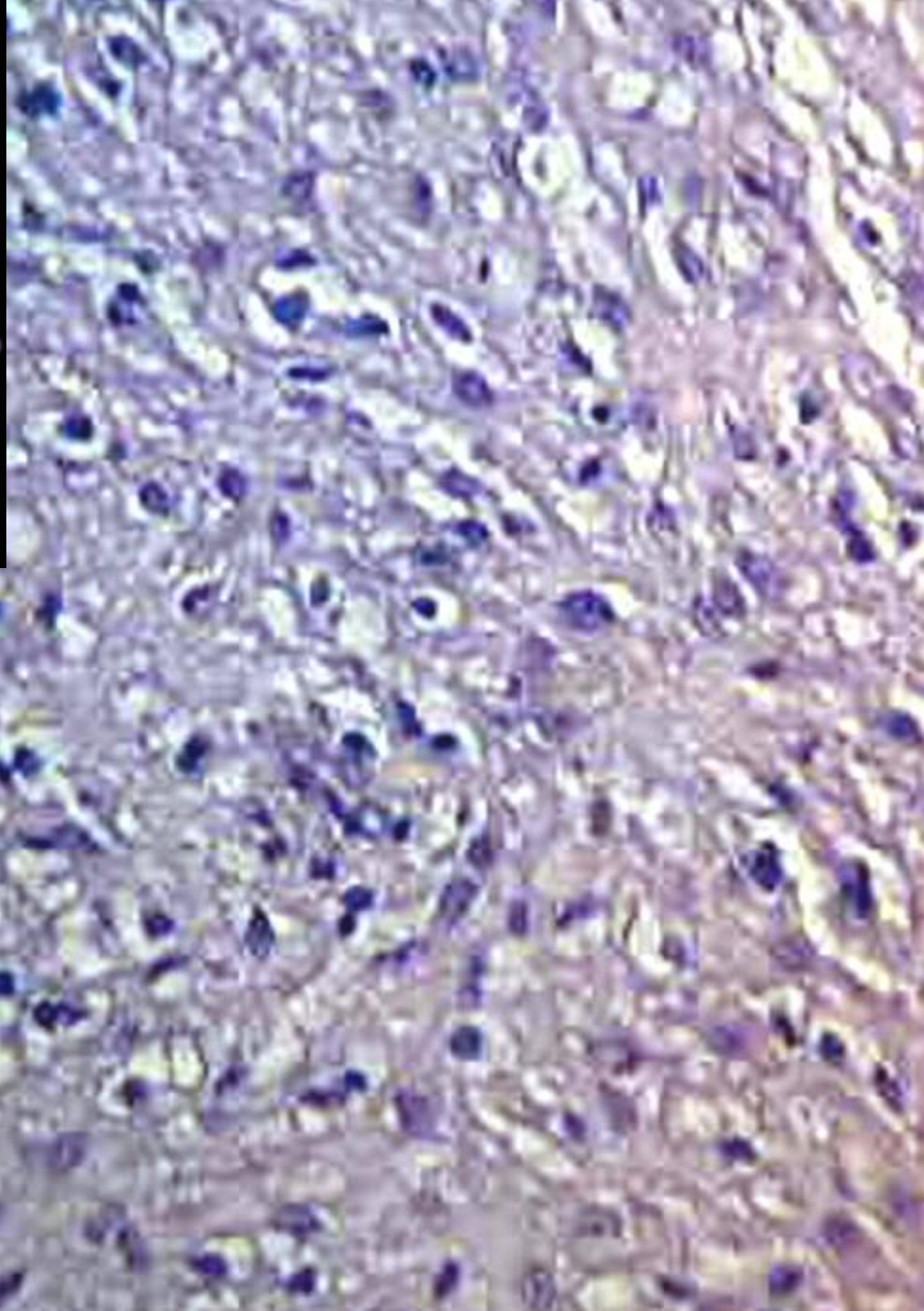
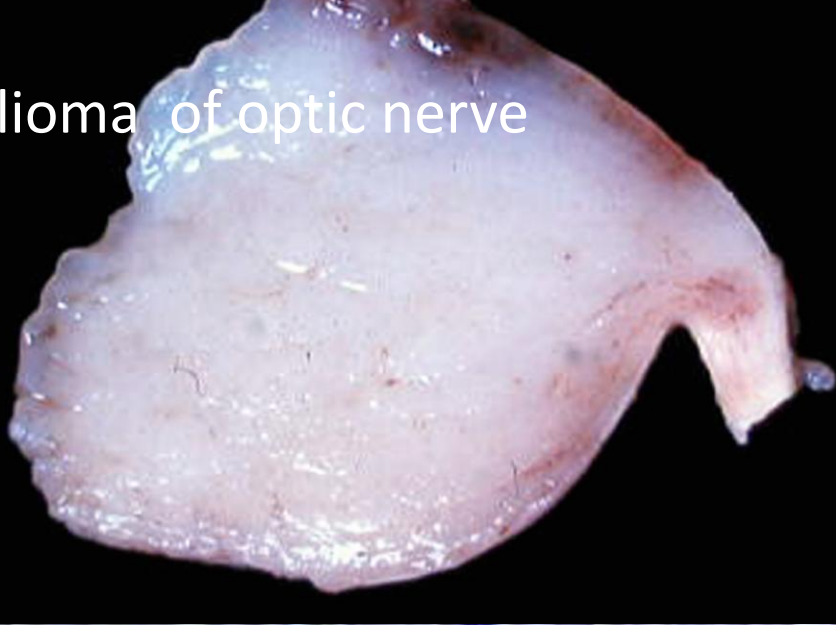
= brain



Optic glioma



glioma of optic nerve

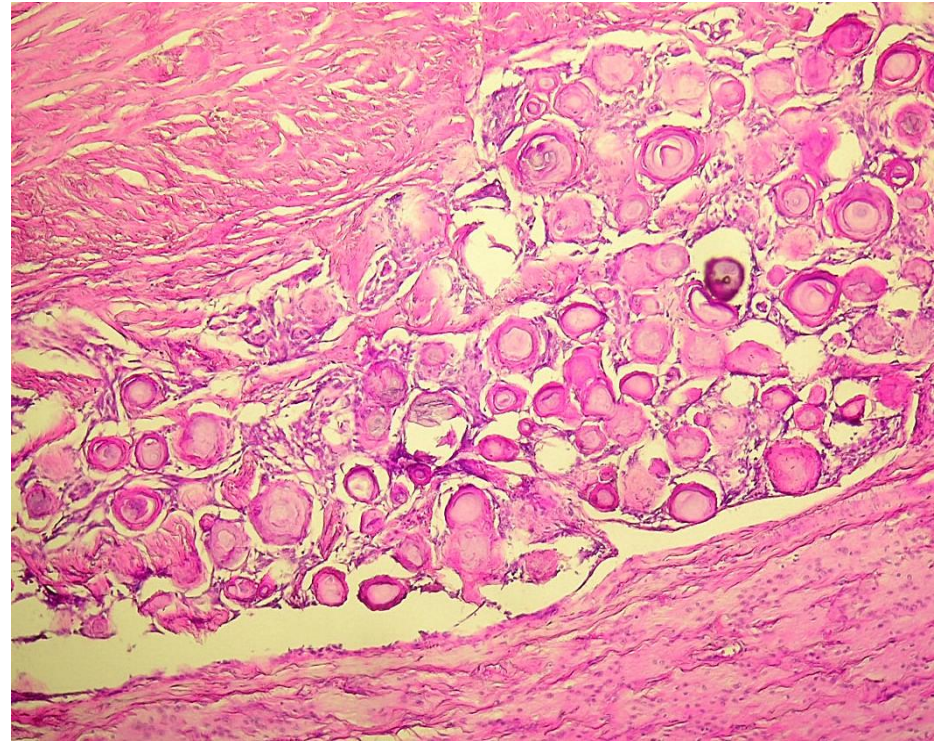
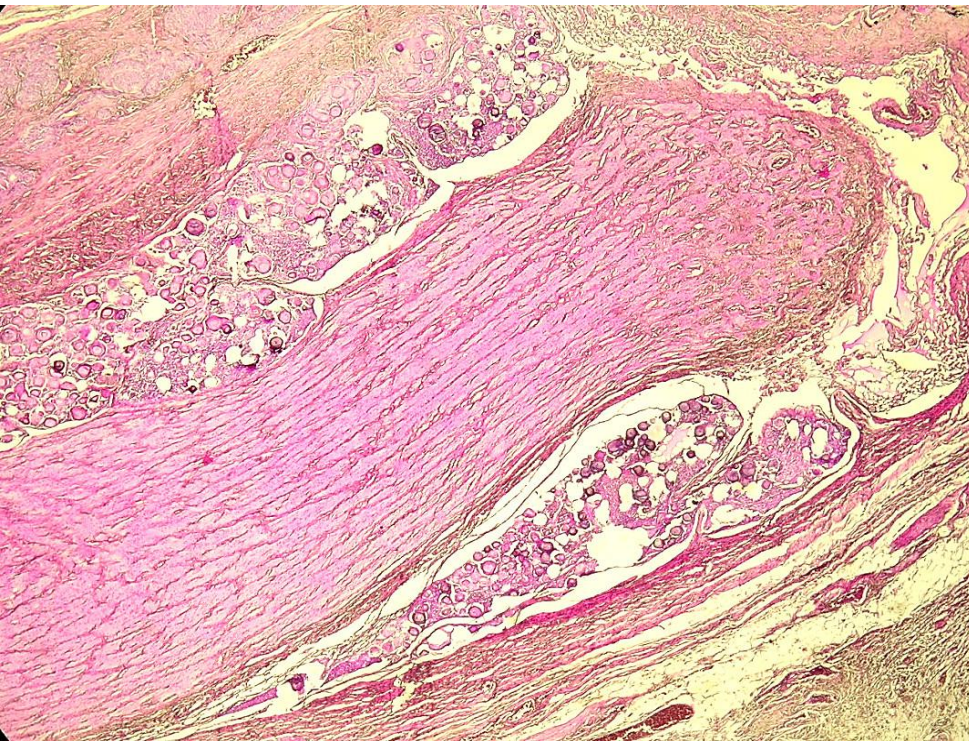




M 16

Meningioma

(psammomatous) of the
optic nerve



Conjunctiva:

Pinguecula

Pterygium

Trachoma

Squamous cell carcinoma

Anterior segment:

Glaucoma

Cataract

Orbit:

Thyroid ophthalmia

Infection (sinuses)

Rhabdomyosarcoma

Metastases

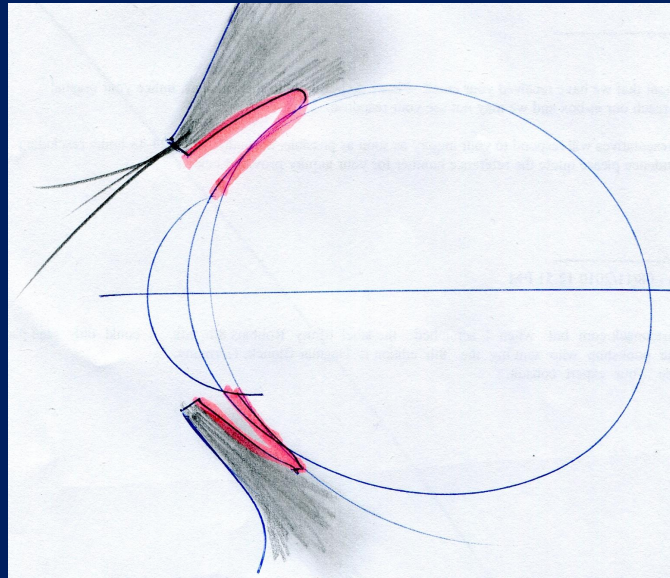
Lymphoma

Retina:

Diabetes, Hypertension

Retinoblastoma

Detachment



Optic:

Papilledema

Glioma

Meningioma

Uvea:

AIDS: Infections

Melanoma

Sarcoidosis

